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Road-Conditioned Diffusion for Traffic Trajectory Synthesis

Complete Journeys, Class-Conditioned Motion and Hilbert–GAF Feedback

Supervisor
[Supervisor name]

Candidate
[Candidate name]

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Abstract

This thesis studies complete traffic trajectory synthesis under road constraints and the use of trajectory-derived class representations as feedback. A training-side prior samples duration, a masked temporal diffusion model generates progress on a physical clock, and a route decoder produces timestamped positions through arrival. The output is synthetic data in the form `time,id,x,y,type`.

Spatial feedback inherits the original position–Hilbert–GASF/GADF/MTF construction. Velocity feedback encodes speeds derived from the same positions and timestamps. Both participant classifiers are trained within this project. Source comparisons motivate native Monaco SUMO Traffic demand, preserving service routes and activity structure. Recognition, candidate ranking and physical generation quality are evaluated separately.

A matched temporal-resolution pilot supports retaining the quarter-second physical clock: native-time diffusion reduces native-acceleration distribution discrepancy by 81–82% relative to a 257-point model. The current training corpus contains 29,895 complete journeys from 24,073 parent vehicles, with 7,230 development legs. Spatial and velocity GAF readers attain development macro F1 of 76.47% and 90.34% across bus, motorcycle, private car, bicycle and taxi/ride-hailing.

Two new SUMO episodes evaluate 100 prospective requests each with two model seeds and eight candidates per supported condition. Physical quality is paired on 95 and 94 requests. All five classes are examined through generated journeys, exact image representations and independent movement measures. The class-level analysis finds representation-dependent gains and costs: spatial feedback reduces motorcycle acceleration-distribution error by 10.3%, while velocity feedback reduces bicycle acceleration error by 9.8% but raises its speed error by 5.4%. Bus stopping errors improve under several policies, whereas taxi/ride-hailing retains adverse motion outcomes. These are descriptive scene/seed summaries; no universally superior selector is established.

The thesis delivers an executable road-conditioned generator and a controlled investigation of class-related motion feedback. It identifies class- and motion-specific feedback patterns for independent follow-up. The evidence concerns terminal selection on shared public road templates, not denoising-time guidance, unseen-city generalization or proven downstream augmentation value.

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Chapter 1

Introduction

A synthetic traffic trajectory must describe both where a participant travels and how that journey unfolds in time. A bus can follow its permitted route and reach the correct destination while exhibiting implausible movement between the endpoints. It may omit sustained stops, alternate too rapidly between acceleration and braking, or replace a long journey with an almost constant speed. Conversely, a plausible speed curve does not determine which connected streets the participant can use. These two requirements meet in the output of this thesis: a complete sequence of timestamped positions for a requested vehicle class on a specified road route.

The intended product is a dataset in the familiar `time,id,x,y,type` format. Each generated identifier represents a new sampled journey, with a sampled duration and positions through its destination. The road network and request determine its spatial support; a learned generative model determines its progression along that support. Repeated requests can therefore produce different temporal realizations of the same route. The usefulness of this output depends on the resulting motion distribution, its diversity and its coverage of the requested conditions. A visually convincing route or a high class probability alone does not establish these properties.

This work brings together two lines of the project. The first is conditional diffusion for trajectory synthesis, situated within research on diffusion models and road-constrained trajectory generation [4, 6, 7]. The second is the group's use of images derived from trajectories for participant recognition [1, 2]. Their connection is a concrete empirical question: once a diffusion model already receives the requested class, does feedback from a spatial or velocity Gramian Angular Field representation improve the trajectories that it delivers? Establishing this connection requires examining what the representation can distinguish within a generated candidate set, and whether its preference agrees with physical quality measured independently of the classifier.

1.1 A complete-trip synthesis problem

Let c denote a participant class and let r be a directed, permitted route with specified endpoints. Write its coordinate interpolation as $P_r(s)$ for normalized progress $s \in [0,1]$. A generated trip takes the form

$$\mathcal{T} = \{(t_i, P_r(s_i), c)\}_{i=0}^{n}, \qquad t_0 = 0, \quad t_n = T, \quad 0 = s_0 \leq s_1 \leq \cdots \leq s_n = 1. \tag{1.1}$$

The duration T and the intervening progress sequence vary across samples. Equal successive progress values represent a stop. A change in progress rate represents acceleration or deceleration along the route. The sequence length follows the sampled duration, rather than truncating every vehicle after a fixed observation window.

The implemented generator factors duration and motion. A training-side class- and route-length-conditioned prior samples T, and a conditional diffusion model samples the temporal progress profile. A deterministic decoder then obtains coordinates from the route. This division makes the learned and imposed properties explicit. Route continuity and endpoint agreement come from the common map construction and decoder. The diffusion model must learn how movement is distributed over the trip. Duration is generated by the full system, but it is not attributed to the diffusion component. The empirical motion-transfer comparator uses the same route and duration prior, so the comparison isolates the contribution of learned motion synthesis more clearly.

Training is supervised in the sense that observed route–motion pairs provide the denoising targets. The resulting task is nevertheless generative: the model receives corrupted motion at different noise levels and learns a conditional distribution from which new sequences are sampled. A class label does not supply the particular speed curve, stop sequence or arrival time of a reference vehicle. Nor does supplying a label certify that the generated movement is characteristic of that class. This must be assessed on the samples, including variation among journeys of the same class.

SUMO supplies the learning observations and independently simulated references [3, 47]. It also gives the project a controlled way to retain a road network, public transport services and background demand. The trained model does not implement SUMO's joint traffic evolution. It samples individual trajectories without simultaneously updating all surrounding vehicles, traffic signals or interactions. Consequently, matching the distribution of observed journeys does not prove collision avoidance or interactive traffic consistency. The contribution is a conditional trajectory-data generator with an explicit spatial interface, not a replacement for a microscopic simulator.

1.2 Why examine GAF feedback after class conditioning?

The inherited spatial pipeline maps ordered positions to a Hilbert sequence and then to a three-channel image containing Gramian angular and Markov transition representations [12, 10]. It was developed for recognition: an image of a recorded trajectory is used to predict its participant class. Here, separately trained readers apply that representation to generated trajectories. A second reader uses a velocity sequence derived from timestamped positions. Both readers predict the same five participant labels in the current experiment, while their input information remains separate.

These transformations introduce no new physical observation. Velocity is derived from positions and time, and each image is a deterministic encoding of its input. Their possible value is therefore as a representation and learned constraint: they may make different regularities easier for a classifier to recognize than for a sequence generator to reproduce from a class embedding alone. Spatial encoding can reflect the geographic and ordered route structure associated with a class. Velocity encoding can emphasize relationships among faster, slower and stationary portions of a journey. The existence of class information in either representation motivates testing feedback; it does not logically guarantee an improvement in generation.

The distinction matters especially when a route is fixed. Across a whole dataset, spatial recognition may benefit from buses repeatedly using service corridors. Within one candidate pool, all trajectories follow the same route. That source of class information may then vary little, even when the motion profiles differ substantially. Conversely, a velocity reader may distinguish candidates but prefer an overly extreme class-associated speed pattern. Combining the two scores may help when their preferences are complementary, or may retain the limitations of both. These are properties of the actual synthesis setting, not questions answered by classification accuracy alone.

The main feedback comparison therefore holds the candidate pool fixed. Eight samples from one conditioned generator are available to a uniform selector, a spatial selector, a velocity selector and an equally weighted joint selector. The selected coordinates are evaluated against independent simulator observations using motion statistics that do not use either reader. This defines an incremental role for GAF beyond class input: improving the delivered dataset by choosing among already class-conditioned realizations. It also provides a necessary reference for any later intervention during denoising. Terminal selection is reported as terminal selection; it is not described as completed sampling-time guidance.

1.3 Data and temporal resolution as design requirements

The project's initial CNR-associated tables contain simulated observations, but they were delivered without the corresponding route and simulation configuration files. Simply adding more random routes on the same map did not preserve the original relationship between participant class and spatial behaviour. A matched source comparison showed that the inherited spatial classifier behaved very differently on those random-demand trajectories and on the supplied observations. Reintroducing public MoST services and native activity demand recovered much of that recognition structure. This finding motivates the data used for the present synthesis study. It is evidence about the learning domain, rather than a result of the diffusion model.

Complete trips introduce a separate representation requirement. A fixed number of relative-time control points provides a different physical resolution for a short bicycle trip and a long bus service. The same 257-point representation can express a short transition densely while smoothing several seconds of movement in a long trip. Reconstruction checks motivated retaining the simulator's 0.25-second clock and using variable-length sequences. They establish what the representation can retain when the target is known. Whether a diffusion model can learn that more detailed target is addressed by a matched training and generation comparison, not assumed from reconstruction accuracy.

These decisions support the two central questions rather than creating separate data-cleaning objectives. Credible demand is needed to learn meaningful class associations; sufficient temporal resolution is needed to preserve the behaviour on which feedback might act. Both requirements are fixed before reading the new generation comparison. All accepted training vehicle trips are retained, including long trips and zero-motion intervals, instead of admitting only the examples that a manually constructed map adapter can easily reproduce.

1.4 Research questions and scope of the contribution

The first question is whether a road-conditioned diffusion system can produce complete variable-duration trajectories with credible class-specific movement, and how its motion distribution compares with empirical transfer and a matched relative-time model. Answering it requires more than travel-distance agreement. On a prescribed route, distance is largely determined already. The evaluation therefore considers speed, acceleration, stopping, sustained movement and changes over several seconds, together with within-request diversity and generation coverage.

The second question is what spatial and velocity GAF feedback add beyond the

class condition. The experiment separates reader performance on observed trips, differences among candidate scores, and independent quality of the selected output. These are three distinct links in the proposed mechanism. A reader that recognizes observed classes can still be unable to rank a fixed-route pool; a reader that changes the selection can still worsen the delivered distribution. Reporting all three links allows the thesis to explain an effect or its absence without equating the classifier's confidence with realism.

The resulting contributions are an implemented complete-trip diffusion pipeline with an explicit physical clock, and a controlled evaluation of two inherited or related image representations as feedback on that pipeline. A source-controlled data comparison explains the choice of native MoST demand. A matched temporal representation comparison tests the cost and quality consequences of preserving finer movement. These supporting studies make the generation experiment interpretable; they are not claims of a universally superior encoding, a new traffic simulator or proven cross-city generalization.

The evidence is organized in three stages. The pilot establishes complete physical-time generation and a controlled temporal-resolution comparison. The current five-class model uses 29,895 training legs, with both readers evaluated on 7,230 development legs. Finally, two new SUMO episodes provide 100 frozen requests each for a paired generator and teacher comparison. Their quality intersections contain 95 and 94 requests; all failures remain visible. The main results combine complete generated journeys, both image views and the full five-class motion comparison. Earlier/current corpus contrasts support the data choice in the appendix; two jointly changing training settings do not constitute a scaling law.

Public bus lines, source templates and the MoST map remain shared. New simulator randomness does not establish unseen-city or unseen-line generalization. The class-specific feedback analysis is explicitly exploratory. It supplies motion-specific hypotheses for independent validation while retaining all classes and adverse endpoints. The following chapters develop the data, generation and feedback methods before presenting this sequence of controlled comparisons.

Chapter 2

Background and related work

2.1 Trajectories as geometric and temporal objects

A sampled trajectory is an ordered sequence $X = ((t_i, p_i))_{i=0}^{T-1}$ with $p_i \in \mathbb{R}^2$. The polyline through its points defines geometry. The assignment of time to those points defines motion. Even on a straight road these properties can be separated. A vehicle covering one hundred metres in ten seconds can travel at constant speed or spend half the window waiting. Its endpoints and total displacement are identical in both cases, but its sequence of increments differs. A trajectory synthesizer therefore needs both a spatial support and a distribution over movement on that support.

A directed lane graph supplies information that a background map cannot convey by itself. For example, a bridge may lie directly above a second street in plan view. The distance between projected centerlines can be zero even though a vehicle cannot turn between them. At a junction, the permitted successor also depends on direction and participant class. A small distance to the nearest road is useful evidence about position. A connected assignment of road states throughout the sequence establishes a stronger property. Both depend on the accuracy and version of the map.

Microscopic simulation offers a practical test bed. The MoST scenario represents Monaco and its surroundings in SUMO, including multimodal roads and mobility demand [3]. Simulation exports can be sampled frequently and can carry participant types without manual annotation. They inherit the modelling choices of the simulator. A generated trace resembling the export can therefore reproduce a simulated mobility pattern even when its performance on traffic measured in the field is unknown.

2.2 Diffusion models for trajectories

A denoising diffusion model learns to recover structure from samples corrupted at known noise levels [4]. The representation of the clean sample is a design choice. When the representation is a sequence of positions, denoising acts on a coordinate plane. In the present work it acts on the temporal sequence of distances travelled along a sampled path. Either representation can produce coordinates as output. Their intermediate states and their routes to enforcing feasibility differ.

DiffTraj models spatial and temporal structure in GPS trajectory synthesis [6]. Its use of diffusion demonstrates the value of modelling a distribution of trajectories when multiple samples can be valid. A coordinate sample does not by itself name a permitted successor lane. Diff-RNTraj addresses road-constrained generation by representing positions through segment information and continuous moving ratios, and learning to denoise a vectorized representation [7]. This is particularly relevant prior work: the general concept of road-constrained diffusion generation predates this study.

A related design, RNTrajGen, separates road-level movement from the production of finer trajectories [8]. These studies make a useful comparator at the level of modelling assumptions. In the primary experiment here, a permitted planned route is supplied before synthesis. Holding it fixed lets us swap in an empirical motion process on identical paths and exposes which part of road consistency is determined by construction. The experiment compares that matched empirical baseline directly. Published systems trained on different cities, sampling rates and definitions of trajectory quality provide context rather than numerical head-to-head results.

Scenario Diffusion generates agents and their trajectories jointly under map and scenario conditioning [9]. Its output represents a different unit: a traffic scene. In our experiment the same starting position can generate several independent participant traces. Simultaneous occupancy and interaction are therefore outside the output distribution. This is consequential for interpretation. A bus trace can follow connected lanes while failing to be compatible with the timing of a particular traffic signal or the trajectory of another vehicle.

2.3 Imaging a time series

Wang and Oates encode time series using Gramian angular fields and Markov transition fields [10]. A pixel represents a relation between temporal indices. In a Gramian field the relation is computed from angular transforms of the underlying scalar values. In a Markov transition field it depends on the estimated transition probabilities between discretized states. Convolutional networks can process the resulting arrays as images. The construction gives spatially local filters access to structured patches of pairwise relations.

For a traffic trajectory the input has two spatial coordinates. The inherited implementation maps each point to a scalar using a Hilbert traversal of a spatial grid. The mapping changes the representation of a location while preserving the order in which observations were made. A CNN therefore sees temporal relations between locations collapsed to a scalar index. Nearby points along the Hilbert traversal occupy neighboring spatial cells, but some neighboring cells can be far apart in traversal order. The mapping is directional in this sense. It can also lose distinctions between slightly different positions in one cell.

Salvadori investigated such images for traffic participant recognition, including choices of encoding and classifier [1]. Ferreira and colleagues report a related classification study on simulated traffic [2]. The available historical repository provides an implementation of an image construction [12]. We use that source to specify the transform. The historical publication, the university thesis, the inherited code and the weights trained for this experiment are distinct pieces of evidence. Differences in splits, sampling horizon and data preparation can change reported scores even for closely related representations.

2.4 Evaluation under multiple objectives

A generative model can satisfy one objective and miss another. A route decoder can enforce topology for any input sequence of speeds. A generator that copies one speed profile can pass the geometric test while providing little new training information. A very diverse generator can produce implausible behavior. We therefore evaluate geometric consistency, one-dimensional motion distributions, candidate diversity and training utility separately. The common route and candidate budget make the comparison interpretable along each axis.

For distributional evaluation we use the first Wasserstein distance. In one dimension this is the area between two empirical cumulative distribution functions. Its units are those of the measured quantity. A difference of twenty metres in the travel-distance comparison has a direct interpretation as a reduction in distribution discrepancy; it is not the error of a predicted endpoint for a particular participant. For downstream evaluation we use macro-F1 on held-out participants. The averaging treats a rare class as a separate classification task instead of letting the most frequent class dominate.

DDIM sampling provides a deterministic sequence of reverse updates conditional on the initial noise [5]. Reusing paths, request order and random seeds reduces incidental changes in a comparison. Different temporal representations still have different noise dimensions, so a common seed is not an identical perturbation of the same vector. The primary full-trip comparison reports both fixed generator seeds separately on one new SUMO episode. It does not treat requests within that episode as independent simulation repetitions or attach a population confidence interval to their aggregate difference. Earlier resampling studies are retained as

supporting evidence with their own conditional scope.

2.5 What a trajectory generator learns

A route model and a motion model face different ambiguities. TrajGAIL learns sequential vehicle decisions by adversarial imitation on a road network [15]. Such a policy can choose among candidate links, while the rate at which a vehicle travels along a link remains a separate quantity. Diff-RNTraj jointly represents segment identity and movement within segments [7]. The two-stage design of RNTrajGen likewise separates road sequences from fine movement [8]. These distinctions motivate the choice of comparator here. Transferring an empirical speed sequence to the very same path gives the baseline every geometric advantage of the proposed model. Because both candidates complete the prescribed route, the relevant improvement is in motion distributions rather than in reaching a different travel distance.

Map-conditioned forecasting offers transferable representation ideas but uses different inference information. VectorNet encodes polylines, whereas LaneGCN explicitly propagates information over lane connectivity [16, 17]. Knowledge of map structure can improve predictions without guaranteeing that each returned sample admits a directed lane assignment. Trajectron++ models heterogeneous agents and feasible dynamics, and Scene Transformer jointly predicts agent futures [18, 19]. Social GAN explicitly treats pedestrian interactions as a source of multimodality [20]. Motion Transformer separates intention localization from local movement refinement [21]. These distinctions matter when interpreting a benchmark. A predictor exploiting an observed history or surrounding agents has information absent from the present synthesis request. Conversely, this thesis supplies a planned route, which removes route-choice uncertainty. The class-conditioned motion network samples movement along that route without observing the reference participant's future.

2.6 Guiding a trajectory while it is being generated

In sequential decision problems, Diffuser treats planning as denoising a trajectory, while Diffusion Policy generates action sequences conditioned on observations [22, 23]. MotionDiffuser is a closer comparison for map-based movement: it represents joint agent futures in a compressed space and guides samples using differentiable costs [24]. Our inherited Hilbert transform quantizes coordinates and changes sequence length. A small change to an increment can leave every encoded pixel unchanged until a cell boundary is crossed. Consequently, scoring complete clean trajectory estimates at intermediate updates is more direct here than differentiating through the original encoder.

Classifier guidance uses a classifier gradient during the reverse process, while

classifier-free guidance combines conditional and unconditional model estimates [25, 26]. Universal Guidance extends the range of external guidance functions used at inference [27]. These methods make different assumptions about the information available at noisy states. In particular, a class score trained on clean trajectories has a physical interpretation only after a provisional motion sequence has been decoded. Inverse-problem samplers face a related distinction. DDRM uses structure in linear measurement models, whereas diffusion posterior sampling approximates a measurement term for more general observations [28, 29]. A participant classifier supplies neither measured future positions nor a known noise likelihood. Using its probability as a selection weight is therefore an experimental preference, rather than a statistically calibrated observation model.

Twisted Diffusion Sampling uses weighted particles to approximate conditional distributions [14]. Feynman–Kac steering makes explicit how intermediate rewards, resampling times and potentials affect reward-guided generation [30]. A terminal scorer can choose only among trajectories already produced. An intermediate scorer can change which partial solutions receive subsequent computation. That opportunity also permits premature loss of viable solutions. If an early estimate of the final class preference is noisy, resampling can discard good trajectories before their motions become recognizable. At a very late update, a surviving state may have too little stochastic evolution left to move appreciably away from a terminal candidate. These are reasons to distinguish terminal feedback from in-sampler guidance. The primary experiment in this thesis isolates the former on a common candidate pool; historical intervention experiments do not turn its results into evidence for the latter.

Sampling budget itself can change conclusions. Improved DDPM studies learned reverse variances and sampling efficiency [31]; EDM separates choices of noise schedule, preconditioning and solver [32]; DPM-Solver constructs efficient numerical updates for diffusion sampling [33]. These works motivate explicit cost accounting. Equalizing network calls controls one cost, but a denoiser call on a long physical-time sequence is more expensive than one on 257 bins. The present comparison holds optimization steps and candidates fixed while reporting measured time; it does not label the two temporal models equal-compute. GAF selection reuses exactly the same eight candidates and adds only encoding and scoring work.

2.7 Representation, fidelity and utility

External representations can also enter diffusion training. REPA aligns projected hidden states from noisy denoiser inputs with clean-image features from a pretrained encoder [50]. Its image-generation results motivate asking whether a trajectory representation could provide useful supervision; they do not establish that a small class-trained GAF reader is an equally effective teacher. Representation similarity is another diagnostic: centered kernel alignment compares relationships

between learned feature representations [49]. Similarity, class separability and improvement of generated motion are distinct empirical questions. A high similarity score cannot replace the trajectory-quality comparison needed here.

A diffusion model over time does not uniquely specify a trajectory task. TimeGrad uses an autoregressive probabilistic model for multivariate forecasting, while CSDI conditions on observed entries for imputation [34, 35]. Those problems have observed context that a synthesis request may lack. DiffWave shows that an entire ordered waveform can itself be the generated object [36]. For road motion the temporal target used here is more structured: nonnegative increments give progressively ordered positions on a chosen path. A plausible marginal distribution of increments can still give an implausible sequence if their temporal ordering is wrong.

Recognition offers another perspective on that loss of information. Temporal convolutional models offer an alternative to recurrent sequence encoders [37, 38]. InceptionTime applies an ensemble of deep convolutional networks to time-series classification; ROCKET and MiniRocket show that convolutional transforms with simple classifiers can also be competitive [39, 40, 41]. Unsupervised multivariate sequence representations can learn reusable embeddings without training an image classifier [42]. These alternatives show why agreement among multiple copies of an image classifier cannot uniquely establish a class-specific motion effect. Raw geometry and motion readers serve as information diagnostics in the data study. In the generation experiment, independent physical measurements are the primary outcome, and neither GAF teacher is an oracle of realism.

Classifier two-sample tests ask whether a learner can distinguish populations [43]. Distributional precision and recall instead separate sample quality from coverage [44, 45]. A selector can increase concentration in a recognizable part of the data distribution while decreasing the range of plausible behaviors available to the user. For that reason we compare variation across candidates, time near rest and steady movement in addition to speed and acceleration. Esteban and colleagues use synthetic time series to train predictors evaluated on real data [46]. Downstream augmentation asks a further practical question whose result need not track either classifier confidence or a single distribution distance. Earlier augmentation results are retained separately; the current full-trip experiment does not claim a downstream benefit without measuring it.

Chapter 3

Data and complete-trip representation

This chapter establishes the observations used to learn traffic motion, the independent requests used to evaluate synthesis, and the representations presented to the generator and GAF readers. The supplied CNR-associated exports provide the original application context. Native MoST demand supplies the complete simulated trips for the pilot and expanded training experiments. A new, prospective SUMO run supplies reference observations only after the generation requests have been fixed. Keeping these roles separate prevents a simulator reconstruction or a classification result from being mistaken for evidence about generated trajectories.

3.1 Original exports and coordinate system

The project began with two supplied tables. The principal export is `midimost_with_class2.csv`; the secondary export is `vehicle_positions_TS_New.csv`. The principal MoST export contains 4,840,789 observations from 2,685 identifiers and six participant classes. The secondary TS export contains 2,189,591 observations from 12,061 identifiers but only one recorded class. They differ in spatial support, trajectory length and class composition. Pooling their rows would consequently change several aspects of the learning population at once. The current study uses the principal export to characterize the original domain and does not treat the secondary table as an additional source of interchangeable labelled trips.

The canonical record is `time,id,x,y,type`. Time is expressed in seconds and positions use local planar coordinates in metres. The principal table also contains simulator-provided speed and a label named `class2`; the class is mapped to `type` for the common interface. For comparisons between observed and generated motion, velocity is calculated from positions and elapsed time on both sides. This avoids comparing a simulator's internal speed field with a different numerical derivative

of generated coordinates.

No original SUMO network, route file or configuration accompanied the exports. A geometric check against the public Monaco SUMO Traffic network supported using its unchanged projected coordinate frame [3]. All 4,841 systematically sampled principal positions were within five metres of a lane centreline; only 11.2% of the corresponding 2,190-point secondary sample met that threshold. This check supports the map alignment of the principal source. It neither recovers its exact simulation settings nor proves that all original observations satisfy the current route decoder. The coordinates are not interpreted as raw longitude and latitude.

3.2 Choosing a learning domain that preserves class structure

An initial expansion generated class-permitted random routes on the same network. This increased the number of simulated observations, but a road permission does not recreate a transport service. In particular, permitting a vehicle to behave as a bus does not specify the public line, stops or activity pattern associated with bus observations in the original domain. A source comparison was therefore conducted before continuing the complete-trip generator.

The comparison used the same six classes and matched class counts: 1,983 training and 257 validation entities in each source. The sources were the supplied CNR observations, random-demand SUMO trajectories, and native MoST demand. The inherited spatial encoding, ResNet18 architecture and fixed training budget were held constant. Two training seeds were used, with the final epoch retained rather than a validation-selected checkpoint. These were classifiers trained within this project; the group did not supply classifier weights. Table 3.1 summarizes the relevant diagnostic evidence.

Table 3.1. Matched six-class source comparison. Values are validation macro F1 in percent. The GAF row averages two fixed classifier seeds. These are recognition results on observed trajectories, not diffusion-generation scores.

Readout	CNR	Random demand	Native MoST
Spatial geometry	79.12	34.85	76.95
Physical motion	88.08	80.57	90.35
Inherited Hilbert–GAF	59.88	30.15	58.69

The spatial classifier recovered from 30.15% macro F1 on random demand to 58.69% on native demand, compared with 59.88% on CNR. The geometry readout changed in the same direction, while the motion readout showed that the random-demand data still contained substantial class-associated movement information.

Thus, low spatial GAF performance in that domain could not be assigned to the image encoder alone. The evidence supports a mismatch in the demand domain. It does not isolate the causal contributions of service routes, stops, traffic density and activity chains, and the similar CNR and native averages are not an equivalence test.

The native source was fixed to public MoST revision `b29b2f65`, with a network file identical to the one already used by the project. Its demand includes 24 public bus services and 181 stops, together with activities, parking and background traffic. This is a documented new use of public demand, not a reconstruction of the unavailable original CNR configuration. The matched source study also shares many demand templates with CNR: 204 of 257 validation entities have the same source identity, and the native validation bus lines are represented in training. Its scope is therefore recovery of domain structure under controlled sources, not external validation on new lines.

The practical decision is to train the present generator on native-demand observations. The six-class 58.69% result is not assigned to the new five-class feedback reader, whose complete-trip cohort differs. Nor does recovery of recognition imply a benefit from GAF selection. That benefit is assessed only after generation, using the new reference population and independent physical metrics.

Participant labels and simulator vehicle classes are separate metadata. SUMO vehicle types specify physical and behavioural parameters whose defaults vary with class [51]. The native source types are retained rather than replacing their permission classes with label names. In particular, assigning the output label taxi or ride-hailing does not itself change a source passenger vehicle into a taxi-permission vehicle. Differences learned from this source remain simulator- and demand-dependent.

3.3 Pilot: event-supported complete vehicle trips

The initial pilot training source is a single SUMO realization, identified as pilot 300001, with a 0.25-second simulation step. Its augmented output records the events needed to distinguish a completed leg from an arbitrary observation segment. For ordinary vehicles, a leg extends from departure or the end of an activity stop to the next activity stop or final arrival. For buses, intermediate service stops remain inside the leg. They are part of the movement to be learned, rather than boundaries at which a bus trip is shortened. The earlier observation study's longest continuous segment and the present event-supported leg consequently refer to different data units.

The event-based extraction retained 1,644 legs across six classes after rejecting 15 coordinate-jump cases above the fixed 100 m/s check. The present vehicle experiment excludes the pedestrian walking stages and keeps all 1,051 accepted vehicle legs. It does not additionally filter them by success of the manually implemented map constructor. A route observed in training can legitimately form a supervised

route–motion pair even when constructing a comparable route from public map inputs requires an engineering adapter.

The inherited parent-level partition gives 925 training legs and 126 development legs, with counts shown in Table 3.2. Every retained vehicle leg in this cohort has a distinct parent. Known identities retain their earlier training or development assignment, and source parent identities assigned to the original test partition were excluded using the stored split metadata. The new experiment does not read the frozen test trajectories for fitting, selection or evaluation. Distinct parents nevertheless share one traffic realization and public demand; the 1,051 legs must not be treated as 1,051 independently simulated worlds.

Table 3.2. Complete vehicle-leg cohort for the generator and both GAF readers. The same 925 training and 126 development examples are used for the three learned components.

Participant class	Training	Development	Total
Bus	81	12	93
Motorcycle	226	43	269
Private car	457	50	507
Bicycle	92	14	106
Taxi or ride-hailing	69	7	76
Total	925	126	1,051

Completion is supported by simulation events, while the numerical trajectory ends at its last recorded coordinate. In the six-class extraction, 1,123 legs have a last observation 0.25 seconds before the ending event and 521 coincide with it. No coordinate is invented between the last observation and the event. Durations in the learning table therefore describe the observed extent of a completed leg. This distinction is retained when comparing generated endpoint arrival with sampled simulator output. The extraction also preserves short legs, long bus services and zero-displacement intervals; it does not discard difficult durations to simplify sequence modelling.

3.4 Separating route geometry from physical time

For a training trajectory with positions $\mathbf{x}_i$, cumulative observed arc length is

$$a_0 = 0, \qquad a_i = \sum_{j=1}^{i} \|\mathbf{x}_j - \mathbf{x}_{j-1}\|_2, \qquad s_i = \frac{a_i}{a_n}. \tag{3.1}$$

For nonzero-length legs, (s_i, t_i) gives the progress target. Repeated positions create zero progress increments and retain their elapsed time. The observed spatial

polyline retains its nonduplicate vertices in double precision, so reconstructing the coordinates at their recorded progress values does not require a fixed 2,049-point spatial approximation. A 129-point equal-arc summary is used only to condition the neural network. That summary is not the polyline used for final coordinate interpolation.

The choice of temporal target was informed by a controlled reconstruction comparison on the 1,644-leg six-class cohort. With the same target-derived spatial approximation, 257 relative-time points, one-second controls and quarter-second controls were decoded at the original observation times. The average native-speed reconstruction MAE was respectively 0.2614, 0.1960 and 0.0093 m/s; native-acceleration MAE was 1.0514, 1.1542 and 0.0314 m/s^2. Improving one-second summaries did not ensure preservation of subsecond movement. These numbers concern known-target reconstruction, including pedestrians and a common spatial approximation. They are not generated-trajectory errors or a result of the new five-class model.

This comparison motivates a variable-length 0.25-second target and a matched 257-point relative-time control. The final generation clock includes a fractional last interval when the sampled duration is not divisible by 0.25. Both models use continuous cumulative progress; they do not normalize disconnected temporal blocks and join them afterwards. Their supervision uses the same route–motion pairs and the same class-conditioned duration prior. Masking and length-aware batches make different sequence lengths practical, and the training loss averages over physical time within a trajectory before averaging trajectories. A long bus service is therefore not assigned extra loss weight solely because it contains more samples.

Retaining a finer target removes one representational limitation, but it also increases the learning problem and computation. Equal optimization steps do not mean equal numbers of processed temporal values. A model can preserve the possibility of subsecond variation and still learn an inaccurate or excessively variable distribution. For this reason the generation comparison reports both one-second motion summaries and native-interval speed and acceleration, rather than using reconstruction fidelity as a substitute for synthesis quality.

3.5 Spatial and velocity images from the same observations

The spatial reader inherits the original project's position-only transformation [12, 1, 2]. Coordinates are normalized with the frozen spatial bounds, mapped through the existing Hilbert routine, and processed by its sequence squeezing and duplication rules to obtain 224 values. The inherited image pipeline produces GASF, GADF and MTF channels, with frozen channel scaling. The shorthand "spatial GAF" in

this thesis refers to that complete inherited RGB pipeline; the MTF channel is not itself a Gramian angular field. No explicit velocity, acceleration or elapsed-duration feature is appended to the spatial reader.

Position-only input does not imply that all effects of movement disappear. Recorded position order and the pattern of repeated locations depend on traversal. However, they are filtered by the spatial grid and sequence processing. A whole-dataset class distinction can therefore coexist with weak differences among same-route candidates. The comparison retains the original transformation so that its feedback value is measured under a documented lineage, rather than silently replacing it with a displacement-based or velocity-assisted classifier.

The velocity reader starts from the same timestamps and coordinates. One-second increments of cumulative coordinate arc length provide a speed sequence, summarized at 128 relative-time locations. The existing velocity-image procedure uses a fixed 35 m/s scale and GASF, GADF and MTF channels, followed by the inherited resizing and normalization. It is a separate representation, not the original spatial method. Values exceeding the scale and spatial coordinates exceeding frozen bounds are counted. In particular, 128 samples of one-second speed summaries do not retain every quarter-second acceleration fluctuation, even though the new generator can produce such fluctuations.

In the pilot, both readers are newly trained ResNet18 networks [11], initialized from ImageNet and fitted to the same five vehicle classes on the same 925 training legs. Each uses the fixed final epoch of a 50-epoch run; the 126 development legs describe recognition and probability quality without choosing a best epoch. The velocity reader in this experiment predicts participant class, not the six speed-mode labels used in an earlier development study. Their matched cohort and labels support a fair representation comparison, while the separate inputs preserve the question of what spatial and velocity information each contributes.

3.6 Pilot prospective requests and simulator references

Formal generation uses a different interface from supervised training. Before a new simulation, the experiment freezes 100 requests: one for each of the 24 public bus services and 19 for each of the other four vehicle classes. Non-bus requests use eligible public single-leg plans from development-side parents, with no activity parking inside the requested leg. The fixed request specifies the planned edge sequence, participant type, start and end lane positions, and any public bus stops. These are available conditions, not facts recovered from a realized future trajectory.

The independent reference run uses seed 330201, retains native background demand, and inserts the targets at fixed new departure times between simulation seconds 18,000 and 18,600. Target rerouting is disabled to match the fixed-plan task,

while background traffic remains active. SUMO directly produces the reference coordinates. All 100 targets depart and arrive, and their realized edge sequences match their prescribed plans. Predetermined continuity and stop checks accept 97 references: 22 buses, 18 motorcycles and 19 in each remaining class. Two bus requests fail, one for a stop-count mismatch and one for a coordinate jump; one motorcycle fails the continuity or validity check. None is replaced with an easier request.

The generation-side map constructor independently supports 97 of the original 100 requests. Its three failures and the three reference failures are distinct, leaving 94 requests with both a supported map condition and an accepted reference. Thus 97/100 describes generation geometry coverage, 97/100 describes reference acceptance, and 94/100 is the paired quality population. The generator is run without consulting reference acceptance: eight candidates per supported request produce 776 candidates per full-pool model or baseline, of which 752 belong to the 94 paired requests. Selection later yields one trajectory per generated request while retaining the same coverage accounting.

The event record also limits interpretation. The new simulation contains 24 background-and-target teleport starts in total. The one directly affected target bus is already outside the accepted reference cohort under the frozen extraction checks. An accepted ride-hailing target has an emergency-stop warning and remains included. No post-hoc resimulation or removal is used to improve the reference distribution. The reference population is therefore a documented simulator realization with retained limitations, not a claim that every accepted movement is free of simulator anomalies.

The model receives only the frozen map request and a duration sampled from training-side statistics. Reference coordinates, realized duration and future lane choices become available to the evaluator after generation. Their role is to characterize the distribution of outcomes for comparable conditions; they do not identify a unique correct generated sequence. Physical distances are consequently reported as class-wise distribution comparisons with equal request weight, accompanied by support counts. When a class lacks support, the fixed five-class macro is undefined rather than silently recomputed over a smaller set.

Finally, independence of this new simulation has a limited meaning. It has a new stochastic realization, target identities and departure instances, but shares the map and public demand families with training. Bus services in particular are familiar templates, not held-out route families. The experiment is a same-map prospective development evaluation on one new episode. It supports controlled comparisons of the generator and GAF feedback under those conditions; it does not supply independent city-level replication or justify confidence intervals that treat every candidate as a separate traffic simulation.

3.7 Expansion with native demand and complete activity legs

The expanded corpus extends the observation interval to the full ten-hour native-demand simulation, from 14,400 to 50,400 seconds. Two training realizations and one development realization retain the same public map, service plans, class definitions and quarter-second time step. This is a new controlled replay of public demand, not recovery of the exact simulator configuration behind the original supplied CNR tables. The original exports continue to provide the source-domain reference described earlier in this chapter.

A parent vehicle is assigned a training realization before inspecting its realized trajectory; a second seed therefore does not count as a second independent vehicle. Known source identities inherit the existing partition, and original-test parents are excluded using identity metadata without opening their held-out trajectories. Activity stops delimit ordinary vehicle legs, while bus service stops remain inside their service leg. A parent may consequently contribute more than one completed leg. This distinction matters for both data volume and statistical uncertainty.

Table 3.3. Expanded native-demand corpus. Activity legs and distinct parent identities are separate units.

Class	Train legs	Train parents	Dev. legs
Bus	685	685	155
Motorcycle	6547	5226	1591
Private car	18381	13880	4436
Bicycle	2744	2744	668
Taxi/ride-hailing	1538	1538	380
Total	29,895	24,073	7,230

The retained dataset contains 29,895 training legs from 24,073 parents and 7,230 development legs from 5,842 parents. It contains approximately 65.1 million training coordinate observations and 15.8 million development observations. Median training duration is 549 seconds, the 95th percentile is 1,152.75 seconds and the longest retained leg lasts 3,037.25 seconds. The representation preserves native samples and in-leg zero-motion intervals. Very short completed events remain in the cohort; complete does not imply long. In particular, 2,454 training and 527 development legs use the declared subsecond fallback in the velocity summary. These cases and constant spatial sequences remain reported rather than being removed to improve classification.

The expanded source is still imbalanced: private cars account for 18,381 training legs, whereas buses contribute 685. Macro recognition metrics and class-wise motion evaluation are therefore necessary beside overall accuracy. Twenty-three

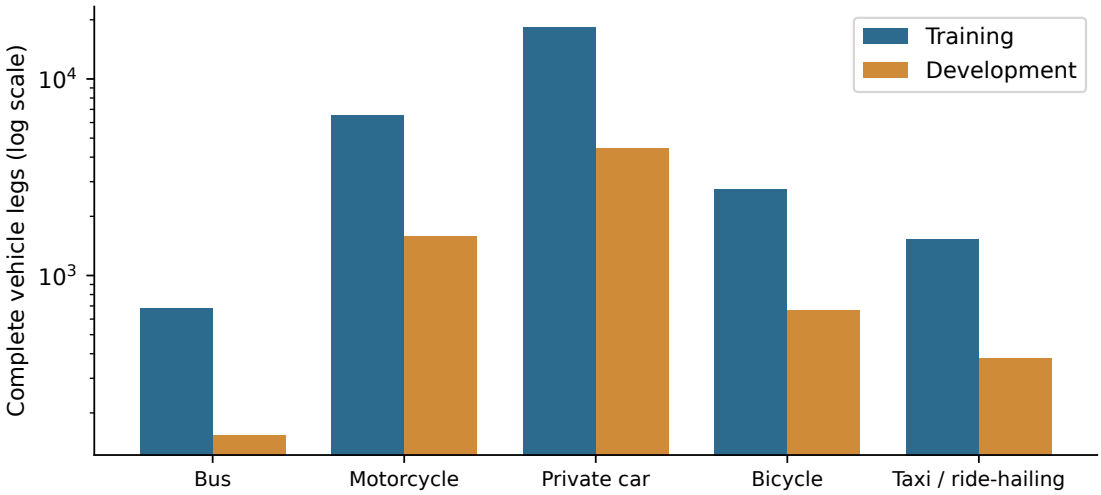


Figure 3.1. Expanded complete-leg counts for the five classes. The logarithmic axis exposes minority support while retaining the substantial private-car majority. These are simulator legs, not independent traffic scenes.

source bus lines occur in each split; familiarity with those services is not evidence of unseen-line generalization. Native-demand expansion increases vehicle and temporal coverage within the existing public domain, rather than adding a new city.

The two expanded GAF readers retain the pilot architecture, pretrained initialization, fixed final epoch and encoding rules. Fifty epochs over the larger corpus entail more optimizer updates. The comparison therefore concerns the combined data-and-training expansion; it does not isolate sample count at equal optimization exposure. Four expanded diffusion fits retain 10,000 steps, two model seeds and the two temporal representations. Independent generation replication below uses the native-time models; the expanded 257-point fits are completed artifacts but have not been included in that replication's physical-quality comparison.

Chapter 4

Road-conditioned synthesis of complete journeys

4.1 The conditional generation task

The required output is a complete timed journey, rather than an image or a continuation of a short observed window. A request specifies a road network M, a participant class c, and a planned route R with explicit starting and ending positions. For a scheduled bus service, the request also retains its public stop sequence. The generator returns a sequence $X = \{(t_i, x_i, y_i, c)\}_{i=0}^{n}$, which can be exported as `time,id,x,y,type`. The task is conditional synthesis: the planned route is available before generation, whereas the realised trajectory, travel time and intermediate speeds are not.

Diffusion has been applied to geographic trajectories and road-constrained trajectory generation [6, 7]. Here, the learned component models how a participant progresses along an already specified route. Geometry and timing are separated to make their contributions identifiable. Let $R(s)$ denote interpolation along the route at arc distance $s \in [0, L]$. The sampling program is

$$T \sim p_{\mathrm{dur}}(\cdot \mid L, c), \qquad p \sim p_\theta(\cdot \mid R, c, T), \qquad X_i = R\big(Lp(t_i)\big). \tag{4.1}$$

The duration prior and motion diffusion together form the generator. Duration itself is not a diffusion output. The decoder supplies route adherence; the neural network must learn useful motion within that constraint. This design does not predict route choice, traffic signals, surrounding agents or collision avoidance.

Training and prospective evaluation use different sources of route information. An observed training journey supplies its paired spatial path and motion target, as in conditional supervised learning. Evaluation receives only a route assembled from the frozen public request and map. An observed validation path can support an explicitly labelled reconstruction diagnostic, but cannot replace the prospective

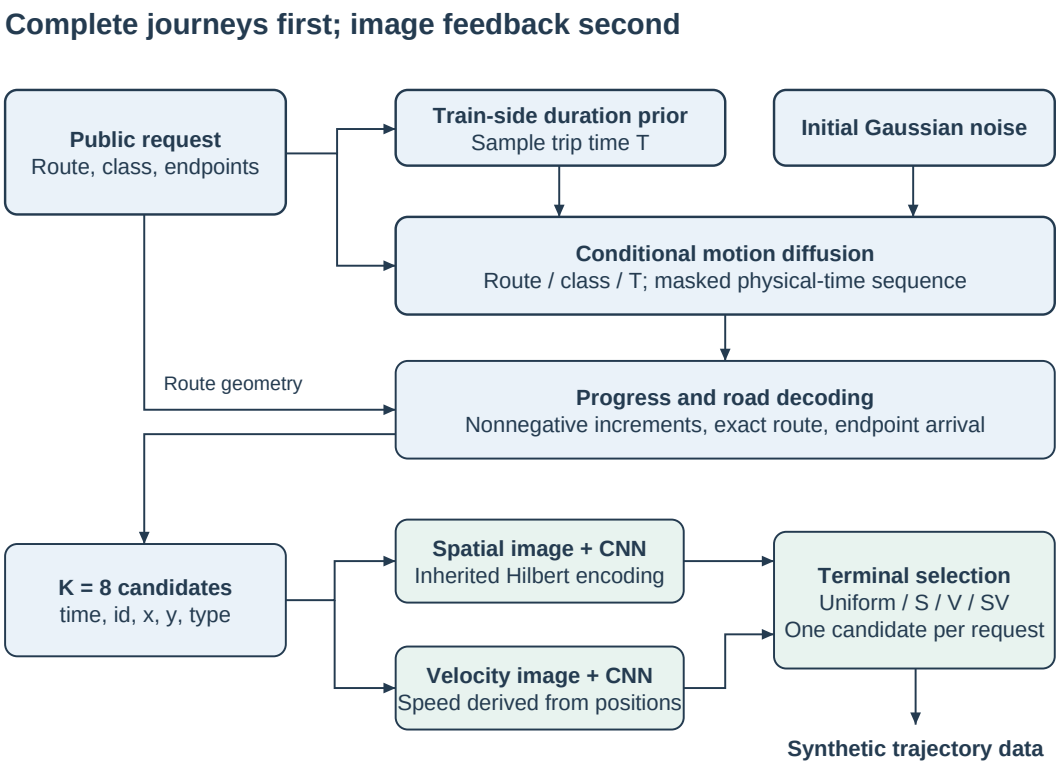


Figure 4.1. Implemented generation and feedback pipeline. Road geometry constrains coordinate decoding, diffusion produces motion, and the two self-trained image readers rank a completed candidate pool. No arrow sends reference future observations into generation. The present feedback is terminal selection, not a gradient inside denoising.

route. In particular, a failed map construction is retained as a failed request rather than repaired with future reference coordinates.

4.2 Representing motion on physical time

For a training trajectory, let $\ell_i = \|X_i - X_{i-1}\|_2$, $s_i = \sum_{j \le i} \ell_j$ and $p_i = s_i/L$, where $L = s_n > 0$. Repeated positions give repeated progress values and therefore retain stopped intervals. All nonduplicate spatial vertices are stored for exact piecewise-linear reconstruction; a separate set of 129 equally spaced arc positions provides the network's route description. Compressing the conditioning input therefore does not compress the coordinate decoder.

Two temporal representations are compared. The relative-clock version uses 257 points uniformly distributed over $[0, T]$. The physical-clock version uses

0,0.25,0.50,…,T, appending the exact final time when it is not a multiple of 0.25 seconds. Its number of intervals varies with duration. The distinction matters because a fixed number of controls gives a long trip much coarser physical resolution than a short one. Preserving fine resolution is a representation choice to be tested in generation, not proof of better generated motion.

With $\Delta t_i = t_i - t_{i-1}$, define the dimensionless relative speed and its transformed target as

$$r_i = \frac{p_i - p_{i-1}}{\Delta t_i / T}, \qquad u_i = \log(1 + r_i), \qquad z_{0,i} = \frac{u_i - \mu}{\sigma}. \tag{4.2}$$

Thus $r_i = 1$ is constant progress at the journey's average rate, and $r_i = 0$ represents a stop. The logarithm moderates the upper tail without removing zero motion. Training-only μ and σ assign equal mass to each trajectory and weight its intervals by $\Delta t_i / T$. Both clock versions use this target definition, with their own fitted normalization. They form a matched comparison on the current network, rather than a byte-for-byte reproduction of an earlier model.

4.3 Generating duration without future information

Within each class, a ridge regression relates log travel time to centred log route length. For training journey j of class c,

$$\log T_j = a_c + b_c(\log L_j - \bar{\ell}_c) + e_j, \qquad T^* = \exp\{a_c + b_c(\log L - \bar{\ell}_c) + e^*\}, \tag{4.3}$$

where e^* is sampled uniformly from that class's training residuals. The slope has ridge penalty 10^{-3}; the intercept is unpenalized. Residual sampling retains variation beyond the conditional mean without imposing a Gaussian residual law. A numerical lower bound gives $T = \max(T^*, 0.25, L/55)$, with metres and seconds as the physical units. This is a broad duration guard, not a class-specific speed limit or a guarantee about instantaneous speed.

The same prior and request-keyed random draws are used by both diffusion versions and the empirical comparator. Consequently their unselected duration distributions are paired by construction. Differences in generated speed variation can be attributed to the motion mechanisms under this shared prior, whereas a common duration result cannot establish a diffusion advantage. Terminal feedback may select different durations from the shared pool; that is a selection effect and is reported as such.

4.4 A masked temporal denoiser

The denoiser is a one-dimensional U-Net with channel widths 64, 96 and 128 and 891,969 trainable parameters. Two downsampling stages, skip connections

and nearest-neighbour upsampling connect the resolutions. Residual convolutional blocks use dilations 1 and 2, with dilations 1, 2, 4 and 8 in the bottleneck. Each block also receives a masked global feature summary, allowing information to pass beyond its local receptive field.

The conditioning vector adds a sinusoidal diffusion-step embedding, a learned class embedding, a convolutional route embedding and a five-component context embedding. The context contains normalized starting coordinates, log route length, log sampled duration and log mean travel speed. Route coordinates are expressed relative to the start and scaled by 1,500 metres. Three channels describe each interval's physical midpoint, relative midpoint and actual width. These allow the network to distinguish physical time from relative trip position. Bus stop times and background traffic states are not separate network inputs; geometric passage through a stop does not enforce a dwell there.

Sequences are padded to a multiple of four. The mask is reapplied through the network, pooling divides by the number of valid entries, and normalization acts across channels at each position rather than across padded time. Padding therefore has no semantic role as a stationary suffix. Batches are sampled before length sorting; sorting and microbatching manage memory without removing long journeys.

Following noise-prediction diffusion [4], a uniformly sampled noise index k corrupts a target by

$$z_k = \sqrt{\bar{\alpha}_k} z_0 + \sqrt{1 - \bar{\alpha}_k}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{4.4}$$

on valid intervals. The 1,000-step schedule increases β_k linearly from 10^{-4} to 0.02, with $\bar{\alpha}_k = \prod_{j \le k}(1 - \beta_j)$. For an effective batch of B trajectories, the loss is

$$\mathcal{L} = \frac{1}{B} \sum_{b=1}^{B} \sum_{i} \frac{\Delta t_{b,i}}{T_b} \left[\epsilon_{b,i} - \epsilon_\theta(z_{b,k}, k, R_b, c_b, T_b)_i\right]^2 . \tag{4.5}$$

Padding has zero weight. Physical-time weighting within a trajectory prevents the fractional final interval from receiving a full interval's influence; averaging trajectories prevents long trips from dominating merely through their larger point counts.

4.5 Sampling and coordinate decoding

Generation uses 100 deterministic DDIM updates [5] between equally spaced indices of the training schedule. At a current index k, the predicted clean target and next state at $j < k$ are

$$\hat{z}_0 = (z_k - \sqrt{1 - \bar{\alpha}_k}\, \hat{\epsilon})/\sqrt{\bar{\alpha}_k}, \tag{4.6}$$

$$z_j = \sqrt{\bar{\alpha}_j}\, \hat{z}_0 + \sqrt{1 - \bar{\alpha}_j}\, \hat{\epsilon}. \tag{4.7}$$

The final update uses $\bar{\alpha}_j = 1$. Different initial Gaussian draws produce candidates; paired request and candidate identifiers reproduce duration draws across methods. Different clock dimensions do not imply identical noise arrays.

After undoing standardization, the decoder clips u_i to [0,8] and constructs non-negative progress mass:

$$m_i = [\exp\{\mathrm{clip}(\mu + \sigma \hat{z}_{0,i}, 0, 8)\} - 1] \, \frac{\Delta t_i}{T}, \qquad \hat{p}_i = \frac{\sum_{j \le i} m_j}{\sum_j m_j}. \tag{4.8}$$

The initial progress is zero and the final progress is one. If total mass is below 10^{-12}, the decoder falls back to constant progress in physical time. Negative-value truncations, upper clipping and this fallback are counted rather than hidden. The normalization couples the whole journey: a local mass change can alter progress elsewhere. It guarantees endpoint completion for a decoded path, but does not guarantee realistic acceleration or stop timing.

Coordinates are interpolated along all retained vertices of the constructed map path. For the relative-clock model, progress is first interpolated onto the same quarter-second output clock; the exact fractional endpoint remains. Thus both methods deliver comparable timestamped coordinates. Permitted lanes, directed connections and source stop anchors are handled by the route constructor. This geometric constraint is shared with the empirical baseline and all GAF selectors. It cannot be credited to diffusion or feedback, and it does not imply vehicle-body clearance, signal compliance or interactive traffic safety.

The resulting generator is complete before any classifier is consulted. Its learned contribution is a distribution of whole-journey motion profiles under the declared conditions. The feedback module in Chapter 5 subsequently tests whether image-based class information provides a useful preference among those generated profiles.

Chapter 5

Spatial and speed GAF feedback

5.1 Two representations of the same journey

The inherited work represents trajectories as images for participant classification [1, 2, 12]. This thesis reuses that spatial encoding and trains its own classifiers; no group CNN weights were supplied. A second view represents the journey's speed sequence. Both views predict the same five participant classes. Their role is to test whether class information learned from source trajectories can improve synthetic data, rather than to generate images and invert them into coordinates.

For a scalar sequence $a_1, \ldots, a_q$, let $u_i \in [0,1]$ be its normalized value and $\phi_i = \arccos u_i$. The two angular fields used here are

$$G_{ij}^{\text{sum}} = \cos(\phi_i + \phi_j), \qquad G_{ij}^{\text{diff}} = \sin(\phi_i - \phi_j). \tag{5.1}$$

A third channel is a Markov transition field. If b_i denotes a sequence value's bin and P_{ab} is the row-normalized count of adjacent transitions from bin a to bin b, then $H_{ij} = P_{b_i b_j}$. The angular construction originates in time-series imaging [10]; it is not a new transform proposed by this thesis. For brevity, "GAF feedback" refers to the implemented three-channel GASF/GADF/MTF representation, including the transition channel.

5.2 Inherited spatial encoding

The spatial input consists only of the ordered coordinate samples. Frozen coordinate bounds map them into the unit square, with out-of-range values clipped and counted. The inherited recursive Hilbert function assigns a scalar index to each point. Its precision parameter is 0.0009, corresponding to a 64×64 spatial grid. The temporal order is preserved: points are not sorted by Hilbert index.

The original squeeze procedure compresses repeated-index runs, followed by mirror duplication when necessary, to obtain 224 values. This is not uniform time resampling or a learned encoder. If the inherited procedure cannot reach the required length, the case is reported as an encoding failure. The scalar sequence is mapped to GASF, GADF and an eight-bin uniform MTF. The angular channels use per-sequence scaling to [0,1], while the frozen channel bounds convert the three fields to an unsigned eight-bit RGB image. Coordinate and channel parameters are inherited from the earlier training-only encoding metadata; they are not refitted on the new references or generated candidates.

This branch receives no additional velocity, acceleration, duration or participant feature. Nevertheless it is not invariant to motion. Different speeds change the number and order of sampled coordinates within quantized cells, and the run compression preserves some of that structure. Conversely, quantization and compression can map distinct motion profiles to the same image. On a fixed route, spatial feedback may therefore have little ability to rank the candidate pool even when it classifies trajectories from different routes well. Image hashes and the number of distinct spatial images in each pool make this limitation observable.

5.3 Speed encoding and its temporal limits

Speed is computed from the same coordinates and timestamps. Let $d(t)$ be the piecewise-linear interpolation of cumulative planar distance. At integer seconds, the speed summary is

$$v_j = d(j+1) - d(j), \qquad j = 0, \ldots, \lfloor T \rfloor - 1. \tag{5.2}$$

These one-second increments are expressed in metres per second and resampled to 128 equally spaced relative sequence positions. The encoder requires at least five one-second increments. Shorter candidates remain in the generation accounting and invoke the shared feedback failure rule; they are not silently removed to improve a classifier's coverage.

The angular normalization is $u_i = \mathrm{clip}(v_i/35,0,1)$, with a common absolute scale across classes. Values above 35 metres per second are counted. The MTF uses eight fixed bins on this normalized interval. The two angular channels are mapped from $[-1,1]$ to $[0,1]$; the resulting image is resized from 128×128 to 224×224. Both representations then use the same ImageNet channel normalization before entering a ResNet18 [11].

The speed image retains absolute speed scale but compresses temporal detail. One-second averaging and the 128-point summary do not preserve all quarter-second acceleration. Explicit duration and location are not supplied to this classifier. The representation is consequently a particular view of whole-trip motion, not a complete alternative encoding of the trajectory.

5.4 From a generated journey to the two image views

Figure 5.1 makes the implemented representation concrete using an actual completed diffusion sample. The top row starts from its ordered Hilbert sequence; the lower row starts from speed computed from the same timestamped coordinates. Each row ends in the three channels that the corresponding class reader receives. The image is an analysis of a generated journey, not an object generated and then inverted into a journey.

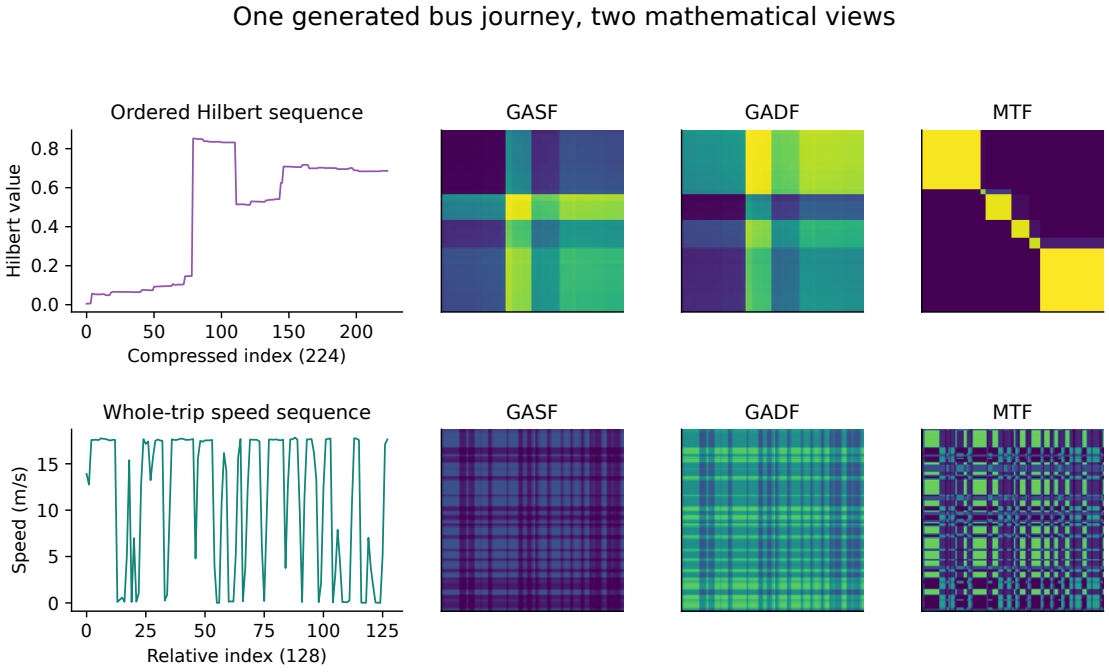


Figure 5.1. Exact representation channels of one generated bus journey from the expanded model. The spatial row uses original Hilbert compression and frozen channel conversion; the speed row uses the 128-value absolute-speed summary and the implemented resize. Each field is shown with a common display range of [0,1] after its implemented conversion. These are channel intensities, not a common physical scale across the two encoders. No per-image contrast enhancement is applied.

The two sequences have different coordinates on their horizontal axes. The spatial sequence index follows inherited compression, whereas the speed sequence index follows relative-time resampling. Their off-diagonal image patterns relate pairs of sequence positions. Consequently, a visible image difference need not imply a better trajectory: it first establishes that the representation distinguishes the inputs. The empirical question is whether the resulting class scores rank motion

quality usefully. Chapter 7 therefore links complete generated journeys, their exact image inputs and the selected candidate identities before presenting population-level quality statistics.

5.5 Frozen class readers and terminal selection

Two independent ResNet18 classifiers are fitted on identical training journeys and class labels, one per view. The fixed training protocol is given in Chapter 6. Neither reader is trained on synthetic candidates, and no classifier is used inside the current denoising process. Let $f_S(c \mid E_S(X))$ and $f_V(c \mid E_V(X))$ be their target-class probabilities. For each request, all methods receive the same $K = 8$ generated candidates and use

$$s_k = \log \max\{f_S(c \mid E_S(X_k)), 10^{-8}\}, \tag{5.3}$$

$$v_k = \log \max\{f_V(c \mid E_V(X_k)), 10^{-8}\}. \tag{5.4}$$

Uniform selects the first candidate in a fixed request-keyed random permutation. The S and V strategies maximize s_k and v_k, respectively; SV maximizes $(s_k + v_k)/2$. Exact score ties use that same permutation. Equal weights are fixed before evaluation rather than selected to favor the better validation reader. The probabilities are not temperature-calibrated, so equal log weights should not be interpreted as equal reliability [48].

Every requested candidate slot is retained. If a pool contains a failed candidate or an unsuccessful encoding in either view, all four strategies use the same predetermined Uniform slot. A failed selected slot remains a failure. There is no replacement sampling, alternate classifier or method-specific exclusion. Nonfinite logits and inconsistent model or data hashes are implementation errors that stop scoring. This policy keeps scoring failures from becoming a hidden selection advantage.

5.6 What would demonstrate useful feedback?

Three kinds of evidence must remain separate. Classification measures whether the source representation contains information about participant labels. Ranking measures whether its scores vary meaningfully among candidates for one fixed request. Generation utility measures whether the selected trajectories improve under independent physical evaluation, with coverage and diversity retained. A stronger classification result alone establishes neither ranking nor utility.

The two views have common physical origins but different losses of information. Their predictions may be complementary, redundant or jointly biased by the simulator's route and vehicle distributions. An SV advantage over Uniform is

insufficient to establish complementarity if one branch already explains the entire improvement; comparison with the better single-view selector is also required. Conversely, a constant spatial image within a fixed-route pool limits this experiment's ability to test spatial feedback, rather than proving that spatial GAF is generally uninformative.

These experiments evaluate terminal feedback. Classifier guidance during denoising is a distinct intervention [25, 27]: it changes the sampling trajectory and its computational budget. The present implementation does neither, and no in-loop guidance benefit is claimed. Likewise, without a matched non-GAF class reader, a positive result supports this implemented feedback mechanism but cannot isolate the superiority of angular encoding over generic class-based selection.

Chapter 6

Protocol for complete-journey generation

The protocol first defines the pilot and its fixed-duration control. Sections 6.7 and 6.8 then specify the expanded replication and distinguish its predeclared endpoints from the later exploratory audit.

6.1 Pilot data, partitions and observation units

The pilot experiment uses native trajectories from the public Monaco SUMO Traffic scenario [3]. These are simulator observations, not measurements from vehicles on physical roads. The source CNR data motivates the inherited representation and class definitions; the current complete-journey models are fitted on directly extracted SUMO vehicle legs. A bus leg is a completed scheduled service including intermediate stops. Ordinary vehicle legs are bounded by departure, arrival or activity events, rather than selected as the longest convenient moving segment.

The retained vehicle cohort contains 1,051 legs: 925 for training and 126 for development. Each vehicle leg in this cohort has a different parent identity, and parent identities do not cross these partitions. All nevertheless originate from one traffic realization; distinct participants are not independent simulation episodes. Pedestrians are outside the declared five-class vehicle generation task. Table 6.1 reports the actual imbalance instead of implying a balanced dataset.

Native quarter-second samples, stopped intervals and long journeys are retained. An event endpoint can occur up to one sampling interval beyond the last observation; no coordinate is invented there. The observed span defines the supervised duration. Map-constructor success is not a training admission criterion: supervised route–motion pairs come directly from the accepted observations. Training-only normalization, duration regression and empirical donors are fitted after selecting

Table 6.1. Complete vehicle legs used by both diffusion versions and both GAF readers. The source spelling `bycicle` is retained in machine-readable labels.

Participant class	Training	Development
Bus	81	12
Motorcycle	226	43
Private car	457	50
Bicycle	92	14
Taxi or ride-hailing vehicle	69	7
Total	925	126

training rows. The older frozen test trajectories are not reopened for this experiment's model or parameter selection.

6.2 Fixed fits and comparison budget

The two clock versions each have two independent model fits, with seeds 42 and 43, giving four fits in total. Each receives 10,000 AdamW updates with an effective batch of 64, learning rate 2×10^{-4}, weight decay 10^{-4} and gradient norm clipping at one. Classes are drawn uniformly, then a training journey is drawn within the chosen class. Length-sorted microbatches obey a 32,768-token budget and a maximum of 32 trajectories. The final exponential moving average, updated with weight 0.001 per step, is the only generation checkpoint; development quality does not select an earlier checkpoint.

Both GAF readers use ResNet18 with ImageNet initialization, seed 42, 50 epochs, batch size 64 and the same AdamW learning rate and weight decay. Cross-entropy class weights are proportional to inverse square-root training frequency. The epoch-50 model is fixed, with no validation checkpoint search, temperature fitting or image augmentation. Accuracy, macro-F1, per-class recall, negative log likelihood, Brier score and 15-bin expected calibration error characterize these components [48]; none is a substitute for generated-trajectory quality.

Each model generates eight candidates per map-ready request using 100 DDIM steps. The empirical comparator selects the eight same-class training routes closest in absolute log length ratio, breaks equal distances by stable identity, and samples one of their whole motion profiles for each candidate. It transfers normalized progress to the common sampled duration and requested route. This preserves training motion structure and supplies a substantive non-neural comparison. The same request and candidate identifiers couple duration draws across all generators. Equal updates or candidate counts do not imply equal computation: valid tokens, memory, elapsed training and sampling time, and teacher calls are reported

separately.

6.3 A prospective simulator reference

One new SUMO run provides a reference independent of model fitting. Before simulation, a manifest fixes 100 requests: all 24 public bus services and 19 requests for each of the other four classes. Non-bus requests are selected deterministically from explicit public vehicle plans without activity stops, using source parents assigned to the development partition; source test parents are excluded. They are not class-labelled random origin–destination routes. Concrete vehicle types are drawn deterministically from their source families, preserving actual SUMO vehicle-class permissions: an Uber-labelled passenger vehicle does not acquire taxi-lane access merely through its label.

Endpoints specify a permitted lane and position on the first and last planned edges. The starting position is $\min(20\,\mathrm{m}, L_{\mathrm{lane}}/2)$ and the endpoint is the end of the chosen final lane. Public bus stop order and relative schedule semantics are retained. New target departures lie between 18,000 and 18,600 seconds; simulation runs from 14,400 to 23,000 seconds with step 0.25 seconds and seed 330201. Native background demand is retained, while rerouting is disabled only for the new targets. FCD output is restricted to those targets [47]. The controlled departures and endpoints make this a new reference experiment, not an exact reconstruction of the source CNR run.

All 100 targets arrived, and their realised edge sequences matched their plans. Predeclared continuity and event checks accepted 97 references: two bus services and one motorcycle were rejected. Independently, the map interface constructed 97 of the 100 planned geometries. These are different success sets: their intersection contains 94 requests before any candidate-level generation or metric-support failure. Coverage therefore always starts at 100, while each quality statistic reports its own paired support. Reference failures are not filled by a new simulation, and unsupported map requests are not replaced by easier ones. One accepted reference contains an emergency stop; it remains in the declared cohort rather than being removed after inspecting quality.

Generation does not consult reference trajectories, arrival times or realized future paths. Reference data are opened for evaluation only after a completed candidate archive is saved. Nevertheless the experiment is a development replication on the same map and public demand templates. A new random seed is not evidence of unseen-line or unseen-city generalization, and shared templates can overlap training geometry.

6.4 Physical quality beyond travel distance

Evaluation computes motion from the returned coordinates and timestamps, independently of the GAF readers. Cumulative planar distance is interpolated onto integer seconds, giving speeds $v_j = d(j+1) - d(j)$ and accelerations $a_j = v_{j+1} - v_j$. Table 6.2 defines the ten fixed journey summaries. Only complete one-second intervals contribute to these summaries; full observed duration and planar distance retain the final fractional interval.

Table 6.2. The ten fixed physical summaries. Thresholds define descriptive states rather than asserting vehicle-specific physical limits.

Summary	Definition
Duration; planar distance	Full timestamp span; sum of consecutive planar displacements.
Mean; 95th-percentile speed	Mean and empirical 0.95 quantile of v_j.
Mean absolute acceleration	Mean of $\lvert a_j \rvert$ on the one-second clock.
Stopped fraction; longest stop	Fraction and longest run with $v_j < 0.3\,\mathrm{m/s}$.
Stop episodes per minute	Number of stopped runs divided by observed one-second time, multiplied by 60.
Longest steady interval	Longest run with $v_{j+1} \geq 0.3\,\mathrm{m/s}$ and $\lvert a_j \rvert < 0.5\,\mathrm{m/s^2}$.
Four-second speed change	Mean of $\lvert v_{j+4} - v_j \rvert$.

For each class and summary, generated and reference distributions are compared by one-dimensional Wasserstein distance,

$$W_1(F, G) = \int_{-\infty}^{\infty} |F(x) - G(x)| \, \mathrm{d}x. \tag{6.1}$$

The unit is the summary's original physical unit. Metrics with different units are not added into an arbitrary overall score. A short journey that cannot support a summary remains in coverage and is marked unsupported for that metric. For N_c jointly supported requests of class c, with K_r supported generated candidates in request r, candidate mass is $1/(N_c K_r)$ and reference mass is $1/N_c$. Partial candidate failures therefore cannot give a request less weight merely because fewer candidates survived. A five-class macro average is defined only when all five classes have support; available-class summaries are labelled separately.

Native-resolution distributions provide a complementary check. For an interval Δt_i, speed is $\|X_i - X_{i-1}\|/\Delta t_i$; acceleration divides successive speed differences by $(\Delta t_i + \Delta t_{i-1})/2$. Their Wasserstein distances first weight requests equally, then candidates within a request equally, then observations by their physical interval length. This exposes rapid fluctuations that one-second summaries and the speed GAF can suppress. Generated and reference durations differ, so framewise coordinate error is not treated as the primary synthesis metric.

6.5 Diversity, selection and interpretation

Within a request, candidate pairs are compared using 129 relative-time coordinate positions, 257 relative-time progress values and duration differences. Pair distances are averaged within each request before summarizing across requests. The nearest same-class training progress profile provides a descriptive memorization check, with its associated duration difference. Geometry overlap alone cannot establish memorization when the route was supplied as a condition, and no unvalidated privacy threshold is inferred from the nearest-neighbour statistic.

The main comparison has two panels: all unselected candidates from the empirical and two diffusion generators, followed by Uniform, S, V and SV outputs from the shared pool of each native-clock model. Terminal selection cannot change the quality of its shared pool; its effect concerns the delivered samples. With one selected sample per request, within-pool diversity is no longer estimable, so the selected class distributions, nearest-training comparisons and selection frequencies must be read alongside the original pool diversity.

Both model seeds are reported separately. The single reference episode does not justify confidence intervals that pretend requests are independent simulation replications. Failures, support, coverage and cost accompany quality results. Improved teacher probability alone supports only successful score optimization; useful GAF feedback requires independent motion improvements with any coverage or diversity tradeoff disclosed. These chapters specify that comparison without presuming its outcome or claiming an unperformed downstream augmentation or in-loop guidance result.

6.6 A subsequent fixed-duration development control

After examining the G33 comparison, a bounded follow-up, G34, tests feedback when candidates cannot compete through different total durations. Its protocol is fixed before generating the new pools. For each request and model seed, the predetermined G33 Uniform permutation identifies one candidate slot. Replaying that slot's training-prior draw supplies a single duration T for all eight new candidates. Neither a reference duration nor a teacher score chooses T. The same native-clock models, seeds 42 and 43, route conditions, initial-noise keys, $K = 8$ and 100 DDIM steps are retained. Other candidates can have a different sequence length because their former duration draws have been replaced; equal candidate and denoising-call counts do not imply equal tensor computation.

The original Uniform slot provides an exact consistency control: its timestamp–coordinate array must remain bitwise identical to the corresponding G33 output. The completed run verifies this for all 97 map-ready requests in each seed, with

zero maximum absolute difference. It also verifies that every candidate and each of the four selected outputs uses that request's common duration. No diffusion model, CNN or duration prior is refitted, and no scoring weight or guidance step is searched. Uniform, S, V and SV operate on the same new pool using the unchanged terminal-selection rule.

The follow-up comprises two pools of 776 candidates and eight selected-set evaluations. All retain the original 100-request denominator, 97 generated requests and 94 paired references. The same physical metrics, support rules and diversity diagnostics apply; constant duration across strategies is a control, not a feedback improvement. Because the reference was already analyzed in G33 and motivated this question, G34 is a development mechanism comparison, not a new blind replication. It is reported alongside the original experiment rather than replacing it. Both comparisons concern terminal feedback; neither supplies an in-loop guidance or new downstream-augmentation result.

6.7 Expanded-model replication and fixed candidate controls

The expanded classification audit first evaluates pilot and expanded readers on exactly the same 7,230 development legs. Saved expanded-model predictions are verified against their checkpoint and input hashes; old readers only perform inference. Training parents of both reader versions are disjoint from these development parents. Some old development identities reappear as new stochastic realizations, so this is not a wholly unseen source-template population. Reported parent-bootstrap intervals condition on the existing public domain and fixed models, with 2,000 paired resamples. Leg-weighted results are accompanied by equal-parent sensitivity results.

The generation replication freezes 100 requests per scene before simulator output is available: 24 public bus services and 19 held-out source plans for each other class. The same underlying plans appear in both scenes, with distinct target identities, independent simulator seeds and departure windows of 18,000–18,600 and 32,400–33,000 seconds. The scenes therefore vary traffic context and realization together; they are not an isolated intervention on clock time. Non-bus requests use complete single-leg plans without internal activity parking. The broader training activity-chain support must not be mistaken for a full activity-chain generation evaluation.

Two pilot and two expanded native-time models generate eight candidates per supported request using 100 DDIM steps. One duration is sampled from the corresponding training-side prior and held fixed across the eight candidates. Each model era retains its own prior; the old/new generator comparison consequently tests the complete factorized system. A reference future path, duration or class score never enters sampling. Model seeds, selection rules and experiment limits are

locked before outcomes.

For each expanded-generator pool, both old and expanded teacher pairs apply Uniform, spatial S, velocity V and equal-weight SV selection to the exact same trajectories. This is the primary teacher contrast: it holds the generator, candidate budget and duration fixed. Pilot-generator pools additionally receive pilot-teacher selection as a baseline. The unchanged Uniform selections are checked array by array across teacher pairs. Eight full-pool and 48 selected-set quality evaluations result, with no success-based replacement of requests.

Table 6.3. Information and controls in the expanded replication.

Quantity	Fixed or varied
Reference	Two new SUMO episodes; shared public plans and map
Generators	Pilot/expanded native models; seeds 42 and 43
Candidates	Eight per route/class; 100 DDIM steps
Duration	Fixed within each pool; sampled from that generator's prior
Teacher contrast	Old/new readers applied to the same expanded pool
Policies	Uniform, spatial, velocity, equal-weight dual
Quality	Independent coordinates; no selecting CNN as ground truth
Coverage	All 100 requests retained in each scene denominator

6.8 Distributional fidelity and request-level agreement

The primary W_1 analysis weights classes and requests equally. Candidates divide their request's mass; native intervals are weighted by duration.

For scalar movement summaries, a second, already recorded quantity compares corresponding requests:

$$\mathrm{MAE}_c(f) = \frac{1}{|Q_c|} \sum_{q \in Q_c} \frac{1}{K_q} \sum_{k=1}^{K_q} |f(\widehat{\mathcal{T}}_{qk}) - f(\mathcal{T}_q)|. \tag{6.2}$$

Here $K_q = 1$ after selection. MAE preserves request identity; marginal W_1 measures population agreement. One stochastic reference is not a route's unique permissible future, so neither measure alone establishes generative quality.

The post-hoc audit retains every endpoint, averaging five-class macro errors equally over four scene/seed cells before taking percentage changes. Two generator seeds are technical replications, not additional traffic scenes. Improving-cell counts are descriptive. Duration, route length and predominantly fixed mean speed remain controls.

Chapter 7

Complete-trip generation results

The initial complete-trip pilot establishes two findings. First, learning motion on the native physical clock substantially improves the generated acceleration distribution relative to the matched 257-point diffusion model, and also improves it relative to empirical motion transfer. Second, both GAF readers recognize classes and change candidate selection, but higher class confidence does not translate into a uniform improvement of the selected trajectory distribution. This chapter presents these findings together with their costs and coverage. The pilot comparisons use the independently simulated reference episode introduced in Chapter 3; no reference path or duration is an input to generation.

The chapter then follows the expanded corpus and the two-scene replication in Sections 7.8–7.11, separating new evidence from the pilot and its reused-reference diagnostic.

7.1 Pilot runs and evaluation population

All four fixed diffusion fits completed 10,000 optimization steps: the relative-time and native-time models, each with seeds 42 and 43. Each model contains 891,969 parameters and is evaluated using its final exponential-moving-average weights. The relative-time runs took 300.6 and 318.8 seconds; the native-time runs took 1,520.0 and 1,529.4 seconds on the available L20 setup. The native runs processed approximately 1.36 billion valid temporal values, compared with 163.84 million for each relative-time run. These are measured execution costs for this implementation and concurrent workload, not a hardware-independent benchmark or an equal-compute comparison.

Generating and exporting 776 candidates took 322.4/309.7 seconds for the relative-clock model and 335.9/324.9 seconds for the native-clock model; each used

77,600 measured denoiser forward calls. Empirical transfer took 16.6/16.4 seconds with no denoiser calls. Encoding and scoring both GAF views for each native pool added approximately 12 seconds. These timings include this implementation's sequential candidate processing and export work. They distinguish computational cost from fidelity and do not establish a general speed advantage over SUMO or a batched alternative.

The prospective request denominator remains 100. The map interface supports 97 requests and the reference extraction accepts 97; their intersection contains 94. Every full-pool method produces eight candidates for each supported map request, giving 776 generated trajectories, including 752 associated with the paired quality population. This gives 3,104 exported diffusion trajectories across the four model runs and another 1,552 empirical-transfer trajectories across the two corresponding baseline runs. Each terminal policy delivers one of the same eight candidates for all 97 generated requests. It neither changes the geometric coverage nor removes the requests without an accepted simulator reference.

The paired population contains 19 buses, 18 motorcycles and 19 participants in each of the other three classes. All reported physical measures have support from these same class counts. The tables use five-class macro Wasserstein distances, with equal request weight inside a class and equal candidate weight inside a request. Native-interval distributions additionally weight observations by their represented time interval. A lower distance means closer distributions; it is not a time-aligned error against the unique future of a particular vehicle. Both model seeds are shown separately. They are two fitted models evaluated against one reference episode, not two independent simulator replications.

7.2 Native-time diffusion improves local motion distributions

The clearest improvement is in native-interval acceleration. Its macro W_1 decreases from 0.8397 to 0.1492 m/s^2 for seed 42, and from 0.8319 to 0.1575 m/s^2 for seed 43. These correspond to reductions of 82.2% and 81.1% relative to the matched relative-time model. The native model also improves over empirical transfer, whose distances are 0.3245 and 0.3588 m/s^2. The reduction relative to empirical transfer is 54.0% and 56.1%. These percentages describe the measured distances in this experiment; no significance level or cross-episode uncertainty is inferred from them.

The class-wise acceleration result is consistent with the macro result: the native model has lower W_1 than both comparators for each of the five classes in both seeds. For example, bus acceleration distances decrease from 0.6174/0.5955 to 0.0837/0.0746 m/s^2, while motorcycle distances decrease from 1.4987/1.5061 to 0.1833/0.2199 m/s^2. The common direction across these classes makes the macro

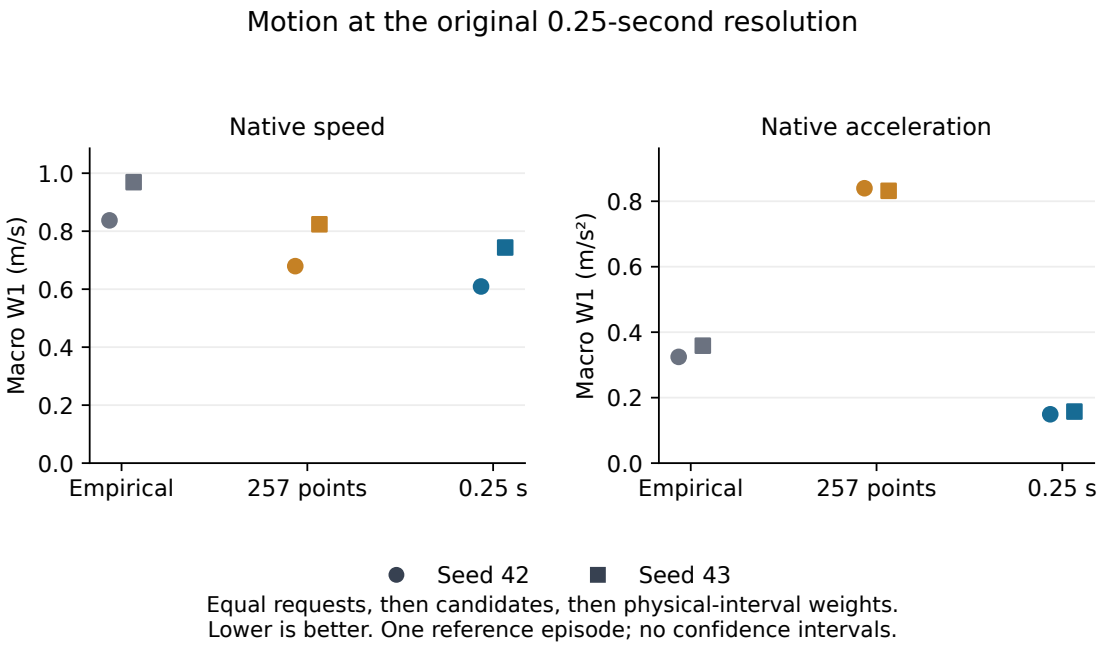


Figure 7.1. Native-interval speed and acceleration distribution distances for the two diffusion clocks and empirical transfer. Both seeds use the same prospective requests and their corresponding shared duration draws. Lower values indicate closer distributions, not pointwise prediction accuracy.

improvement more informative than a result driven only by the largest class. Class-balanced aggregation also prevents the much larger private-car training cohort from determining the headline number by sample count alone.

Native-speed distances improve more modestly. The 257-point values are 0.6794 and 0.8238 m/s; native-time diffusion gives 0.6093 and 0.7437 m/s. Empirical transfer gives 0.8373 and 0.9691 m/s. Unlike acceleration, not every class-level speed comparison improves. Seed-42 motorcycle speed distance, for example, rises from 0.5157 to 0.6876 m/s. Thus the acceleration finding should not be expanded into a claim that every feature of every class is closer to the simulator.

The lower-frequency summaries provide a related but distinct result. The distance for per-trip mean absolute acceleration falls from approximately 0.113 m/s^2 in both relative-time seeds to 0.0661/0.0560 m/s^2 in the native model. This is not the same metric as native acceleration W_1: one compares a summary value per trip after one-second processing, whereas the other compares the distribution of acceleration intervals. Agreement in direction supports an improvement at both resolutions, without making their numerical values interchangeable.

Table 7.1. Representative full-pool quality measures. Each cell gives seed 42 / seed 43; lower W_1 is better. The native speed and acceleration rows compare interval distributions. Other motion rows compare distributions of per-trip summaries on the one-second clock. All rows have 94 paired requests and all five classes.

Measure and unit	257-point	Native 0.25 s	Empirical
Native speed (m/s)	0.679 / 0.824	0.609 / 0.744	0.837 / 0.969
Native acceleration (m/s^2)	0.840 / 0.832	0.149 / 0.158	0.325 / 0.359
Mean absolute acceleration (m/s^2)	0.113 / 0.113	0.066 / 0.056	0.197 / 0.206
Stopped fraction	0.0278 / 0.0268	0.0179 / 0.0171	0.0156 / 0.0128
Longest stop (s)	4.69 / 4.89	7.75 / 7.66	4.73 / 4.15
Longest steady movement (s)	20.42 / 19.26	21.54 / 20.43	22.54 / 20.73
Four-second speed change (m/s)	0.233 / 0.220	0.238 / 0.193	0.581 / 0.595
Duration (s)	48.28 / 46.39	48.28 / 46.39	48.28 / 46.39

7.3 Long-duration structure and diversity retain important tradeoffs

The finer clock does not solve the whole motion problem. Stopped-fraction distance improves relative to the 257-point model, but longest-stop distance becomes worse in both seeds. The difference is especially visible for buses: their longest-stop W_1 rises from 12.03/11.76 to 26.37/25.66 seconds. A closer total proportion of stopped time can therefore coexist with a less accurate distribution of sustained stops. Stop amount and stop arrangement are separate properties of a synthetic journey.

Longest steady-movement distance also increases modestly in the five-class macro. The class breakdown reveals a larger private-car change: 22.57/30.12 seconds becomes 54.82/54.31 seconds. Motorcycle steady-movement distance, by contrast, falls to 9.40/4.81 seconds from 24.47/20.56 seconds. Four-second speed-change distance improves for seed 43 but is slightly worse for seed 42. These patterns show why a single temporal summary would be insufficient to describe the learned motion. Empirical transfer remains competitive on stopped fraction and longest stops, despite its worse native acceleration distribution.

The equality of duration distances is an intended control. For a given seed, the three full-pool methods receive identical duration draws: their macro duration distances are exactly 48.2841 or 46.3904 seconds. This equality does not show that the diffusion models have learned duration equally well; duration is provided by the common training-side prior. Mean-speed distances are correspondingly close because the methods traverse the same prescribed route over those same durations. Their differences in stopping and acceleration describe how movement is arranged within that shared constraint.

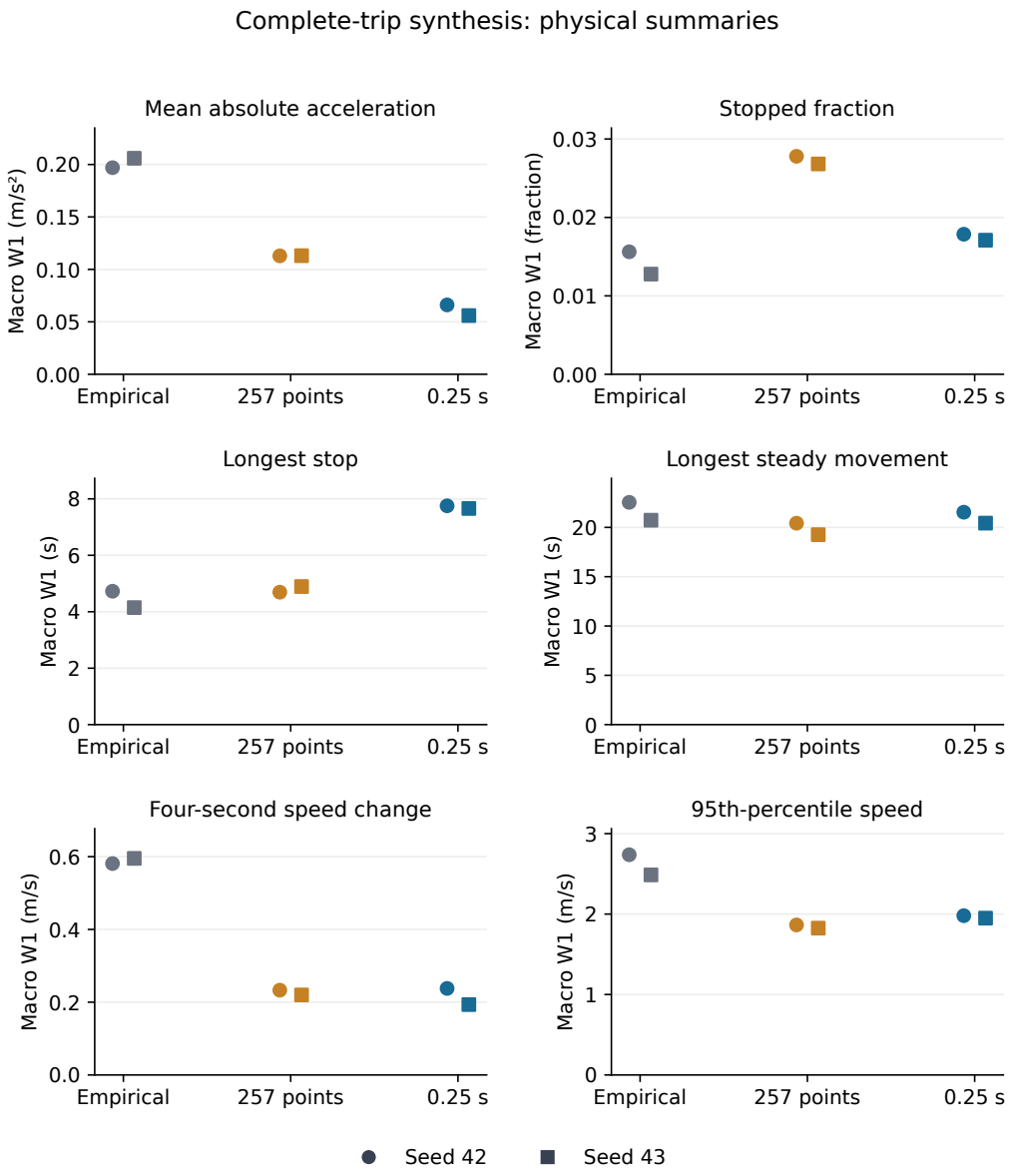


Figure 7.2. Six representative per-trip physical summaries for the complete candidate pools; all ten are tabulated in the appendix. Circles and squares denote seeds 42 and 43. Stop amount, the longest contiguous stop, and sustained steady movement expose different properties of the generated motion. Shared duration and route conditions are accounted for separately in the tables.

The native model retains variation among its candidates. Across all 97 generated requests, mean pairwise position separation at corresponding fractions of trip time is 147.27/145.95 m, compared with 129.90/132.85 m for the relative-time model and 133.07/134.04 m for empirical transfer. The corresponding normalized-progress differences are 0.0402/0.0404, 0.0343/0.0351 and 0.0346/0.0348. Candidate-pair differences are averaged within each request before averaging requests. These are descriptive measures of variation, not objectives to maximize: larger separation can reflect either useful diversity or less accurate movement.

The mean normalized-progress distance to the nearest training shape is about 0.015 for native diffusion and 0.0126 for relative-time diffusion, compared with approximately 0.00013 for the transfer baseline. The empirical result is consistent with explicitly transferring a training progress profile. The diffusion results indicate a different relationship to that reference library under this particular shape measure. They do not prove absence of memorization or a privacy guarantee. The full metric matrix and class-specific values are retained in the appendix and machine-readable experiment tables, rather than reduced to a single cross-unit quality score.

7.4 GAF readers distinguish classes and candidate realizations

The frozen spatial reader achieves 69.80% macro F1 on 126 observed development legs; the velocity reader achieves 84.99%. Their accuracies are 75.40% and 84.13%, respectively. Both fit the training labels perfectly, and their development probability estimates retain calibration error: fixed-bin ECE is 0.1788 for spatial images and 0.1303 for velocity images. These are single-reader-seed results on the current five-class complete-trip cohort. They cannot be compared by subtraction with the six-class source-diagnosis results as if only the network had changed.

The generated candidate pools also show genuine score variation. Among the 194 request–model-seed pools, 193 contain eight distinct spatial RGB images and one contains seven. Neither reader has an exactly constant floored target-class log score in any pool, and no pool uses the shared encoding-failure fallback. Spatial selection differs from Uniform in 177/194 pools, velocity selection in 172/194, and joint selection in 174/194. Spatial and velocity selectors choose the same candidate in only 31/194 pools. These counts describe paired pools from the same 97 map requests, not 194 independent traffic episodes.

It would therefore be incorrect to explain this experiment by saying that a fixed route makes all spatial GAF images identical. The route is shared, but its sampled traversal and the resulting encoding can vary. Image differences and score changes establish that the feedback can act on the candidate set. They do not establish that the changes represent better movement: spatial differences may include effects

of sampling density or skipped grid cells, while confidence can reward a class-distinctive but distributionally atypical realization.

7.5 Independent quality after terminal selection

Table 7.2 compares the delivered samples under the four policies. All policies see the same eight candidates, use the same frozen readers, and return one sample for each of the 97 supported requests. The quality comparison uses the same 94 paired references. Comparing a selected set directly with the entire eight-sample pool would mix selection with sample count; Uniform is the relevant equal-output-size control.

Table 7.2. Physical quality after selecting one trajectory per request. Values are five-class macro W_1; lower is better. Speed and acceleration use native intervals. The complete per-trip and class-specific matrix is retained in the appendix.

Policy	Seed	Duration s	Speed m/s	Acceleration m/s^2	Steady s	Longest stop s
Uniform	42	57.78	0.733	0.1245	20.56	9.28
Spatial	42	58.40	0.578	0.1362	25.48	7.65
Velocity	42	74.42	1.060	0.1233	25.31	7.71
Joint	42	59.01	0.721	0.1280	24.19	8.92
Uniform	43	51.59	0.583	0.1597	21.62	8.18
Spatial	43	51.49	0.762	0.1500	19.12	7.90
Velocity	43	77.68	1.008	0.1314	30.77	9.63
Joint	43	68.06	0.862	0.1373	23.02	10.80

Spatial selection has some consistent, specific benefits: stopped-fraction, longest-stop and 95th-percentile-speed distances are lower than Uniform in both seeds. For example, longest-stop distance changes from 9.28/8.18 to 7.65/7.90 seconds. Other effects depend on the model seed. Native-speed distance improves from 0.733 to 0.578 m/s for seed 42 but worsens from 0.583 to 0.762 m/s for seed 43. Steady-movement and four-second-change distances also change in opposite directions across the two seeds. This supports a differentiated result, rather than either a universal spatial-feedback improvement or a claim of no effect.

Velocity selection illustrates the gap between recognition and generative quality particularly clearly. It slightly lowers native-acceleration distance in both seeds, from 0.1245/0.1597 to 0.1233/0.1314 m/s^2. At the same time, duration distance increases by 16.64/26.09 seconds and native-speed distance by 0.327/0.425 m/s. Steady-movement distance increases by 4.75/9.16 seconds. Mean target-class probability under the selecting velocity reader rises from 0.754/0.778 to 0.986/0.992

across all generated requests. That increase confirms successful optimization of the selection score; it is not evidence that the selected dataset better matches the reference.

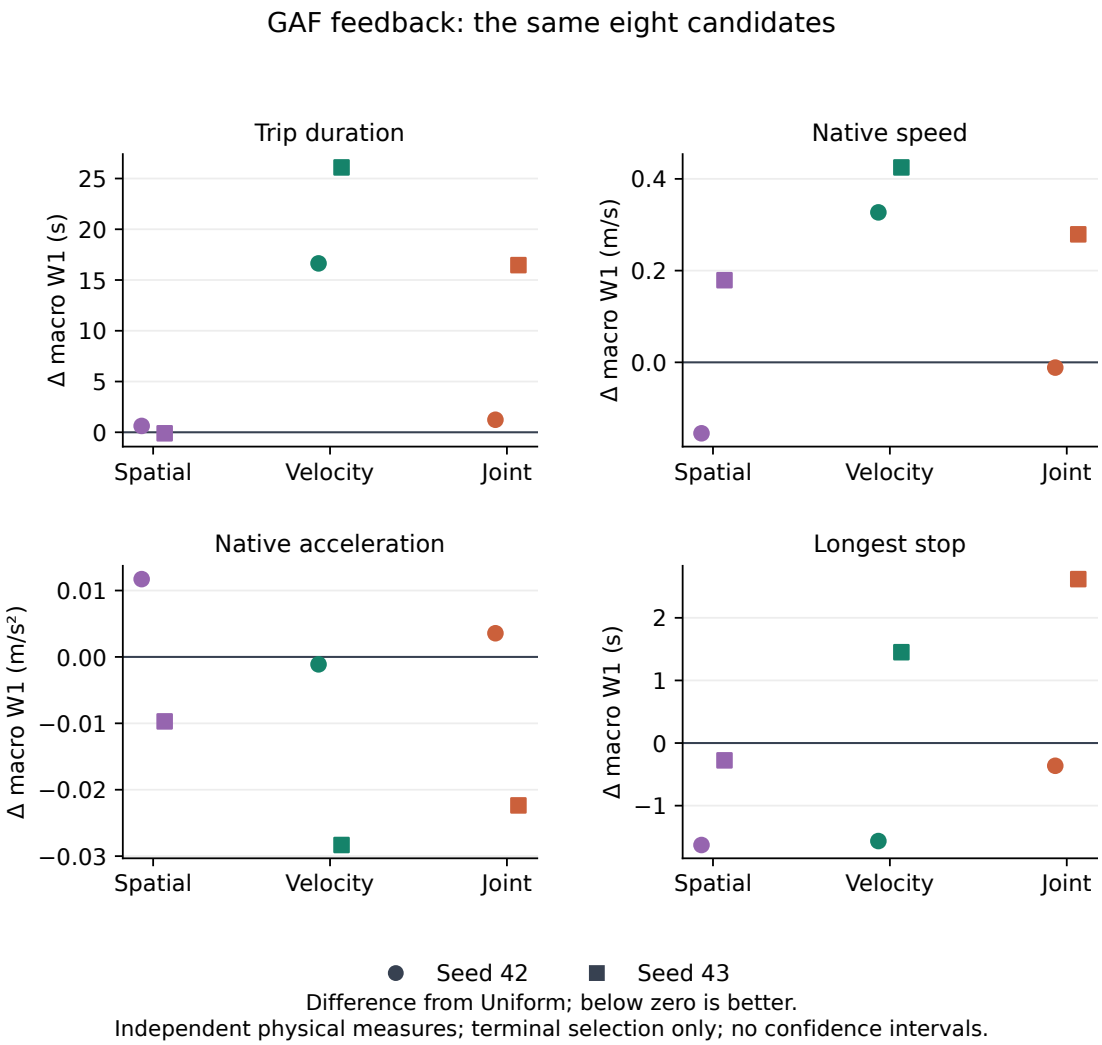


Figure 7.3. Changes in four representative physical distances relative to Uniform for spatial, velocity and joint selection: duration, native speed, native acceleration and longest stop. Negative values favour the GAF policy. Seeds are displayed separately, and every policy uses the same candidate budget. The figure shows criterion-dependent effects rather than a single overall ranking.

Joint selection does not consistently resolve these tradeoffs. Its native-speed distance is close to Uniform for seed 42 and worse for seed 43, while duration and steady-movement distances are worse in both. Averaging two class log scores is therefore not sufficient evidence of useful complementary feedback. Each candidate

in the pool has its own duration draw, so all three GAF policies can change the delivered duration distribution despite sharing a duration prior at the pool level. The present comparison measures that whole selection policy. It does not isolate motion-shape preference at fixed duration.

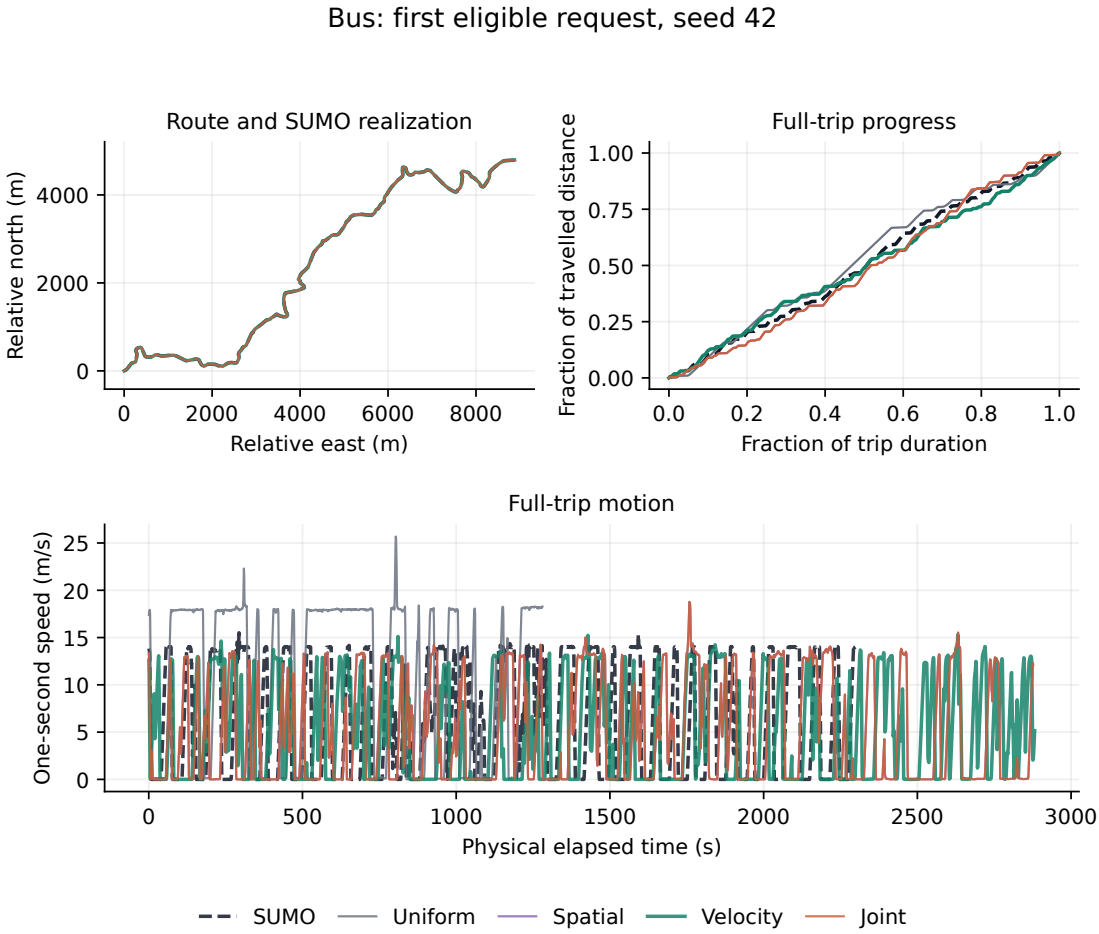


Figure 7.4. The first paired bus request in lexicographic order for seed 42, showing the independent reference and the four selected complete trajectories. The common route, normalized progress and physical-time speed are displayed separately. The example is selected by request order, not by quality, and is not a pointwise forecasting comparison.

The bus example makes the output interpretation concrete: the curves share a declared route but differ in duration and in the placement of movement and stops. A matching map trace alone cannot decide which output distribution is useful. Likewise, a case in which one selector looks closer to the reference cannot overturn

the aggregate comparison. In this original variable-duration comparison, native-clock diffusion improves an important component of generated motion, whereas GAF terminal feedback has selective benefits and measurable costs. A general improvement from spatial, velocity or joint class scoring has not been established on this reference episode.

7.6 A fixed-duration mechanism check

The preceding result leaves an ambiguity: a selector changes both the duration and the motion shape of the delivered trip. A subsequent diagnostic, recorded as G34, removes the first freedom. It was designed after inspecting G33 and reuses the same already-viewed development reference; it is not a new blind evaluation. The diffusion models, GAF readers, prior and selection weights are unchanged, with eight candidates and 100 DDIM steps retained. For each request and model seed, all candidates use the training-prior duration of the previously predetermined G33 Uniform candidate. No reference duration is used and no model is retrained.

This construction also retains a particularly strong baseline check. The old and new Uniform time and coordinate arrays are bitwise identical for all 97 generated requests in each seed. All eight candidates and all four selected outputs share that request's fixed duration. The other candidates retain their noise keys, but their sequence lengths can change when their original duration draw is replaced; the procedure is not a post-hoc rescaling of existing coordinates. The paired quality population remains the same 94 requests, with the same five-class support.

Table 7.3. Fixed-duration terminal selection in the G34 mechanism diagnostic. Within each request and seed, every policy uses the same training-prior duration. Values are five-class macro W_1 on the reused 94-request reference intersection; lower is better. Uniform is bitwise unchanged from G33.

Policy	Seed	Native acceleration m/s^2	Longest stop s	Steady s	Native speed m/s
Uniform	42	0.1245	9.280	20.556	0.733
Spatial	42	0.1372	9.287	22.488	0.812
Velocity	42	0.1124	7.193	23.536	0.745
Joint	42	0.1430	8.797	22.756	0.797
Uniform	43	0.1597	8.181	21.616	0.583
Spatial	43	0.1650	6.459	19.526	0.542
Velocity	43	0.1322	7.647	23.118	0.526
Joint	43	0.1412	7.290	22.025	0.588

With duration fixed, velocity selection reduces native-acceleration distance from 0.1245/0.1597 to 0.1124/0.1322 m/s^2, reductions of 9.73%/17.22%. Longest-stop

distance also decreases, from 9.280/8.181 to 7.193/7.647 seconds. These macro improvements are not universal across classes: acceleration distance improves for only two of five classes in seed 42 and four of five in seed 43; taxi/ride-hailing is the only class that improves in both. The result identifies a conditional aggregate benefit, rather than a claim of uniformly better class-specific motion.

The tradeoffs remain visible. Velocity selection increases longest steady-movement distance by 2.980/1.501 seconds and the 95th-percentile-speed distance by 0.236/0.294 m/s. Native-speed distance is slightly worse for seed 42 and better for seed 43. Spatial selection does not reproduce the acceleration improvement, while its longest-stop and steady-movement results differ across seeds. Joint selection improves longest-stop distance in both seeds but has mixed acceleration effects and worsens steady-movement distance in both. The full ten-summary and class-specific native-interval matrices appear in Tables A.4–A.6; source checks remain in the G34 archive.[1]

Duration distances are now exactly equal across the four policies: 57.7782 seconds for seed 42 and 51.5908 seconds for seed 43. This equality is imposed by the diagnostic, not an improved duration model. It prevents feedback from changing the request's allotted travel time and shows that the measured acceleration and longest-stop gains can occur through selection among motion realizations under that fixed constraint. The remaining adverse metrics show that feedback tradeoffs can persist when duration is held fixed. Because this is a post-G33 diagnostic on the same episode, the repeated directions across model seeds are descriptive evidence and not independent statistical confirmation.

The fixed-duration runs take 381.6 and 331.2 seconds for generation and export, with 77,600 network forward calls per seed. Encoding, scoring and selection add 12.6 and 12.3 seconds. These measured costs preserve the declared candidate and DDIM budgets while recording the actual consequences of changing the duration plan; they do not constitute an equal-wall-time comparison.

7.7 Seeing the generated journeys and their GAF feedback

The delivered object is a timestamped coordinate trajectory. To make that output visible, Figures 7.5–7.14 connect the expanded model's generated journeys to the exact two-view images used for feedback. These are completed G36 outputs, not schematic curves or newly sampled examples chosen to improve a figure. Before rendering, the display rule was fixed to the morning episode, expanded native model

[1]The completed record is under `docs/results/g34-fixed-duration/attempt2/`; its `report/macro.csv`, `report/per-class.csv` and `report/summary.json` retain the full results and provenance.

seed 42, expanded readers and the first lexicographic request in each class jointly supported by the reference, map and four policies. All five classes receive the same paired trajectory/image presentation in the main text. The cases explain the output and feedback mechanism; one fixed request per class is not an estimate of typical quality. Population-level comparisons below retain all classes and physical endpoints.

7.7.1 Geometry and motion are different parts of a trajectory

The two maps in each journey panel display the independent SUMO realization and the Uniform diffusion output, with road lanes in grey and elapsed trip fraction in colour. Start and end markers locate the complete journey. The map geometry is shared by design; it does not demonstrate that GAF has learned road connectivity. Timing along that geometry is the generated quantity. Plotting only the XY curves would conceal differences between candidates that traverse the same route at different speeds.

The lower panels therefore show physical elapsed time, accumulated progress and speed. Pale progress curves retain all eight candidates; the coloured curves identify the four recorded selectors. This separates the diversity available before feedback from the single output delivered by each selector. The reference is an independently simulated realization for the request, not a future trajectory supplied to diffusion or a unique deterministic target that every synthetic sample should reproduce.

The displayed bus also reveals a limitation that a map overlay would hide: its independent reference lasts 2,315 seconds, whereas the shared generated duration is 1,235.64 seconds. All selectors inherit that value, so none can correct the overall travel-time mismatch within this pool. Spatial selection changes the pattern of movement within the allotted duration. That is a different intervention from learning a more realistic duration prior.

7.7.2 The GAF images associated with those outputs

Figure 7.6 shows the actual images obtained from the same five trajectories. Each spatial RGB image consists of the inherited GASF/GADF/MTF channels; the velocity image contains the corresponding speed-derived fields. The generated spatial pixel hashes match the archived images used for scoring. Velocity panels use the exact reader-input transform with its final ImageNet normalization undone for display. No image receives an individually optimized colour range.

In this bus pool, Uniform, V and SV retain the same candidate, while S retains another. The spatial target probability rises from 0.9915 to 0.9992 under S; the velocity probabilities remain very small for these displayed candidates. These

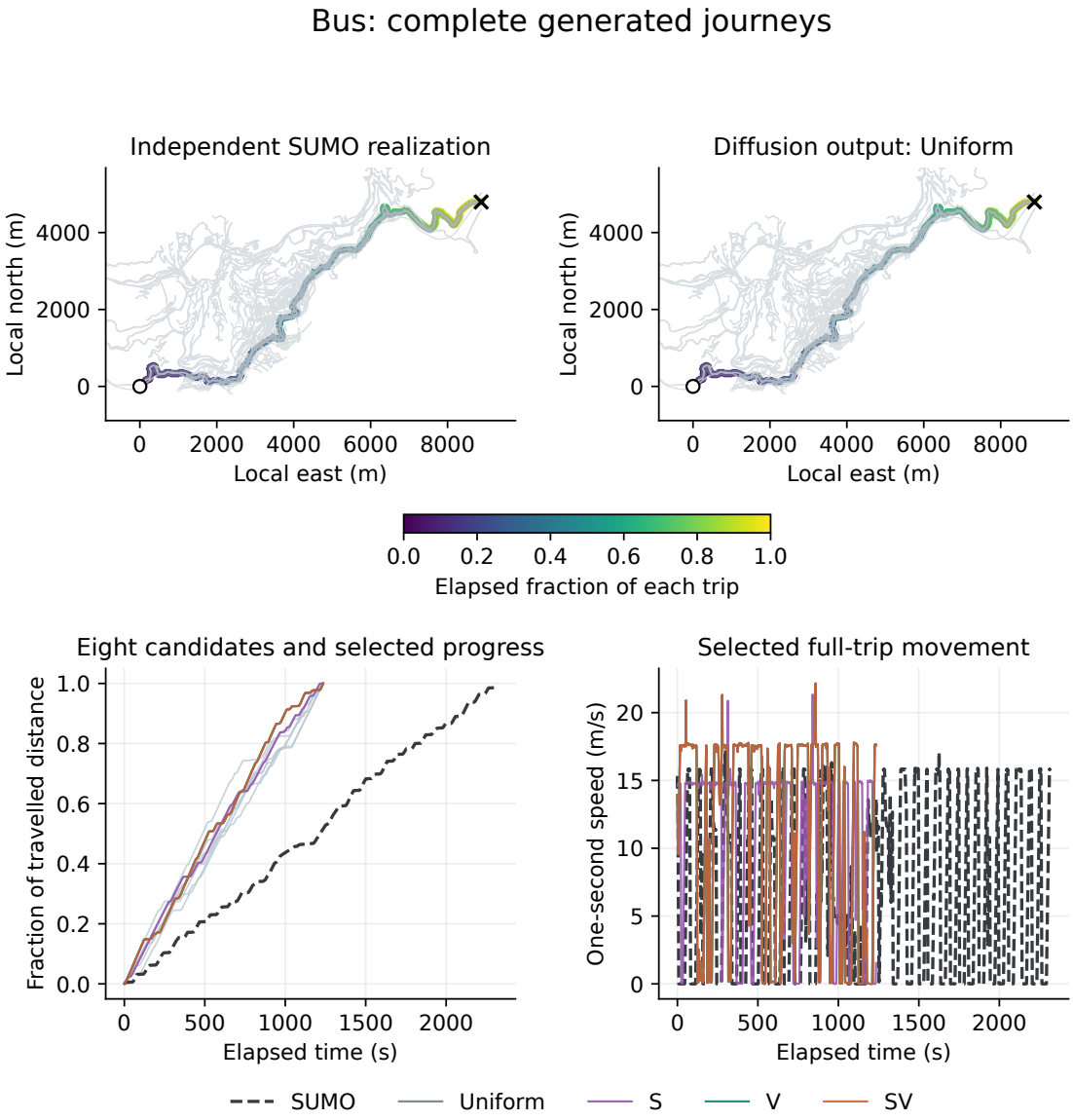


Figure 7.5. Actual complete bus journeys under a fixed road request. Top: independent SUMO reference and Uniform diffusion output, with a shared local coordinate frame and within-trip time colouring; white circle and cross mark start and end. Bottom: all eight candidate progress curves (pale) and the four selected outputs, alongside reference motion (dashed). Speed averages complete one-second intervals. Different reference and generated durations remain visible; the four generated selections share one duration.

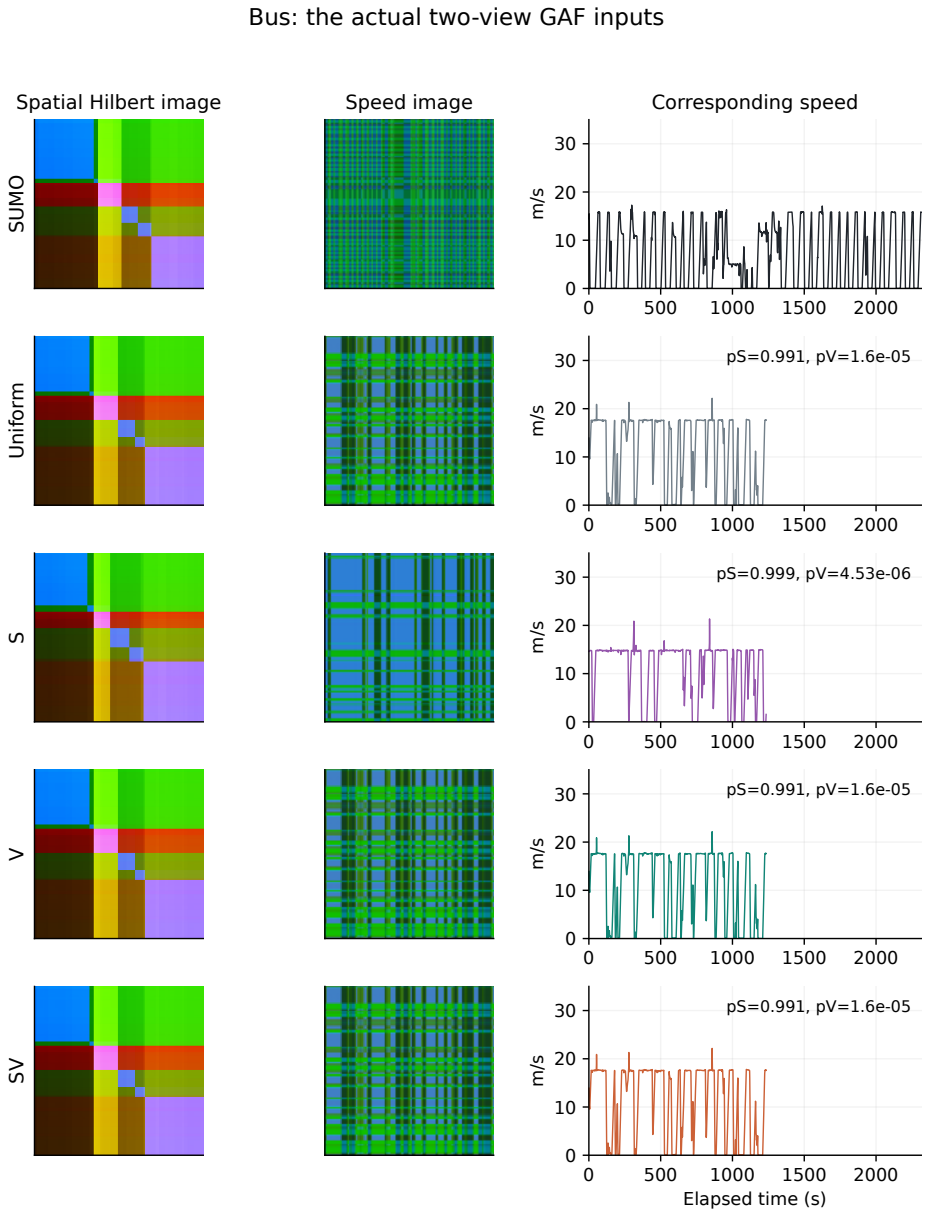


Figure 7.6. Bus trajectories and their exact GAF representations. Rows are the SUMO reference and the recorded Uniform, spatial (S), velocity (V) and joint (SV) selections. The right column ties every image pair back to its generated motion. p_S and p_V are the stored target-class probabilities of the two readers; they are auxiliary scores, not independent realism measures. Repeated candidate choices intentionally produce repeated image rows.

observations are a concrete warning against interpreting high spatial class confidence as sufficient evidence of realistic whole-trip timing. They also explain why apparently repeated image rows are informative rather than a plotting error.

The motorcycle pair follows the same display contract as the bus. Figures 7.7 and 7.8 apply the identical case rule, transforms and display settings to a motorcycle. Comparing image patterns is useful for understanding what changes under selection, but visual texture alone cannot explain participant semantics or establish a latent causal mechanism. Those claims require controlled quantitative evidence.

7.7.3 Private car, bicycle and taxi/ride-hailing

The remaining classes are part of the same synthesis task, not secondary examples. Private cars dominate the training corpus, bicycles supply a distinct low-speed domain, and taxi/ride-hailing retains a service label that need not correspond to a unique local movement pattern. All three therefore matter when interpreting what a participant-conditioned generator has learned and what its GAF readers can add.

The private-car example has very high target probabilities in both readers, yet the reference duration is 747.25 seconds and the generated duration is 834.07 seconds. The bicycle example shows a different representation mismatch: its Uniform spatial target probability is approximately 2.59×10^{-8} while its velocity probability is saturated near one. Spatial selection changes the chosen image but still gives a low bicycle target score, approximately 3.15×10^{-5}. This is one generated-request diagnostic, not a contradiction of the reader's development-set F1.

For the taxi/ride-hailing example, Uniform has spatial/velocity probabilities of approximately 0.307/1.000. Spatial selection raises its own probability near one but chooses a candidate whose velocity probability is only 0.082. Velocity and joint selection choose other outputs. This disagreement makes the proposed two-view mechanism concrete; it does not by itself establish that either view or their combination selects more realistic motion. The five-class physical evaluation is needed to answer that question.

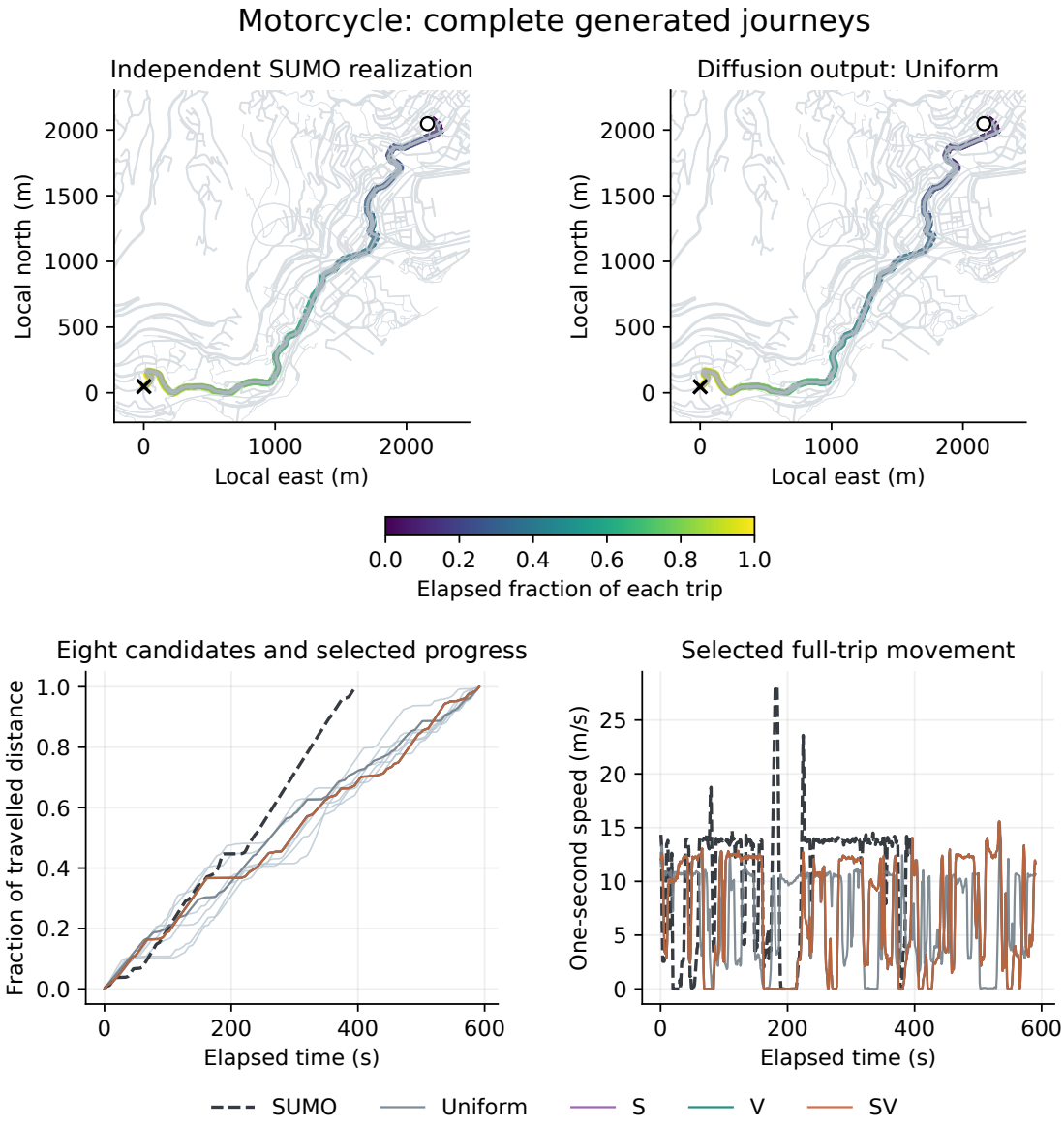


Figure 7.7. Complete motorcycle generation under the same display protocol as the bus. Map agreement reflects the route interface; progress and speed expose the generated timing and motion. All four recorded policies and all eight pre-selection progress curves are retained, regardless of whether they appear closer to the reference.

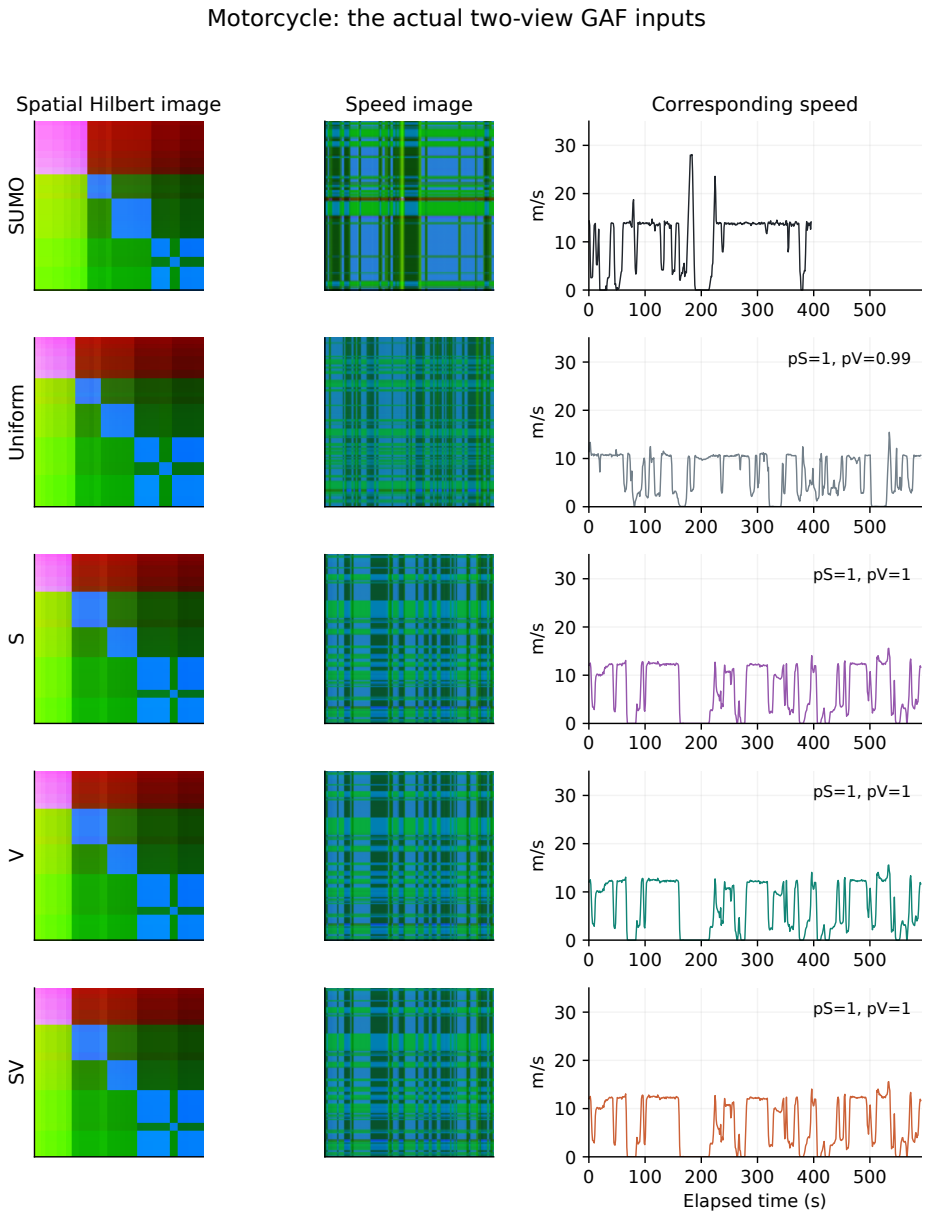


Figure 7.8. The motorcycle's exact spatial and speed image inputs, aligned with the corresponding motion curves. Both image axes follow sequence position, not map coordinates. All channels and candidate choices are produced by the same frozen transformations and selectors as the aggregate experiment. A higher reader score is not presented as a physical-quality guarantee.

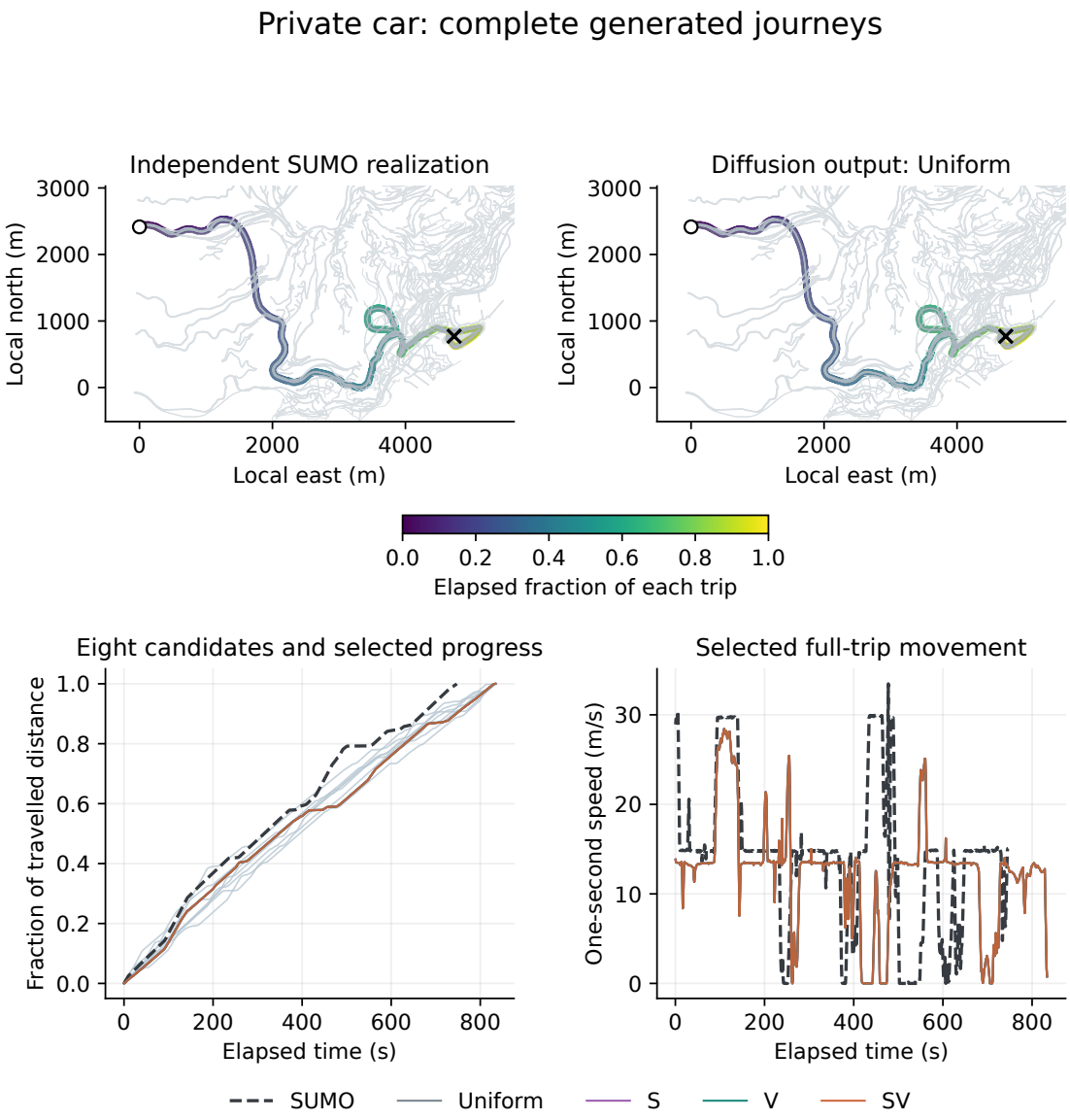


Figure 7.9. Private car: complete independent reference and generated journeys, all eight progress curves and four recorded selections. Same fixed-case rule as the main visual results.

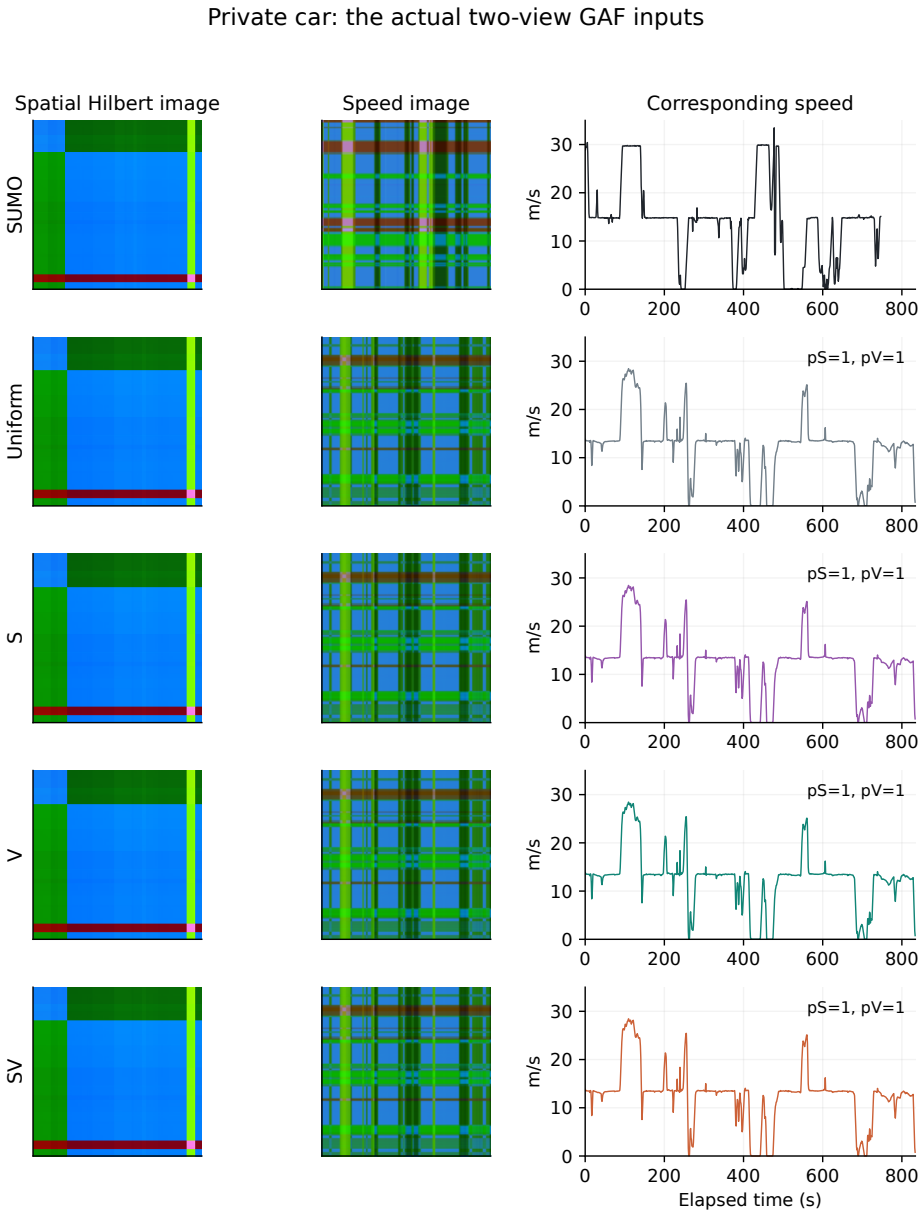


Figure 7.10. Private car: exact spatial and speed GAF images of the corresponding reference and selected outputs, with motion and auxiliary reader probabilities. Repeated selections are retained.

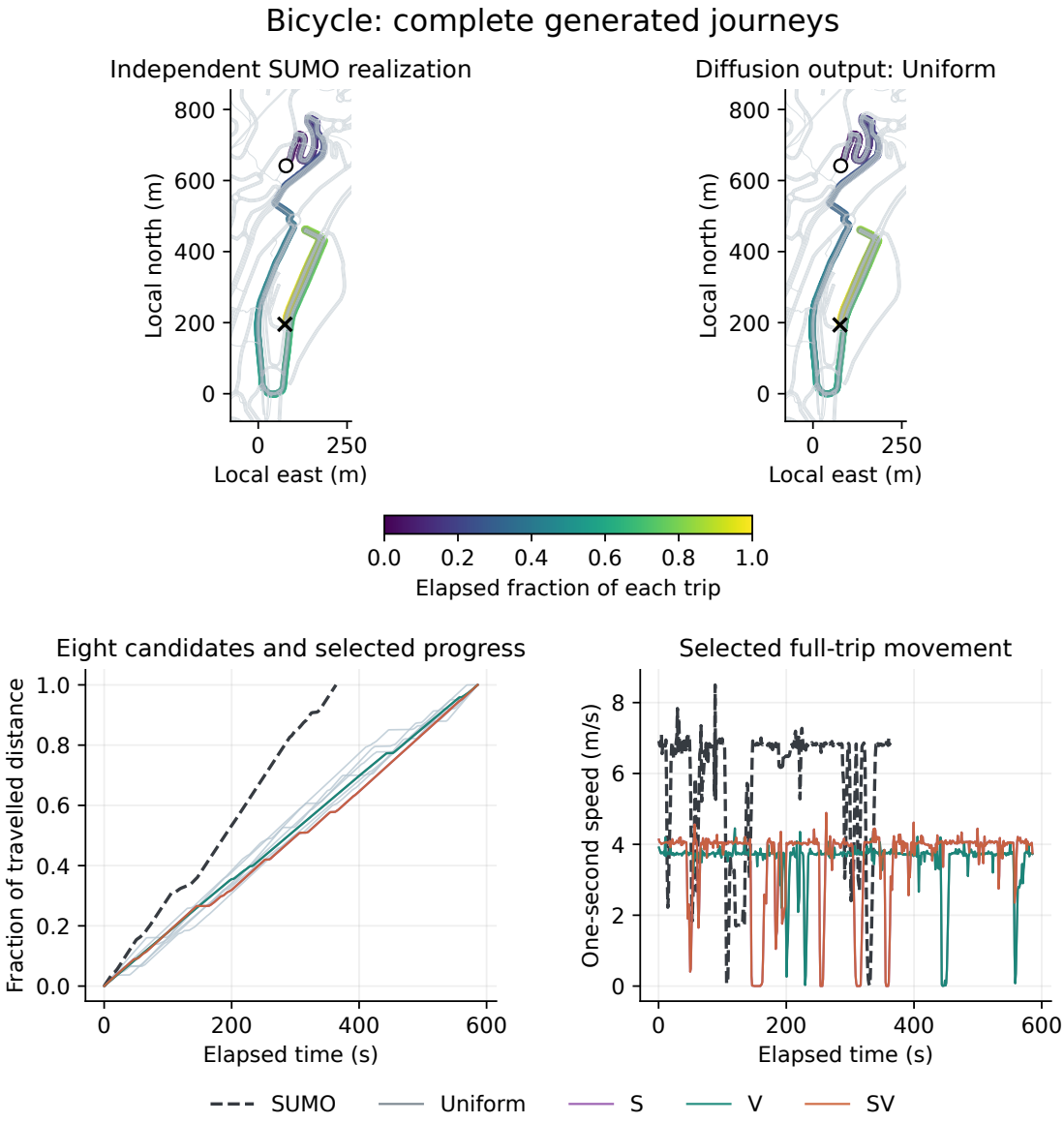


Figure 7.11. Bicycle: complete independent reference and generated journeys, all eight progress curves and four recorded selections. Same fixed-case rule as the main visual results.

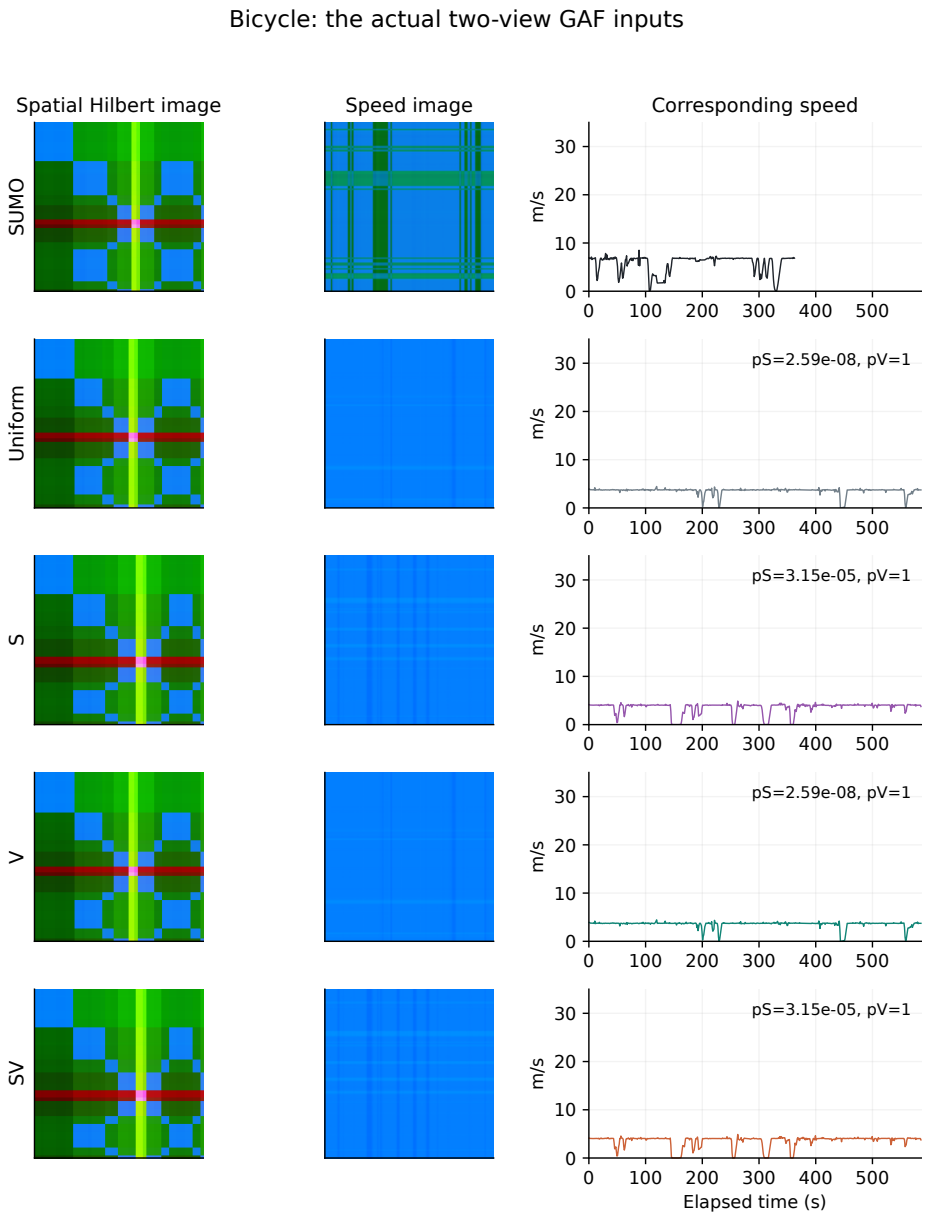


Figure 7.12. Bicycle: exact spatial and speed GAF images of the corresponding reference and selected outputs, with motion and auxiliary reader probabilities. Repeated selections are retained.

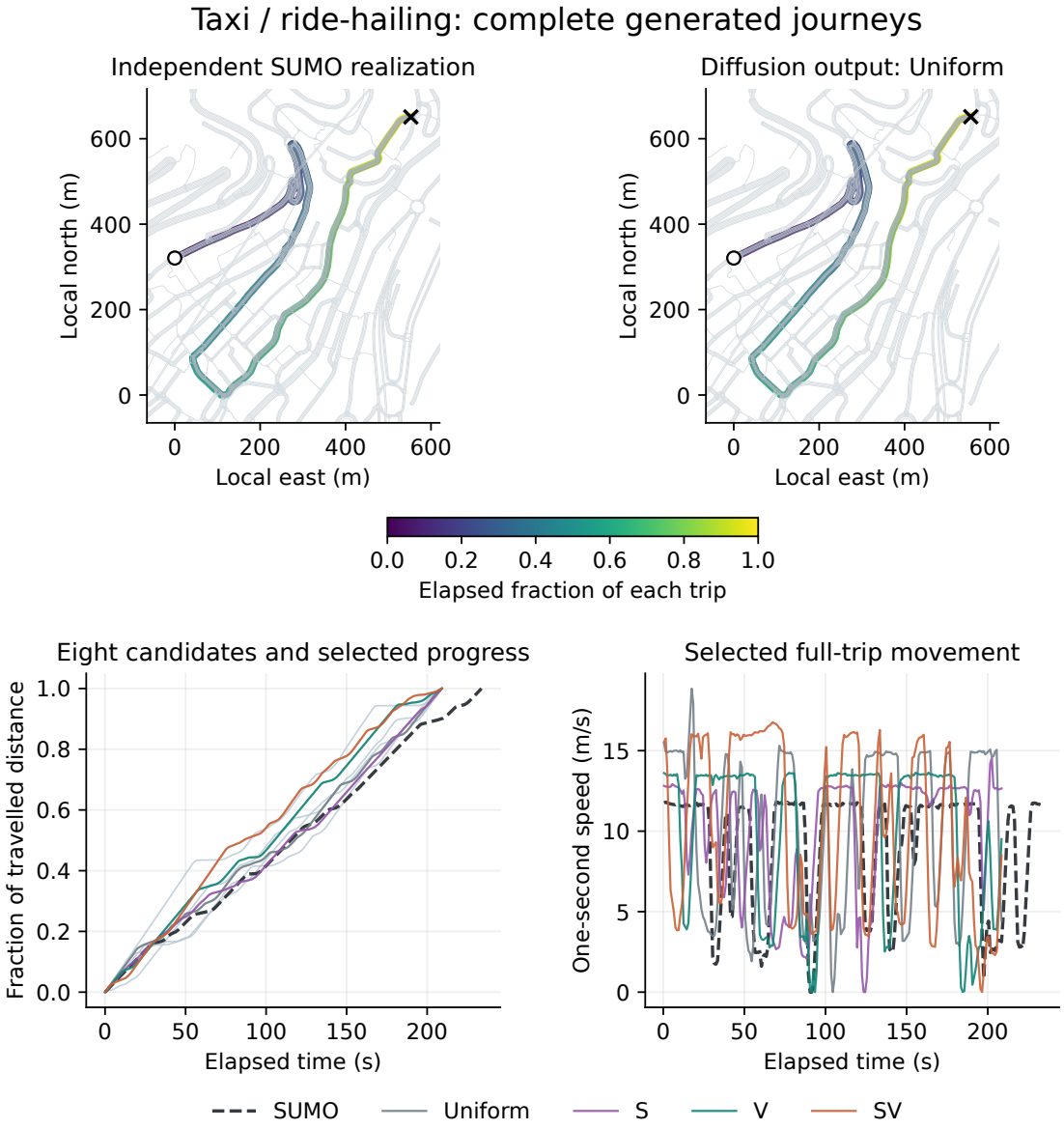


Figure 7.13. Taxi/ride-hailing: complete independent reference and generated journeys, all eight progress curves and four recorded selections. Same fixed-case rule as the main visual results.

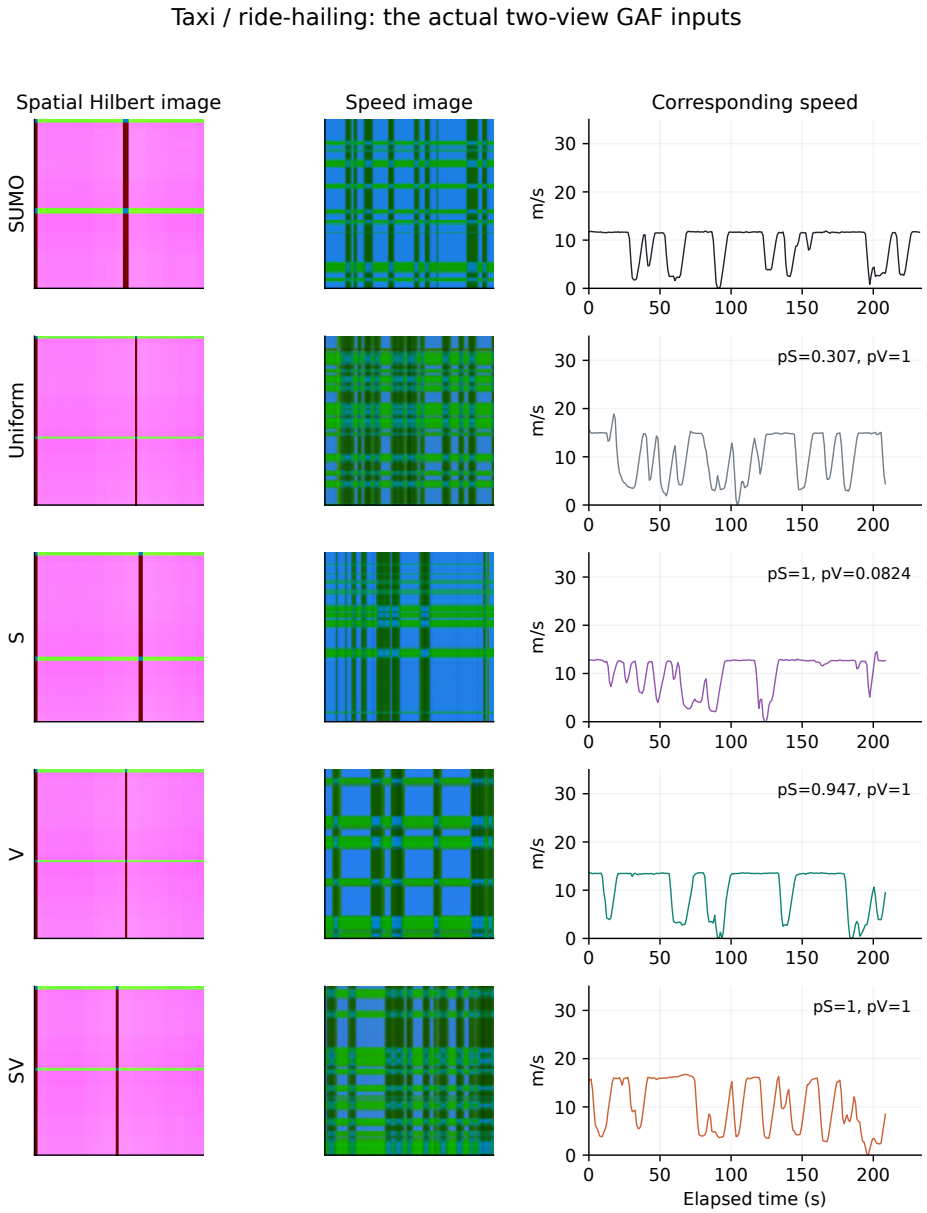


Figure 7.14. Taxi/ride-hailing: exact spatial and speed GAF images of the corresponding reference and selected outputs, with motion and auxiliary reader probabilities. Repeated selections are retained.

7.7.4 What feedback can change in a fixed candidate pool

Figure 7.15 opens the bus pool before selection: all eight generated motions and both image views are shown. Policy tags identify which existing candidate each strategy retained. Terminal feedback cannot create a ninth motion profile or repair a flaw shared by every candidate. It can only alter which available realization is delivered. This is why the following sections evaluate physical properties of the selected outputs and compare them against Uniform under the same candidate budget.

The new visual material changes the presentation, not the experimental outcomes. It adds no model fit, candidate search, teacher scoring or test access. The full scene/seed and class-level results below remain the basis for conclusions about generation quality and feedback utility.

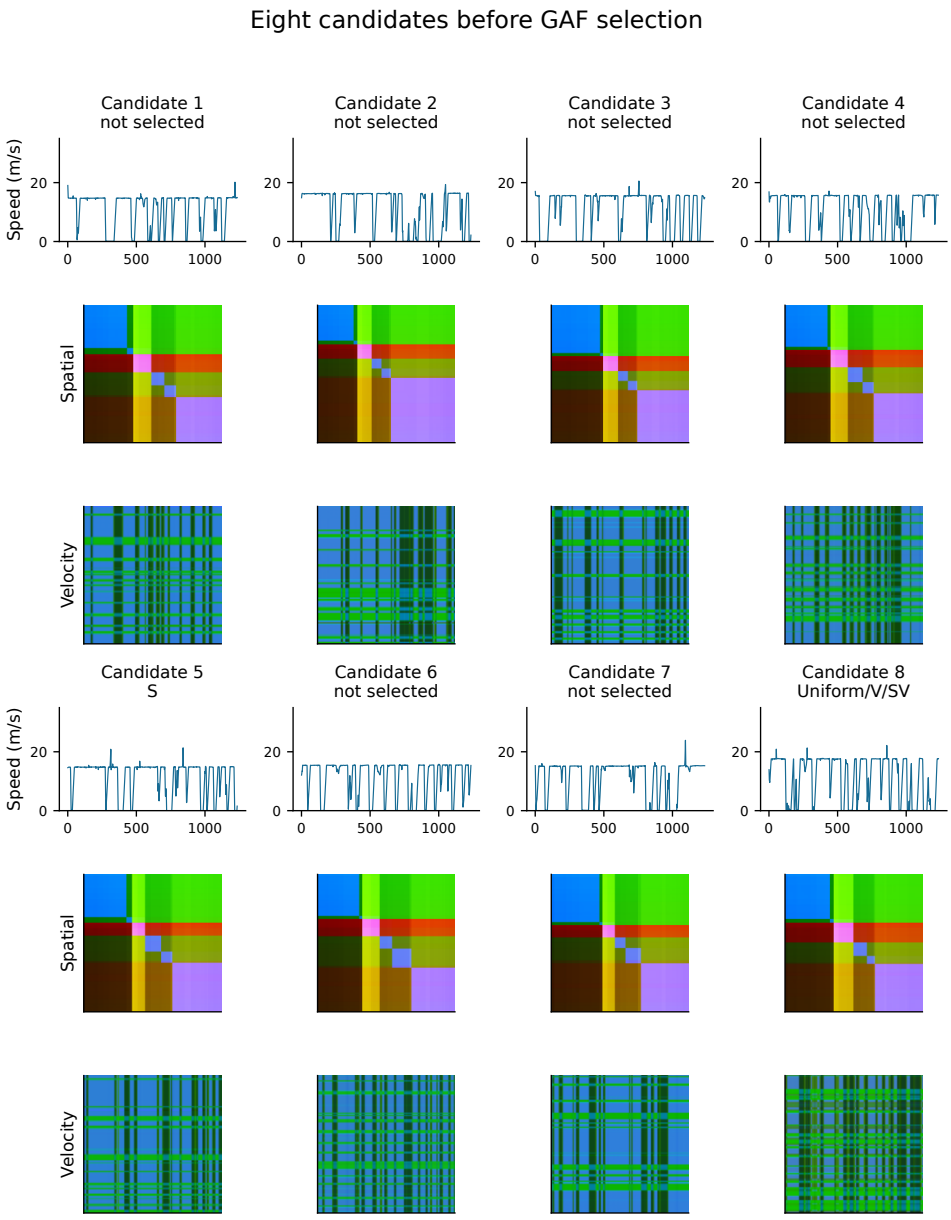


Figure 7.15. The complete eight-candidate bus pool, with one-second speed and both actual image views for every candidate. Tags mark recorded selections; multiple tags mean the same output was retained by multiple policies. All candidates have the same requested route, class and duration. This panel exposes the available alternatives rather than displaying only the winning GAF image.

7.8 Do the two GAF views recognize all five classes?

The current classifiers are trained on 29,895 complete native-demand legs and evaluated on a shared development population of 7,230 legs. The question here is whether each representation supplies usable class information for the synthesis task. Accuracy is 86.17% for the spatial reader and 89.13% for the velocity reader; macro F1 is 76.47% and 90.34%, respectively. Table 7.4 reports every class rather than relying on the more numerous private cars.

Table 7.4. Current GAF readers on the same development population. F1 values are percentages. Support counts are legs, not independent simulator scenes.

Class	Development legs	Spatial F1	Velocity F1
Bus	155	95.65	100.00
Motorcycle	1591	73.57	78.90
Private car	4436	95.20	92.23
Bicycle	668	72.10	99.48
Taxi / ride-hailing	380	45.85	81.07
Macro F1	7,230	76.47	90.34

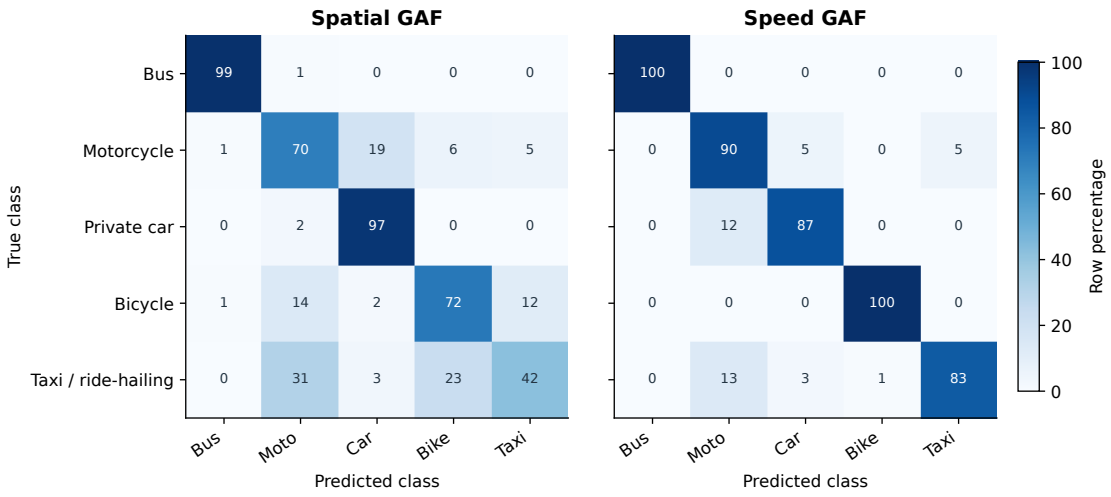


Figure 7.16. Expanded-reader development confusion matrices, normalized within each true class. The common five-class labels allow direct comparison. Spatial taxi/ride-hailing recognition remains weak despite high overall accuracy; speed-based recognition also retains confusion among motorized classes.

Both views recognize buses strongly in this development cohort. Velocity F1

is also high for bicycles, whereas spatial F1 is lower. For private cars, spatial F1 exceeds velocity F1. Motorcycles and taxi/ride-hailing remain less well separated, especially in the spatial view. These differences motivate keeping both representations in the generation comparison, but they do not identify which selector will improve a particular class's movement.

The readers are fixed 50-epoch self-trained networks, not supplied group CNNs. Shared road and service templates, minority-class support and confidence calibration limit the interpretation of their scores. The earlier and current corpus comparisons are retained in Appendix A.6 as data and optimization diagnostics. They are not a scaling-law contribution and do not replace generation evaluation.

7.9 Independent five-class generation evaluation

Both new reference simulations, all eight candidate pools and all 48 selected-set evaluations completed. Each scene has 100 predeclared requests and 97 map-supported conditions, with the three unsupported bus routes retained in its denominator. The intersections with accepted reference trajectories contain 95 requests in the morning scene and 94 in the later scene. The eight full pools contain 6,208 generated trajectories in total. They are repeated conditional realizations, not 6,208 independent road environments. The finite cloud queue, including reference simulation, generation, scoring and evaluation, took approximately 64.9 minutes.

Table 7.5. Prospective replication coverage. Requests are shared public templates across scenes; paired reference support is reported separately from geometric support.

Scene	Requested	Map supported	Quality intersection
Morning	100	97	95
Later	100	97	94

The principal comparison uses the current generator with Uniform, S, V and SV selection. These four policies share the same eight candidates, route, class and duration for each request. Reference motion is opened only for evaluation. Old/new generator contrasts are preserved in Appendix A.7; their scene-dependent effects are a supporting domain-coverage diagnostic rather than the thesis's central result.

7.10 Does GAF feedback improve the generated motion?

The central comparison is between Uniform and the three GAF policies on identical current-generator pools. No selector receives an extra candidate or a different route or allotted duration. This evaluates what the trained representation contributes after class-conditioned diffusion has already generated possible journeys. Auxiliary target probabilities are reported in the visual panels; physical quality below is computed independently of those readers.

The predeclared native-acceleration distribution comparison remains mixed. Expanded velocity feedback versus Uniform changes the four scene/seed errors by +8.7, +0.9, −9.6 and −4.7%. Thus, the two-seed acceleration benefit in the earlier fixed-duration development diagnostic does not reproduce uniformly across the new scenes and expanded components. The replication provides evidence about the boundary of that finding rather than invalidating its original, narrower population.

Table 7.6. W1 change relative to Uniform (percent), with number of improving scene/seed cells in parentheses. Post-hoc equal-cell summary; lower is better. No significance test is implied.

Endpoint	Spatial S	Speed V	Dual SV
Duration [control]	+0.0 (0/4)	+0.0 (0/4)	+0.0 (0/4)
Observed arc [control]	-1.2 (4/4)	+0.3 (2/4)	+0.0 (2/4)
Longest steady movement	-4.9 (3/4)	-5.4 (3/4)	-6.2 (3/4)
Four-second speed change	-2.6 (3/4)	-0.5 (2/4)	-4.0 (2/4)
Mean speed [control]	+0.0 (2/4)	-0.0 (2/4)	+0.0 (2/4)
Mean absolute acceleration	-3.8 (3/4)	+0.3 (1/4)	-3.9 (2/4)
Stopped fraction	+3.1 (1/4)	+1.4 (2/4)	+3.2 (2/4)
95th-percentile speed	+2.0 (2/4)	-1.3 (2/4)	-5.8 (2/4)
Longest stop	-4.6 (3/4)	-6.3 (3/4)	-4.7 (3/4)
Stop episodes/minute	+5.2 (2/4)	+10.5 (1/4)	+6.3 (1/4)
Native speed distribution	+4.2 (1/4)	+1.0 (2/4)	+1.1 (3/4)
Native acceleration distribution	-2.0 (2/4)	-0.8 (2/4)	+0.0 (1/4)

Tables 7.6 and 7.7 display all recorded scalar endpoints as well as the native distribution measures where defined. The mean errors are averaged before their ratio is taken; an average percentage change would answer a different weighting question. W1 is not removed when a paired error appears more favourable. Nor are nearly invariant travel time and mean speed counted as useful feedback wins. Small observed-arc differences arise from temporal coordinate sampling on a common route and do not establish improved route planning.

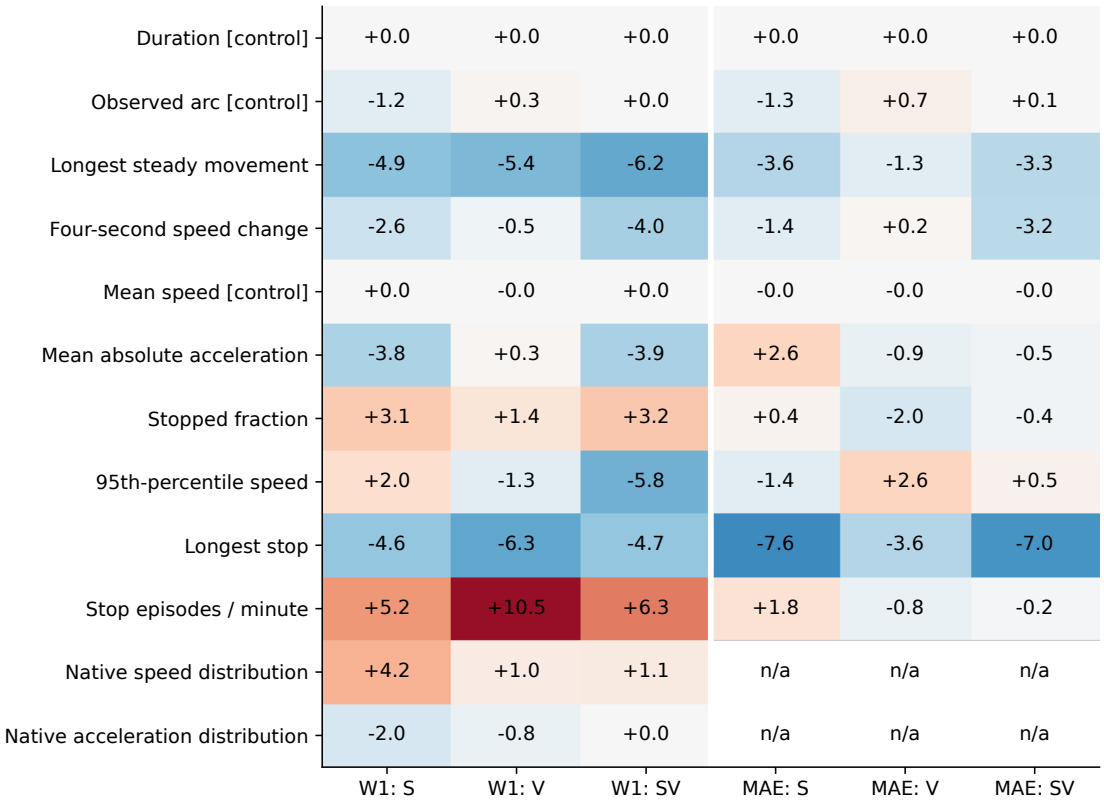


Figure 7.17. Complete metric audit for expanded-teacher feedback relative to Uniform. Entries are percentage changes in equally weighted scene/seed mean errors; blue is lower and red is higher. W1 and request-level MAE address different targets. Controls are retained. This aggregate view is post hoc, with the individual cells and original protocol preserved.

Table 7.7. MAE change relative to Uniform (percent), with number of improving scene/seed cells in parentheses. Post-hoc equal-cell summary; lower is better. No significance test is implied.

Endpoint	Spatial S	Speed V	Dual SV
Duration [control]	+0.0 (0/4)	+0.0 (0/4)	+0.0 (0/4)
Observed arc [control]	-1.3 (3/4)	+0.7 (1/4)	+0.1 (1/4)
Longest steady movement	-3.6 (3/4)	-1.3 (2/4)	-3.3 (3/4)
Four-second speed change	-1.4 (3/4)	+0.2 (2/4)	-3.2 (3/4)
Mean speed [control]	-0.0 (4/4)	-0.0 (4/4)	-0.0 (3/4)
Mean absolute acceleration	+2.6 (1/4)	-0.9 (2/4)	-0.5 (3/4)
Stopped fraction	+0.4 (1/4)	-2.0 (3/4)	-0.4 (2/4)
95th-percentile speed	-1.4 (3/4)	+2.6 (1/4)	+0.5 (1/4)
Longest stop	-7.6 (4/4)	-3.6 (3/4)	-7.0 (3/4)
Stop episodes/minute	+1.8 (2/4)	-0.8 (2/4)	-0.2 (2/4)

The combined view identifies movement-specific effects. All three selectors reduce the equal-cell mean longest-stop and longest-steady-movement distribution errors. At the same time, all three increase mean stopped-fraction and stop-frequency distribution errors. Velocity feedback lowers longest-stop W1 by 6.3% but raises stop-frequency W1 by 10.5%. Spatial feedback lowers native-acceleration W1 by 2.0% while raising native-speed W1 by 4.2%. The equal-weight dual policy does not consistently dominate either single view. These paired benefits and costs belong together in the main interpretation.

7.11 Five-class motion effects and representation tradeoffs

A participant-conditioned synthesis system must account for every supported class. Figure 7.18 therefore places all five classes, all three GAF policies and six motion-distribution endpoints together. Table 7.8 gives the underlying absolute speed and acceleration errors, so a relative improvement can be read against its baseline scale. The aggregate errors average the four scene/seed cells equally; counts of improving cells are descriptive, not independent statistical tests. This class-level reading is post hoc and is not a rule for selecting a different policy for each class.

Table 7.8. Absolute native-distribution W1 errors of the current generator and readers, averaged equally over the four scene/seed cells. Speed uses m/s and acceleration uses m/s^2; lower is better. Units must not be pooled.

Class / endpoint	Uniform	S	V	SV
Bus / speed	0.544	0.594	0.532	0.558
Bus / accel.	0.089	0.093	0.092	0.094
Motorcycle / speed	1.233	1.180	1.205	1.218
Motorcycle / accel.	0.318	0.286	0.329	0.308
Private car / speed	1.475	1.584	1.460	1.465
Private car / accel.	0.183	0.181	0.177	0.174
Bicycle / speed	0.535	0.548	0.564	0.541
Bicycle / accel.	0.198	0.199	0.178	0.207
Taxi / ride-hailing / speed	1.309	1.404	1.385	1.372
Taxi / ride-hailing / accel.	0.140	0.150	0.144	0.146

7.11.1 Bus: stopping structure and complete-trip timing

All three selectors reduce bus longest-stop distribution error in all four cells, with equal-cell reductions of 12.7% for S, 22.5% for V and 13.4% for SV. Stop-frequency errors also decrease on average. Native acceleration errors, however, increase on

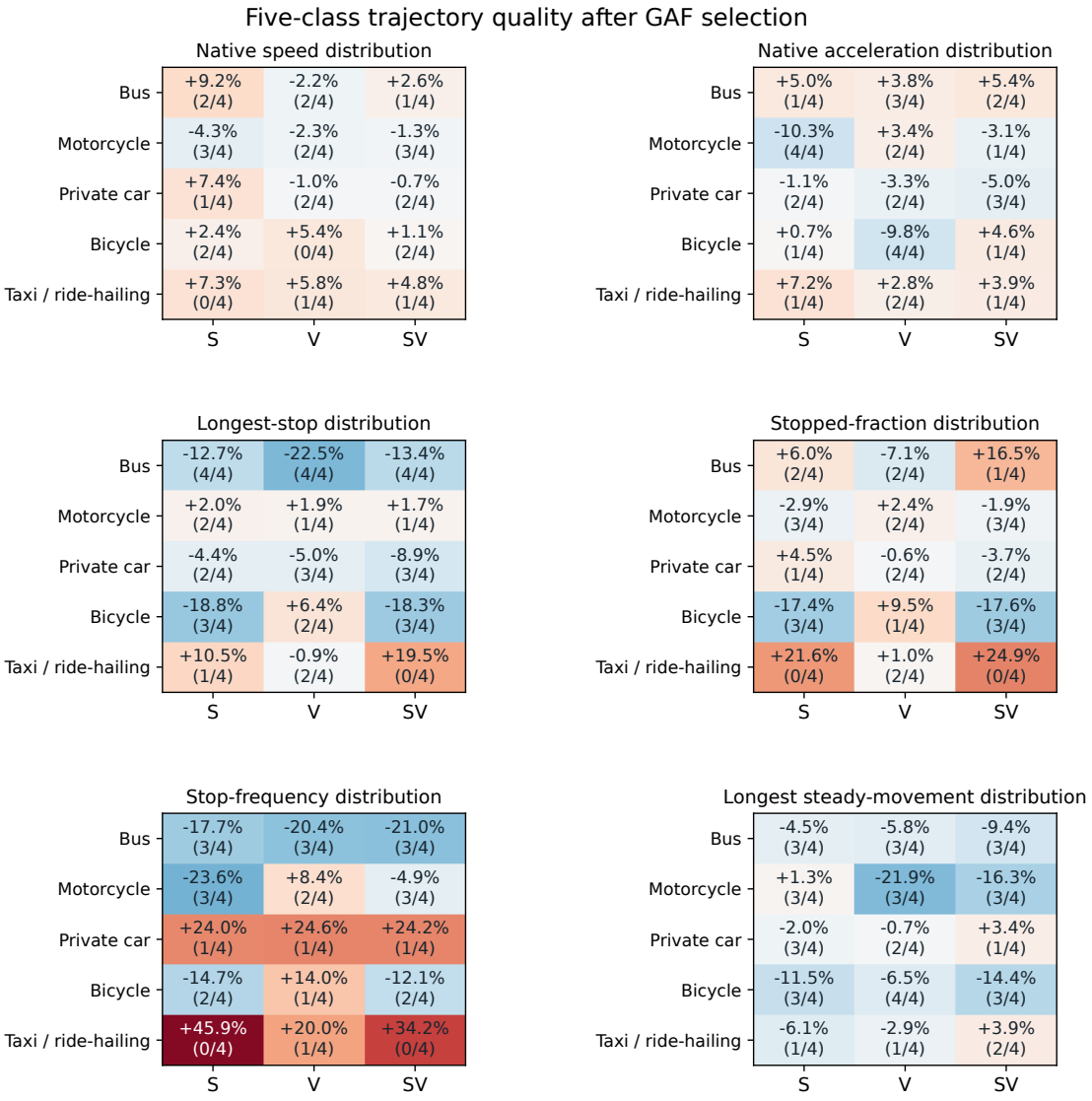


Figure 7.18. Complete five-class motion comparison for current-generator GAF feedback versus Uniform. Entries give the percentage change in mean W1 error and the number of improving scene/seed cells. Negative values are better. The same colour scale is used throughout; all classes and six motion-distribution endpoints are retained. Four cells contain two simulator scenes and two model seeds, not four independent scenes.

average for all three. These findings concern different parts of motion and cannot be merged into a claim that the bus trajectory is uniformly better. The fixed visual example further shows that selection cannot correct a travel-time error shared by the pool. The earlier paired longest-stop analysis is retained in Appendix A.9; it does not establish correct station-stop location.

7.11.2 Motorcycle: acceleration rather than a stopping-only narrative

Spatial feedback reduces motorcycle native-acceleration W1 by 10.3% in the equal-cell summary, with all four cells improving. Native-speed W1 decreases by 4.3%, with three cells improving. Velocity feedback instead increases acceleration W1 by 3.4% on average. These comparisons show why the two representations should be evaluated on motion quality even when the velocity reader has higher development F1. Motorcycle longest-stop W1 increases under all three policies, retaining a concrete cost alongside the acceleration finding.

7.11.3 Private car: common-class performance and stop-frequency cost

Joint feedback reduces private-car acceleration W1 by 5.0% and longest-stop W1 by 8.9%, with three cells improving for each. Native-speed W1 is nearly unchanged on average under V and SV, while spatial selection raises it by 7.4%. All three policies raise stop-frequency W1 by approximately 24%. The numerous private-car training examples and strong reader recognition therefore do not make every physical property easy to improve. This class is essential to the system's coverage and to the interpretation of its macro averages.

7.11.4 Bicycle: a velocity-feedback gain with a speed cost

Velocity feedback reduces bicycle native-acceleration W1 by 9.8%, with all four cells improving, while it increases native-speed W1 by 5.4%, with no cell improving. Spatial feedback reduces longest-stop W1 by 18.8% in the equal-cell summary, but does not reproduce the velocity selector's acceleration benefit. This pattern is especially informative because a nearly perfect velocity development classifier does not produce a uniformly better synthetic bicycle trajectory. The task is to reproduce a within-class motion distribution, not merely a recognizable bicycle image.

7.11.5 Taxi/ride-hailing: representation disagreement and transfer limits

Taxi/ride-hailing is retained despite weaker spatial recognition and adverse generation effects. All three policies raise mean native-speed and acceleration W1. Spatial selection raises stop-frequency W1 by 45.9%, with no cell improving; velocity and joint policies also increase its mean error. The fixed candidate example shows strong cross-view disagreement, but the aggregate physical outcomes do not establish a useful selector for this class. A service label may overlap with ordinary car motion, and the current evidence does not identify which geographical or temporal feature causes the reader's ranking.

7.11.6 What the five-class evidence supports

The spatial-motorcycle acceleration and velocity-bicycle acceleration patterns broaden the mechanism question beyond bus stopping. They are specific observations to replicate with new simulator episodes, alongside the adverse endpoints and every class. They do not justify a post-hoc class-specific policy switch, establish complementary latent factors, or isolate GAF's advantage over a matched non-image class reader. That last comparison remains necessary to attribute utility to the representation rather than generic class-based selection.

Full-pool pair diversity describes the variation available before selection. With one retained sample per request, conditional pair diversity is undefined, not zero; across-request dispersion cannot replace it. Geometric support remains 97 of 100 requests in each scene and belongs to the common route interface. The result is a five-class trajectory-synthesis study with representation-dependent motion effects, rather than a claim that more data alone or one feedback score solves generation quality.

Chapter 8

Discussion and conclusions

The work addresses two linked problems: learning complete movement on a specified road route, and using participant information from GAF representations to improve that movement. The evidence now spans a temporal-resolution pilot, a full-day native-demand expansion and a two-scene independent simulator replication. The resulting contribution is a usable synthesis pipeline and a controlled account of feedback effects. The latter are specific to movement properties and conditions; the experiments do not establish a universal GAF quality score.

8.1 A complete trajectory generator with explicit responsibilities

The delivered system generates a duration, an ordered progress profile and physical timestamps, and decodes them into positions through arrival. It exports the original five-column trajectory format. A class embedding conditions the diffusion network, while route geometry and length describe the requested journey. Class supervision is a condition on a stochastic generator, not a guarantee that every sample displays realistic within-class variation. Distinct noise draws produce different movement realizations for the same planned route.

The decomposition makes attribution possible. The deterministic road decoder supplies spatial support and endpoints; the training-side prior supplies duration; the diffusion component learns how progress unfolds within that duration. Static road conformity therefore belongs to the common geometric interface. It is not an achievement of a high GAF probability, nor does it establish collision or signal compliance in interacting traffic. Similarly, matching trip duration cannot be credited to the diffusion component alone.

The pilot provides a controlled reason to retain the physical clock. Native-time diffusion improves the native-acceleration distribution over both the matched relative-time generator and empirical transfer, with explicit costs in computation

and some long-range movement properties. That comparison remains a pilot result, not a comparison rerun at every subsequent data scale. Four expanded fits were completed, but the new prospective generation replication tests the native models. Keeping these scopes distinct prevents combining the best result from each population into a fictional single experiment.

8.2 Five participant classes as the synthesis domain

The current corpus supplies 29,895 training legs from 24,073 parent vehicles. Native public services and activity demand preserve the relationship between participant labels and movement that motivates the inherited encoding. The supplied CNR-associated tables remain the empirical source audit; the additional observations are new SUMO realizations, not an exact reconstruction of those files.

A five-class system must be assessed beyond a visually distinctive bus. Private cars test the common class, motorcycles and bicycles retain different movement regimes, and taxi/ride-hailing tests a service label whose motion may overlap with ordinary cars. The current evaluation shows every class under the same candidate budget, transforms and physical endpoints. The original label spelling and simulator role mapping remain recorded; the paper does not invent additional class distinctions to make classification easier.

The earlier/current corpus comparisons remain useful implementation evidence, but are secondary to trajectory generation. They change data coverage, optimization exposure and the training-side duration prior together, with only two corpus settings. They establish neither a sample-count law nor a reason to keep increasing data until a favourable score appears. The current five-class reader table and generated-motion matrix are the relevant evidence for the delivered system.

8.3 What GAF contributes beyond a class input

The inherited spatial view and the velocity view are transformations of the same timestamped trajectory, but they emphasize different structure. The spatial pipeline maps ordered coordinates through Hilbert indexing and image transforms. It can vary among fixed-route candidates because dwell repetition, temporal progress and visited grid cells differ. The velocity view emphasizes the motion sequence derived from coordinates and time. Neither view introduces a second independent measurement of the participant.

The experimental chain has three distinct links: class recognition on simulator observations, candidate discrimination within a requested condition, and useful selection under independent physical evaluation. Expansion strengthens the first link. The candidate comparisons test the second and third directly. Applying both old and new teachers to the same expanded pool is particularly informative: any

changed physical output arises from the selector, not from an unobserved improvement to the generator or a larger candidate budget.

A class-discriminative score and within-class realism need not coincide. Features that separate buses from motorcycles can be very different from those that distinguish an ordinary bus journey from an extreme bus-like example. Maximizing target-class probability can therefore favour a subset of class-distinctive behaviours while underrepresenting movements shared across classes. The observed mix of physical gains and costs is compatible with this explanation; the current experiments do not identify the exact causal feature used by each CNN. Calibration error remains relevant, but temperature adjustment alone is not established as a remedy [48].

The fixed-duration diagnostic showed that motion selection can improve acceleration and longest-stop discrepancies without changing the sampled travel time. The independent replication narrows that finding: more accurate velocity readers do not reproduce the acceleration benefit uniformly across new scenes and models. Equal-weight dual feedback also does not consistently surpass the better single representation. Shared origin of the two views is a reason to investigate their relation, not evidence of complementarity by itself.

8.4 Class- and motion-specific feedback hypotheses

The complete class matrix reveals effects that a bus-only analysis would miss. Spatial GAF selection reduces motorcycle acceleration W1 by 10.3% in the descriptive aggregate, with all four cells improving. Velocity GAF selection reduces bicycle acceleration W1 by 9.8% in all four cells, while making bicycle speed W1 worse in every cell. Bus longest-stop distributions improve under all selectors, private-car stop-frequency distributions worsen on average, and taxi/ride-hailing shows adverse native speed and acceleration results. No common winning policy follows from this set.

These findings motivate a representation question rather than a classifier leaderboard: which trajectory relationships are useful for controlling which movement properties? Both views derive from the same observations, but their quantization, temporal compression and normalization differ. The observed effects are consistent with different ranking preferences. They do not identify latent class factors or establish a causal benefit of angular encoding. Matched non-GAF readers, fixed budgets and independent physical evaluation remain necessary controls.

The patterns were examined after the experiment and should be replicated with fixed endpoints and new simulator episodes. A class-specific choice of S or V based on this table would be a newly tuned method requiring separate evaluation. The present paper instead retains all three policies and their costs. Bus service-stop location, motorcycle acceleration structure and bicycle speed/acceleration tradeoffs are concrete follow-up targets; none replaces the full five-class reporting obligation.

More SUMO data should primarily improve coverage of independent episodes, demand and permitted routes. New participant classes should be introduced only with documented simulator roles, appropriate road permissions and complete trajectories that preserve their behavioural context. A future truck or delivery cohort, for example, would be a new simulation-domain experiment, not a class retroactively attributed to the supplied CNR tables. Repeating a few templates under extra IDs would not establish broader class or route generalization.

8.5 Statistical perspective and synthetic-data usefulness

The original W1 weighting already gives equal importance to classes and requests, with time weighting inside trajectories. Its mixed result is not explained by an accidental dominance of long vehicles or the numerous private-car samples. Class-marginal distribution fidelity is an appropriate concern for synthetic data, but it loses request correspondence and does not measure every joint dependency. Matched-request errors restore that correspondence at the cost of comparing against one random reference realization. The two views should coexist.

The statistical audit consequently presents all endpoints under both perspectives, with the original cells and denominators preserved. Temporal controls remain separate from movement outcomes. No composite score adds seconds, metres and acceleration with arbitrary post-hoc weights. The number of improving cells is descriptive, and no significance inference treats the eight candidates or the millions of coordinate samples as independent experiments.

Future distributional assessment can use several reference realizations per condition and proper scoring rules for a fixed motion representation [52]. A feature-space score would complement, rather than replace, the interpretable physical endpoints. Its standardization must come from training data, and equal candidate budgets remain necessary. A single selected sample cannot demonstrate conditional diversity; the present selected sets therefore report that measure as undefined. Across-route dispersion and nearest-training-profile descriptions provide useful context but do not establish privacy or downstream learning utility.

The final application is synthetic trajectory data. Demonstrating improved downstream training would require a separate, controlled augmentation experiment evaluated on held-out observations. The native-demand evidence presented here does not establish such a new benefit. Earlier downstream studies concern different populations and remain separate. The present delivery is a trained generator, a reproducible conditional sampling procedure and independently evaluated feedback policies.

8.6 Scope of the evidence and final conclusions

The public map, bus services and demand families are shared between learning and evaluation. Three new simulator reference episodes are available across the pilot and replication, while the fixed-duration pilot diagnostic reuses its original reference. The expanded replication uses two traffic scenes and two generator seeds per model era. Parent-level partitions prevent fitting/development identity leakage, but shared source templates and road infrastructure remain. No unseen-city or unseen-line generalization is claimed.

Geometric coverage remains 97 of 100 requested conditions in each replication scene; physical quality is paired on 95 and 94 accepted references. Unsupported routes and simulator-reference exclusions stay in the denominator. SUMO provides documented independent realizations, not physical-road measurements or a guarantee that every retained trajectory satisfies every downstream safety criterion. The model generates one requested participant's route-conditioned motion and does not jointly simulate interactions among agents.

The thesis establishes an executable method for complete road-conditioned diffusion synthesis and a matched study of spatial and velocity GAF feedback. Physical-time representation yields a clear pilot acceleration benefit, the five-class evaluation exposes representation-dependent motion gains and costs, and same-pool comparisons separate recognizable class information from useful generation feedback. The motorcycle, bicycle and bus findings provide specific motion hypotheses for independent replication, with the private-car and taxi outcomes retained. This is a coherent connection between trajectory generation and the inherited GAF work, supported by measured outputs and explicit boundaries rather than by classifier confidence alone.

Appendix A

Complete numerical results and fixed cases

A.1 G33: all physical summaries

Tables A.1 and A.2 retain all ten predefined physical summaries for all fourteen completed evaluations. They supplement the representative measures in the main text without selecting columns by the direction of an effect. Every entry is a five-class macro Wasserstein distance, with lower values indicating closer distributions. Each quantity retains its own unit; the columns are not combined into an overall score.

Every row begins with the same 100 prospective requests and uses the same 94 paired requests for these quality measures: 19 buses, 18 motorcycles, and 19 participants in each remaining class. Map construction supports 97 requests and reference extraction accepts 97, but their supported sets differ. Each full-pool row contains 776 generated trajectories, of which 752 belong to the paired requests (94×8). Each terminal-selection row contains 97 delivered trajectories, of which 94 have accepted references. These denominators distinguish generation coverage from quality support; none of the six unsupported requests is replaced. The eight selection rows reuse candidates, so their outputs are not additional independent generations.

In the tables, "257" and "Native" denote the relative and quarter-second diffusion models, and "Empirical" denotes whole-motion transfer from training journeys. N42 and N43 identify the native models; U, S, V and SV denote Uniform, spatial, speed and joint terminal selection. Seed numbers refer to fitted diffusion models and their matched empirical draws, not independent reference episodes. All columns have full five-class support in this completed cohort. The figures and tables remain descriptive for one new simulator episode.

Table A.1. Complete macro physical distances, part I: trip scale and one-second motion summaries. Every row has 94 paired requests out of the original 100.

Model / seed	Duration (s)	Distance (m)	Longest steady (s)	Four-second change (m/s)	Mean speed (m/s)
257 / 42	48.2841	4.6610	20.4234	0.2330	0.4601
257 / 43	46.3904	4.6817	19.2598	0.2199	0.6682
Native / 42	48.2841	4.6980	21.5398	0.2379	0.4602
Native / 43	46.3904	4.7070	20.4322	0.1935	0.6682
Empirical / 42	48.2841	4.6858	22.5400	0.5810	0.4599
Empirical / 43	46.3904	4.8039	20.7270	0.5953	0.6685
N42 / U	57.7782	4.5722	20.5561	0.2155	0.6082
N42 / S	58.4039	4.5503	25.4848	0.3205	0.3985
N42 / V	74.4163	4.5190	25.3082	0.2722	0.9320
N42 / SV	59.0127	4.6159	24.1906	0.2950	0.6988
N43 / U	51.5908	4.5515	21.6164	0.2416	0.3590
N43 / S	51.4898	4.6036	19.1246	0.1354	0.7402
N43 / V	77.6837	4.5177	30.7719	0.2088	0.9148
N43 / SV	68.0578	4.4888	23.0170	0.2141	0.8416

Table A.2. Complete macro physical distances, part II: acceleration, stopped motion and upper speed. Stopped-fraction distances use fractions, not percentage points. Every row has 94 paired requests out of the original 100.

Model / seed	Mean $\lvert a \rvert$ (m/s^2)	Stopped fraction	Speed p95 (m/s)	Longest stop (s)	Stop episodes per minute
257 / 42	0.1129	0.0278	1.8650	4.6934	0.1111
257 / 43	0.1131	0.0268	1.8252	4.8932	0.1184
Native / 42	0.0661	0.0179	1.9802	7.7539	0.1000
Native / 43	0.0560	0.0171	1.9495	7.6598	0.0939
Empirical / 42	0.1970	0.0156	2.7377	4.7270	0.1265
Empirical / 43	0.2059	0.0128	2.4870	4.1470	0.1236
N42 / U	0.0547	0.0221	1.7914	9.2801	0.1185
N42 / S	0.0870	0.0167	1.7201	7.6515	0.1248
N42 / V	0.0700	0.0203	1.8115	7.7146	0.1312
N42 / SV	0.0759	0.0169	1.6402	8.9170	0.1100
N43 / U	0.0693	0.0163	1.5543	8.1813	0.0880
N43 / S	0.0432	0.0159	1.4687	7.9047	0.1039
N43 / V	0.0563	0.0234	1.9131	9.6322	0.1208
N43 / SV	0.0561	0.0198	1.7139	10.7994	0.1169

Duration and distance retain the complete observed span. The remaining summaries use complete one-second intervals; for example, "Four-second change" denotes the mean absolute four-second lag difference in speed, not acceleration over a four-second sampling interval. Precise state thresholds and support rules are defined in Chapter 6. All unselected generators share the duration prior, explaining the identical duration distances within a seed. Selection can change the distribution of durations retained from that pool.

A.2 G33: native-resolution distances by participant class

Table A.3 reports interval-distribution distances for the six full-pool generators. These are computed at the native output resolution, including fractional terminal intervals, and therefore address a different question from the one-second journey summaries. Requests have equal mass within a class; candidate mass is divided within a request, and speed or acceleration samples are then weighted by their physical interval length. A macro improvement does not imply improvement in every class or every other motion property.

Table A.3. Class-wise native speed and acceleration W_1 for every full-pool generator. Each cell lists speed (m/s) above acceleration (m/s^2). The 94 paired requests comprise 19 buses, 18 motorcycles and 19 members of each other class; each row has 776 generated and 752 paired trajectories.

Model / seed	Bus	Motorcycle	Private car	Bicycle	Taxi / ride-hailing
	1.0800	0.5157	0.9632	0.2570	0.5812
257 / 42	0.6174	1.4987	0.6751	0.8113	0.5961
	0.8305	0.5742	1.8238	0.2460	0.6445
257 / 43	0.5955	1.5061	0.6588	0.7934	0.6055
	0.7540	0.6876	0.8473	0.2243	0.5334
Native / 42	0.0837	0.1833	0.1280	0.1632	0.1879
	0.6947	0.5444	1.6986	0.2193	0.5615
Native / 43	0.0746	0.2199	0.1250	0.1563	0.2117
	1.4309	0.7069	1.2088	0.2879	0.5521
Empirical / 42	0.4435	0.4065	0.1947	0.2379	0.3401
	1.2757	0.6626	2.0861	0.2634	0.5578
Empirical / 43	0.4483	0.4308	0.3134	0.2502	0.3513

A.3 Fixed-duration control: complete G34 tables

Tables A.4 and A.5 report all ten predefined physical summaries for G34's two full pools and eight terminal selections. Table A.6 gives the corresponding native speed and acceleration distances for every class. N42 and N43 denote the same native-clock models used in G33; "all" denotes a full pool, and U, S, V and SV retain their earlier definitions. Every row has 94 paired requests out of the original 100. A full pool contains 776 generated trajectories and 752 paired trajectories; a selected set contains 97 and 94, respectively. The class counts remain 19 buses, 18 motorcycles and 19 members of each other class. All reported metrics have that support.

G34 fixes one training-prior duration per request and model seed before scoring the shared candidate pool. Its reference episode is the already analyzed G33 episode, not a new blind sample. The following tables use the completed attempt2 summary and retain each metric's units; all distances are lower-is-better. Values are rounded to four decimal places for presentation.

Table A.4. Complete G34 macro physical distances, part I: trip scale and one-second motion summaries. Each row covers 94 paired requests out of 100.

Model / policy	Duration (s)	Distance (m)	Longest steady (s)	Four-second change (m/s)	Mean speed (m/s)
N42 / all	57.7782	4.5295	20.7243	0.2212	0.6080
N42 / U	57.7782	4.5722	20.5561	0.2155	0.6082
N42 / S	57.7782	4.4864	22.4883	0.3001	0.6079
N42 / V	57.7782	4.6340	23.5357	0.2675	0.6069
N42 / SV	57.7782	4.5187	22.7556	0.3144	0.6076
N43 / all	51.5908	4.4876	21.1753	0.1956	0.3586
N43 / U	51.5908	4.5515	21.6164	0.2416	0.3590
N43 / S	51.5908	4.4406	19.5263	0.2063	0.3586
N43 / V	51.5908	4.4136	23.1175	0.1998	0.3577
N43 / SV	51.5908	4.4905	22.0246	0.2114	0.3587

Table A.5. Complete G34 macro physical distances, part II: acceleration, stopped motion and upper speed. Stopped-fraction distances are fractions, not percentage points. Each row covers 94 paired requests out of 100.

Model / policy	Mean $\lvert a \rvert$ (m/s^2)	Stopped fraction	Speed p95 (m/s)	Longest stop (s)	Stop episodes per minute
N42 / all	0.0591	0.0180	1.9190	8.3425	0.1059
N42 / U	0.0547	0.0221	1.7914	9.2801	0.1185
N42 / S	0.0868	0.0201	2.0436	9.2865	0.1142
N42 / V	0.0710	0.0204	2.0278	7.1930	0.1125
N42 / SV	0.0896	0.0185	1.9874	8.7971	0.1079
N43 / all	0.0577	0.0157	1.7418	8.0993	0.0901
N43 / U	0.0693	0.0163	1.5543	8.1813	0.0880
N43 / S	0.0553	0.0172	1.7112	6.4591	0.0953
N43 / V	0.0562	0.0158	1.8483	7.6474	0.0861
N43 / SV	0.0604	0.0165	2.1182	7.2901	0.0883

Table A.6. Complete class-wise G34 native speed and acceleration W_1. Each cell lists speed (m/s) above acceleration (m/s^2), using the same 94-request paired population as the macro tables. Both full pools and all terminal policies are included.

Model / policy	Bus	Motorcycle	Private car	Bicycle	Taxi / ride-hailing
	0.7412	0.7595	0.7589	0.3107	0.9118
N42 / all	0.0778	0.1877	0.1208	0.1434	0.1945
	0.7129	0.9390	0.7597	0.3934	0.8594
N42 / U	0.0745	0.2073	0.1217	0.0872	0.1316
	0.7579	1.1450	0.8358	0.4140	0.9080
N42 / S	0.0890	0.1356	0.1257	0.1852	0.1502
	0.8729	0.7472	0.7720	0.4152	0.9175
N42 / V	0.0797	0.1405	0.1270	0.1019	0.1128
	0.9038	1.1317	0.6814	0.4000	0.8677
N42 / SV	0.0927	0.1348	0.1235	0.2062	0.1578
	0.6245	0.7224	0.6906	0.2492	0.4653
N43 / all	0.0800	0.2065	0.1103	0.1778	0.1939
	0.5770	0.6701	0.6441	0.2998	0.7232
N43 / U	0.1049	0.1968	0.1223	0.1678	0.2067
	0.6618	0.7591	0.6005	0.2941	0.3924
N43 / S	0.0742	0.2525	0.1146	0.1688	0.2150
	0.5973	0.7557	0.5623	0.2950	0.4191
N43 / V	0.0878	0.2119	0.1125	0.1432	0.1057
	0.6200	0.9857	0.7289	0.2790	0.3258
N43 / SV	0.0828	0.1880	0.1112	0.1660	0.1578

A.4 Five G33 cases selected before visual inspection

For G33 model seed 42, the case for each class is the lexicographically first request with an accepted reference and outputs from all four terminal policies. This rule is based on request order and common support, not physical quality, teacher confidence or visual appearance. The five plots illustrate the same declared conditions and the consequences of selecting different candidates; they are not five independent tests of a method advantage.

Each figure shows coordinate geometry, relative-time progress and a physical-time speed curve. Speed is derived from cumulative planar distance over complete one-second intervals, with the final partial second omitted from that panel. It is not the original quarter-second speed signal and cannot display all fluctuations measured by the native-interval metrics. Different selected durations are allowed in G33. Near-coincident map traces are expected under a common route condition and do not establish that the selected timing is equally realistic.

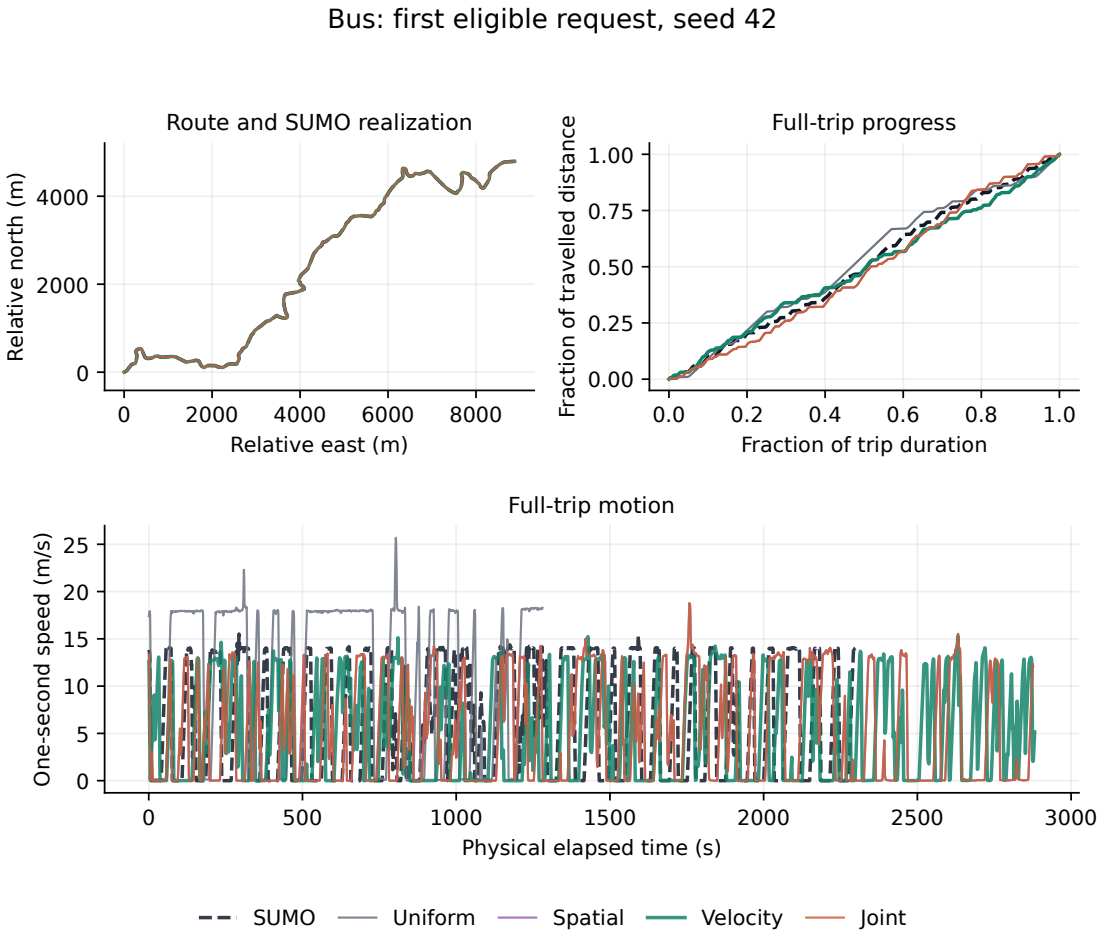


Figure A.1. Bus. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs, not a visually selected success. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

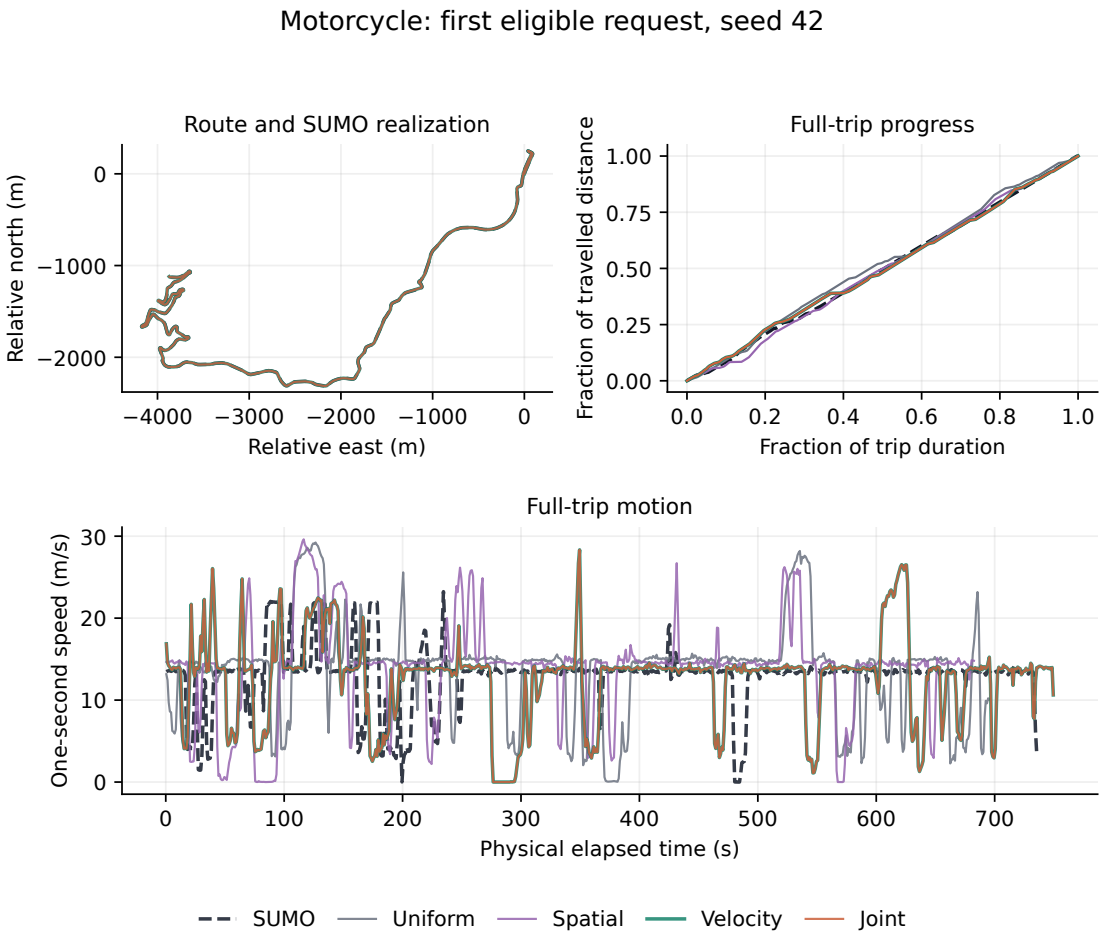


Figure A.2. Motorcycle. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs, not a visually selected success. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

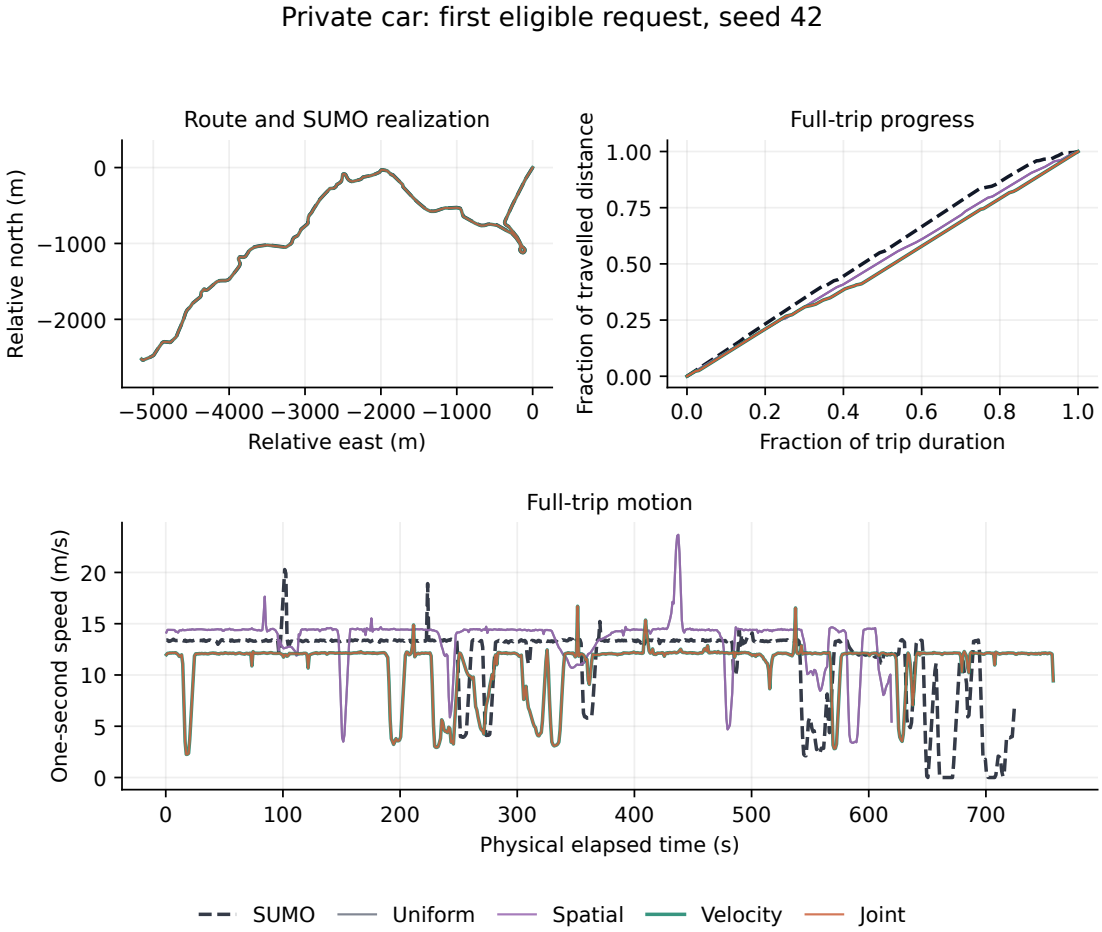


Figure A.3. Private car. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs, not a visually selected success. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

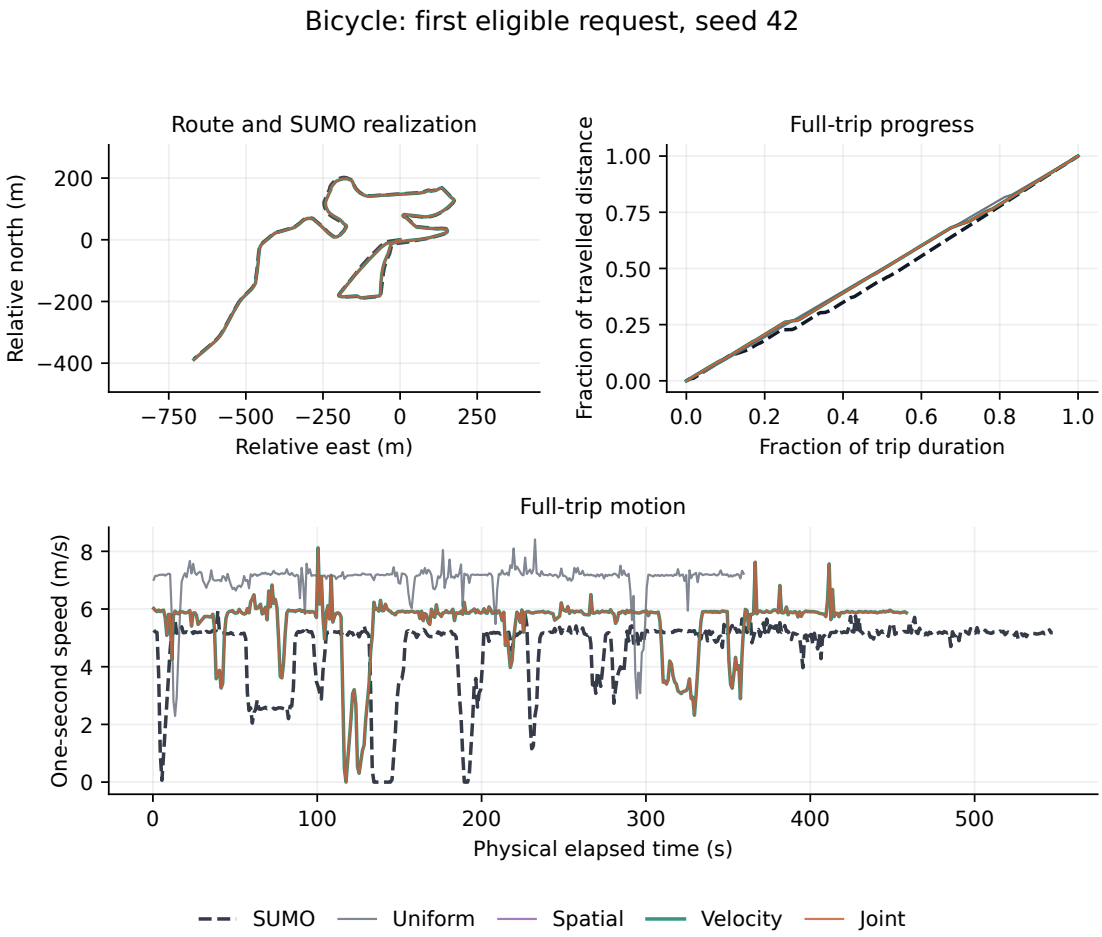


Figure A.4. Bicycle. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs, not a visually selected success. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

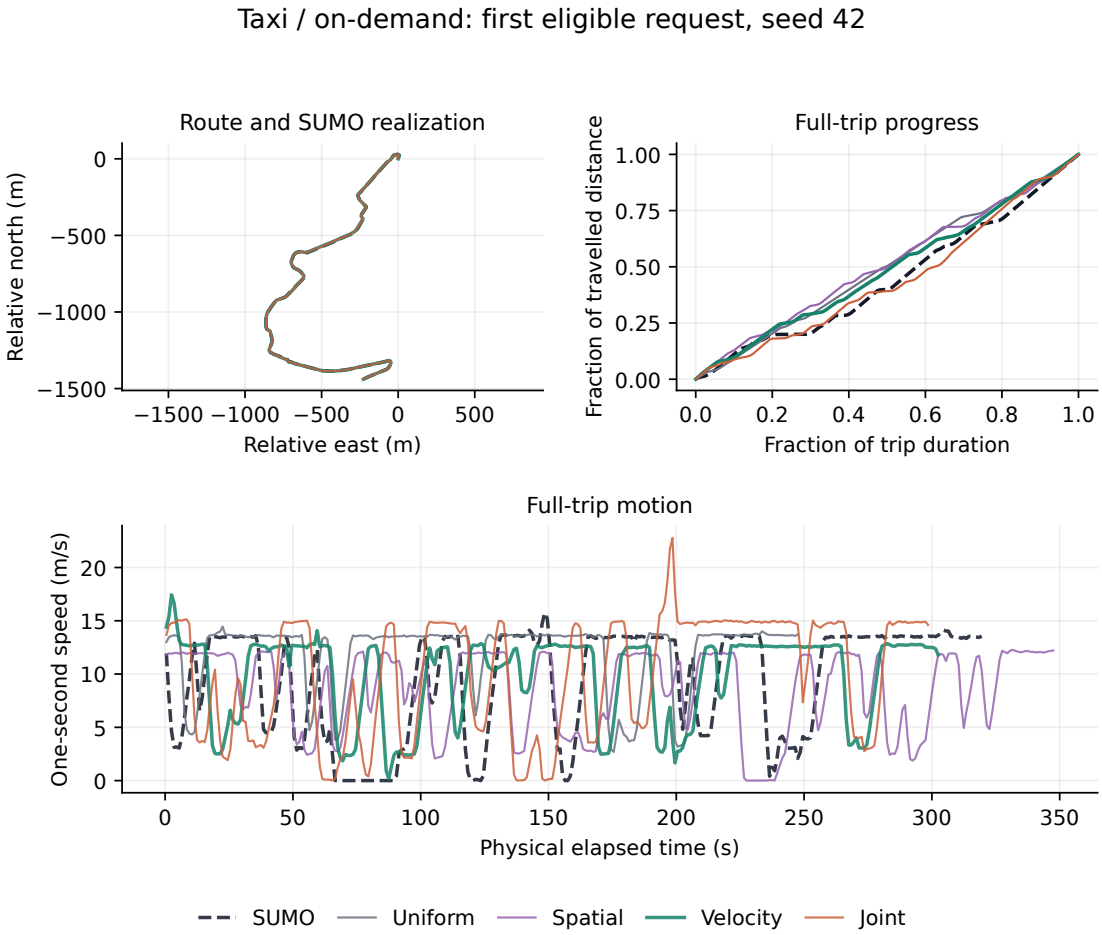


Figure A.5. Taxi or ride-hailing vehicle. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs, not a visually selected success. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

A.5 Historical evidence and reproduction scope

The earlier locked E7 evaluation is preserved in the repository, including its unfavorable feedback and augmentation findings. It began with 270 held-out participant IDs, of which 53 satisfied the strict moving-window generation eligibility rules. Its 512-point task, reference population and evaluation protocol differ from the present complete-journey experiment. The 53 eligible requests must not be presented as coverage of all 270, merged into the present 94-request quality population, or reused to select the current method after their results were exposed. The original report remains at `docs/results/final-test-e7/REPORT.md`.

The paired-representation study G28 is also retained, at `docs/results/g28-paired-latents/REPORT.md`. It trained spatial and speed readers for matching participant labels and tested latent supervision in an earlier eight-class simulator corpus. Its planned comparisons did not establish stable independent physical gains from paired spatial–speed supervision. That result remains evidence about that corpus and intervention; it is neither erased by the current temporal-resolution result nor merged with terminal selection on native-demand trajectories. The present thesis distinguishes classification information, the mechanism used to transmit it, and physical generation quality across these stages.

For the current experiment, `docs/G33_PROTOCOL.md` fixes the conditions and budgets, and `docs/results/g33-generation/report/summary.json` contains the numerical source of the G33 tables, including per-class support and provenance hashes. The G34 tables use `docs/results/g34-fixed-duration/attempt2/report/summary.json` under the separate `docs/G34_PROTOCOL.md`. `docs/results/g33-generation/paper-figures/complete.json` records the fixed case rule and figure hashes. The entry scripts `scripts/g33_train.py`, `scripts/g33_evaluate.py`, `scripts/g33_feedback.py` and `scripts/g33_plots.py` separate fitting, generation, feedback, evaluation and presentation. Reproduction requires the corresponding versioned source observations, public scenario and saved model artifacts; the manuscript does not embed raw participant records or access credentials. These relative repository paths identify the required evidence without relying on a particular server layout.

The original locked-result snapshot remains separately identified in the experiment repository [13]; its test population is not reused to select the present models.

A.6 Expanded recognition on a common development population

The full-day expansion completed all four 10,000-step diffusion fits and both fixed 50-epoch GAF readers. The native models took 2,771.9 and 2,750.8 seconds, approximately 46 minutes each; the 257-point controls took approximately 319 seconds each. Model capacity remains 891,969 parameters. The larger corpus and longer native sequences change data exposure and computational cost, so equal update counts alone do not imply equal computation. No new 257-point generation-quality claim follows merely from completing those fits.

The paired reader comparison is stronger evidence than comparing the pilot's small development set with the expanded set. Both old and new checkpoints see the same 7,230 complete development legs, and the original training populations remain separate from that cohort. Table A.7 and Figure A.6 show a substantial recognition improvement in both representations.

Table A.7. Fixed old and expanded readers on the same 7,230 development legs. Percentages; no checkpoint selection.

Reader	Accuracy	Macro F1	Taxi F1
Old Spatial	60.65	50.10	19.07
New Spatial	86.17	76.47	45.85
Old Speed	63.61	61.82	30.61
New Speed	89.13	90.34	81.07

Spatial macro F1 increases by 26.37 percentage points, with a paired parent-bootstrap 95% interval of 24.54–28.16 points. Velocity macro F1 increases by 28.52 points, with an interval of 27.27–29.73 points. These intervals condition on the available public-demand cohort and fixed model fits; they do not quantify uncertainty over cities or training recipes. The improved recognition supports the decision to preserve native demand and expand complete-trip coverage. It does not identify a pure causal effect of sample count because the time coverage and number of CNN optimizer updates also changed.

The confusion matrices explain why overall accuracy alone would be incomplete. Spatial private-car F1 is 95.20%, but taxi/ride-hailing F1 is 45.85%. The velocity values are 92.23% and 81.07%, respectively. Spatial training accuracy is 97.22%, compared with 86.17% on development; the corresponding velocity figures are 92.39% and 89.13%. Generalization gaps, calibration error and shared route templates remain relevant when transferring these readers from simulator observations to generated candidates.

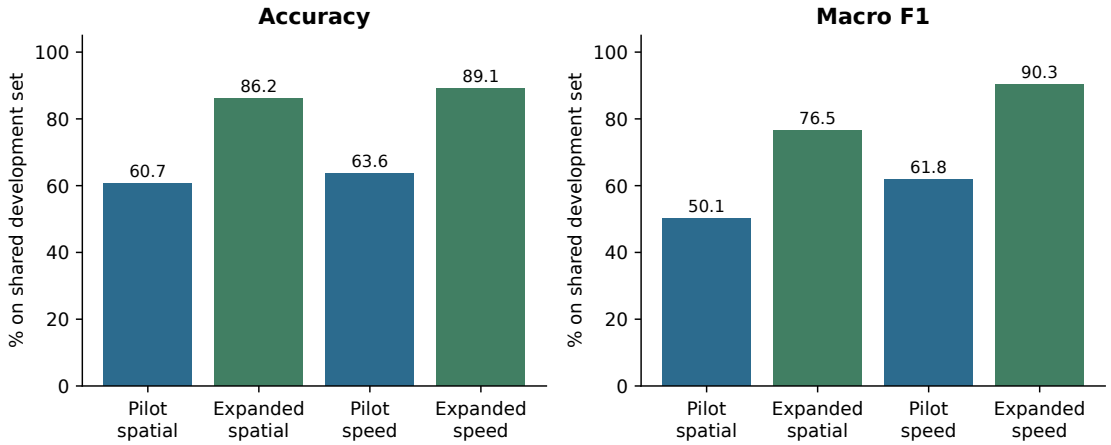


Figure A.6. Pilot and expanded readers evaluated on identical development inputs. Old/new refers to training corpus and optimization exposure, not different evaluation sets. Higher is better.

A.7 Supporting old/new generator comparison

Table A.8. Expanded versus pilot native generators, full candidate pools: class–macro native W1 changes. Each cell uses its own training-side duration prior.

Scene	Seed	Speed change	Acceleration change
Morning	42	+4.7%	+22.0%
Morning	43	+12.2%	+34.4%
Later	42	-57.2%	-15.7%
Later	43	-65.5%	-5.6%

Expansion changes the generator's applicability across the observed contexts. In the later scene, native-speed distribution error falls by 57.2% and 65.5% for the two model seeds, while acceleration error falls by 15.7% and 5.6%. In the morning scene, speed error instead rises by 4.7% and 12.2%, and acceleration error rises by 22.0% and 34.4%. The signs agree across seeds within each scene, but there are only two simulator realizations and the underlying public plans are shared.

This contrast is consistent with sensitivity to demand and temporal coverage, but it does not isolate that cause. Expanded and pilot models differ in training data and duration prior, and neither receives the instantaneous background traffic state. Departure context and simulator random seed change together between scenes. The result establishes a scene-dependent change in the delivered system, not that the diffusion network has learned an explicit time-of-day controller.

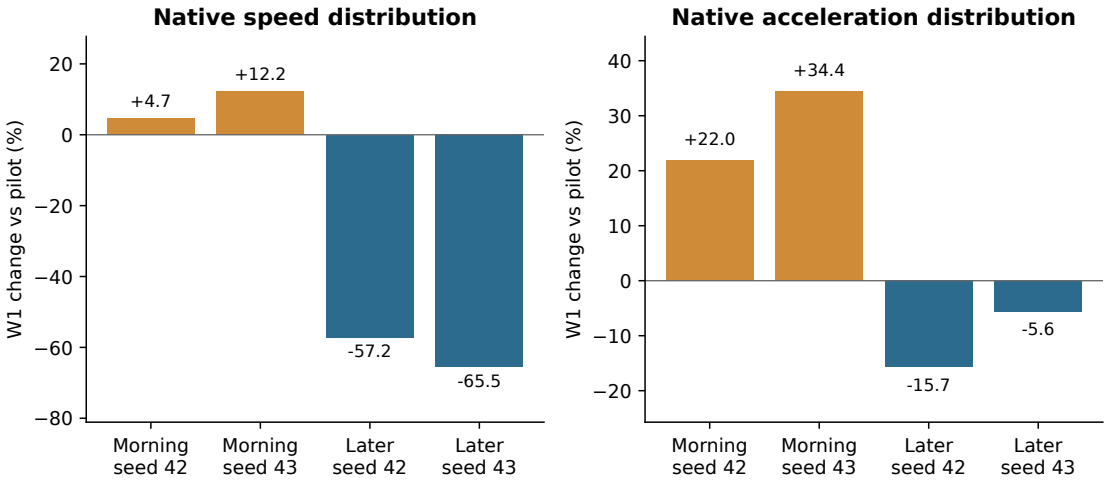


Figure A.7. Expanded versus pilot native-time generation on two new SUMO scenes. Bars show changes in full-pool five-class macro distances; negative is better. Each model uses its own training-side duration prior. The reversal across scenes prevents a universal expansion claim.

A.8 Does a more accurate GAF reader improve selection?

The central comparison applies old and expanded readers to the same expanded-generator candidates. Duration, geometry, candidate noise and the requested class are unchanged by the teacher contrast. Uniform controls are verified identical array by array. The experiment therefore distinguishes improving the generator from improving the criterion used to select its outputs.

A higher recognition score does not induce a uniformly better physical ordering. For example, the expanded spatial reader reduces longest-stop request error by 6.4% relative to the pilot spatial reader on these same pools, but raises stopped-fraction error by 1.1% and stop-frequency error by 2.8%. The expanded velocity reader reduces mean-absolute-acceleration request error by 2.8% but increases longest-stop error by 4.1% relative to the pilot velocity reader. These comparisons concern the selector with a fixed generator; they do not reuse the classifier's own target probability as a quality label.

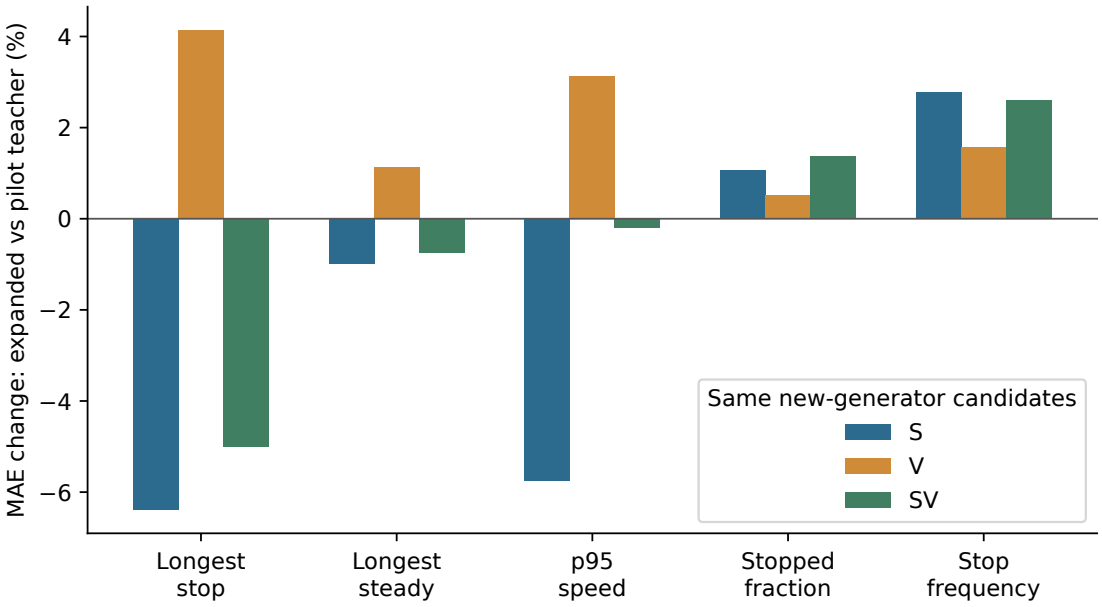


Figure A.8. Expanded versus pilot teachers on identical expanded-generator pools: changes in request-level error for five motion summaries. Negative values favour the expanded reader. These post-hoc equal-cell summaries expose gains and costs; they are not significance tests or a replacement for the full distributional matrix.

A.9 Exploratory evidence for class-specific stopping feedback

The post-hoc request-level analysis suggests a more precise mechanism to test. Spatial feedback reduces longest-stop MAE in all four scene/seed cells, with an aggregate reduction of 7.6%. This differs from its class-marginal longest-stop W1, which improves in three cells and decreases by 4.6% in the aggregate. The distinction matters: the first statement concerns correspondence to the same request's reference, while the second concerns the distribution of that scalar over requests.

For buses, the spatial longest-stop request error falls by 20.7%, with all four cells improving. For motorcycles it rises by 5.3%, with all four cells worsening. The dual policy's corresponding bus reduction is 21.1%; that small difference from spatial-only feedback is not evidence of representation synergy. These observations motivate class-sensitive stopping analysis, but the metric does not determine whether the longest stop occurs at the correct bus station. It also does not distinguish a service stop from waiting in traffic.

A mechanism claim therefore requires a fresh, fixed comparison of stop location, service-stop coverage and non-service waiting, alongside the complete class matrix.

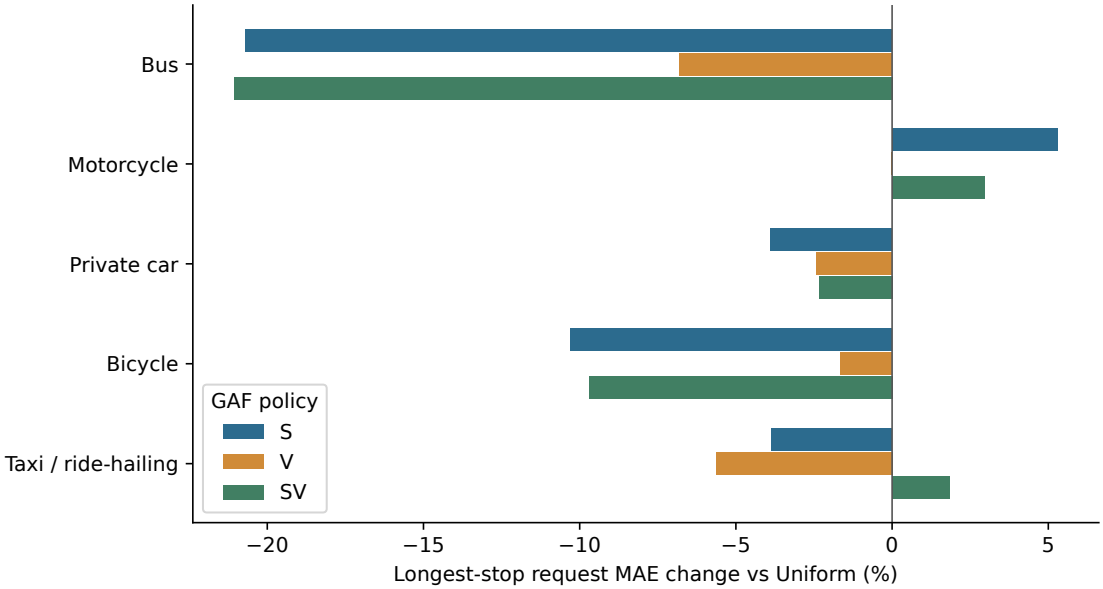


Figure A.9. Class-specific longest-stop request error after expanded GAF selection. The same five classes and all three policies are retained. The bus improvement coexists with a motorcycle cost; percentage reductions across classes are not a single common-unit effect size.

The present finding was identified after inspecting results, and its consistency over two generator seeds cannot supply an independent confirmatory test. It is valuable as a specific link between the spatial trajectory representation and an output motion property, not as proof that the group encoding has become a universal realism objective.

Full-pool pair diversity describes the variation available before selection. With one retained sample per request, conditional pair diversity is undefined, not zero; across-request dispersion cannot replace it. The common 97-request map coverage is unchanged by selection and belongs to the geometric interface.

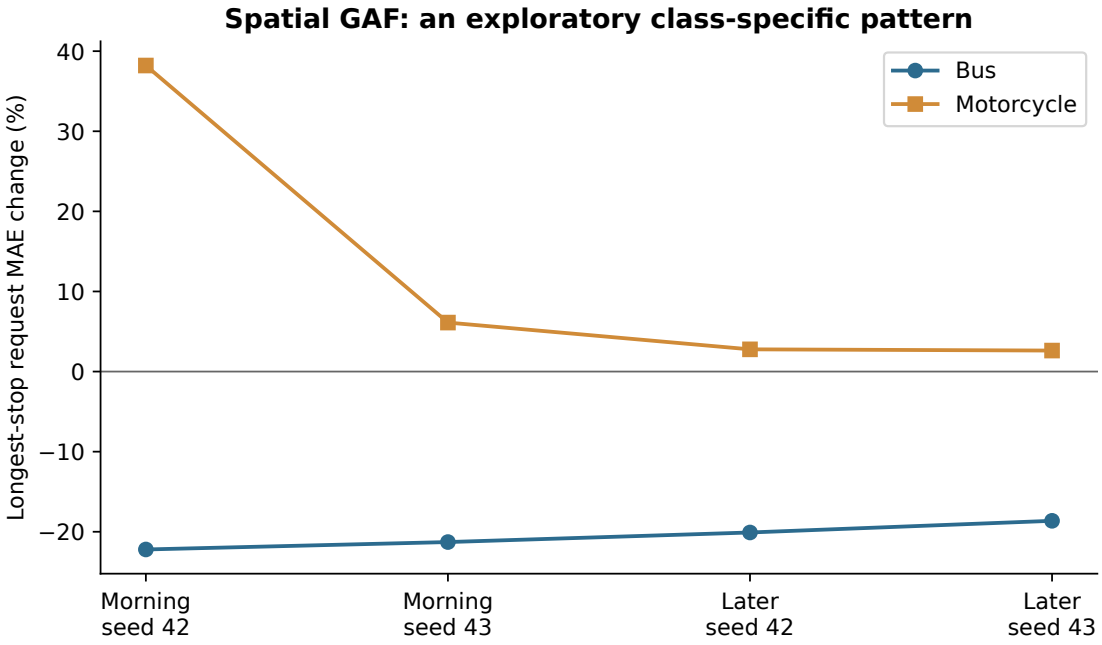


Figure A.10. Expanded spatial feedback versus Uniform for buses and motorcycles, shown separately in every scene/seed cell. The two classes illustrate opposing directions found in the complete class table. Connecting lines aid reading only; the four cells are not a time series or four independent traffic scenes.

Bibliography

[1] E. Salvadori, *Unraveling Urban Mobility: Gramian Angular Fields for Trajectory Classification*. Master's thesis, Politecnico di Torino, 2024. PoliTo WebThesis 30378.

[2] G. O. Ferreira, F. Dabbene, F. Malandrino and C. Ravazzi. Unraveling Urban Mobility: A Domain Knowledge-Free Trajectory Classification Using Gramian Angular Fields. *IEEE Transactions on Intelligent Transportation Systems*, 27, 531–543, 2026. doi:10.1109/TITS.2025.3631275.

[3] L. Codecá and J. Härri. Monaco SUMO Traffic (MoST) Scenario: A 3D Mobility Scenario for Cooperative ITS. *SUMO User Conference*, 2018. Scenario repository.

[4] J. Ho, A. Jain and P. Abbeel. Denoising Diffusion Probabilistic Models. *Advances in Neural Information Processing Systems*, 33, 2020. arXiv:2006.11239.

[5] J. Song, C. Meng and S. Ermon. Denoising Diffusion Implicit Models. *International Conference on Learning Representations*, 2021. arXiv:2010.02502.

[6] Y. Zhu, Y. Ye, S. Zhang, X. Zhao and J. Yu. DiffTraj: Generating GPS Trajectory with Diffusion Probabilistic Model. *Advances in Neural Information Processing Systems*, 36, 2023. Conference proceedings.

[7] T. Wei, Y. Lin, S. Guo, Y. Lin, Y. Huang, C. Xiang, Y. Bai and H. Wan. Diff-RNTraj: A Structure-aware Diffusion Model for Road Network-constrained Trajectory Generation. 2024. arXiv:2402.07369.

[8] Z. Lv, Z. Fan, Z. Zhang and X. Song. Two-Stage Fine-Grained Trajectory Generation Constrained by Road Networks. *IJCAI*, 2915–2923, 2026. doi:10.24963/ijcai.2026/324.

[9] E. Pronovost, M. R. Ganesina, N. Hendy, Z. Wang, A. Morales, K. Wang and N. Roy. Scenario Diffusion: Controllable Driving Scenario Generation With Diffusion. *Advances in Neural Information Processing Systems*, 36, 2023. arXiv:2311.02738.

[10] Z. Wang and T. Oates. Encoding Time Series as Images for Visual Inspection and Classification Using Tiled Convolutional Neural Networks. *AAAI workshop*, 2015. Workshop paper.

[11] K. He, X. Zhang, S. Ren and J. Sun. Deep Residual Learning for Image Recognition. *IEEE Conference on Computer Vision and Pattern Recognition*, 2016.

CVPR proceedings.

[12] Historical trajectory project. *Deep Learning Based Generative Models for Traffic Trajectory Synthesis*, revision `4773ad65`. Encoding source.

[13] Thesis experiment repository. *CNR Road/GAF Trajectory Diffusion*, results snapshot `6031e3f`. Private repository and frozen evaluation report.

[14] L. Wu, B. Trippe, C. Naesseth, D. Blei and J. Cunningham. Practical and Asymptotically Exact Conditional Sampling in Diffusion Models. *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.17775.

[15] Choi, Seongjin, Kim, Jiwon, Yeo, Hwasoo. TrajGAIL: Generating Urban Vehicle Trajectories using Generative Adversarial Imitation Learning. 2020. arXiv:2007.14189.

[16] Gao, Jiyang et al.. VectorNet: Encoding HD Maps and Agent Dynamics from Vectorized Representation. 2020. arXiv:2005.04259.

[17] Liang, Ming et al.. Learning Lane Graph Representations for Motion Forecasting. 2020. arXiv:2007.13732.

[18] Salzmann, Tim, Ivanovic, Boris, Chakravarty, Punarjay, Pavone, Marco. Trajectron++: Dynamically-Feasible Trajectory Forecasting With Heterogeneous Data. 2020. arXiv:2001.03093.

[19] Ngiam, Jiquan et al.. Scene Transformer: A unified architecture for predicting multiple agent trajectories. 2021. arXiv:2106.08417.

[20] Gupta, Agrim, Johnson, Justin, Fei-Fei, Li, Savarese, Silvio, Alahi, Alexandre. Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks. 2018. arXiv:1803.10892.

[21] Shi, Shaoshuai, Jiang, Li, Dai, Dengxin, Schiele, Bernt. Motion Transformer with Global Intention Localization and Local Movement Refinement. 2022. arXiv:2209.13508.

[22] Janner, Michael, Du, Yilun, Tenenbaum, Joshua B., Levine, Sergey. Planning with Diffusion for Flexible Behavior Synthesis. 2022. arXiv:2205.09991.

[23] Chi, Cheng et al.. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. 2023. arXiv:2303.04137.

[24] Jiang, Chiyu Max, Cornman, Andre, Park, Cheolho, Sapp, Ben, Zhou, Yin, Anguelov, Dragomir. MotionDiffuser: Controllable Multi-Agent Motion Prediction using Diffusion. 2023. arXiv:2306.03083.

[25] Dhariwal, Prafulla, Nichol, Alex. Diffusion Models Beat GANs on Image Synthesis. 2021. arXiv:2105.05233.

[26] Ho, Jonathan, Salimans, Tim. Classifier-Free Diffusion Guidance. 2022. arXiv:2207.12598.

[27] Bansal, Arpit et al.. Universal Guidance for Diffusion Models. 2023. arXiv:2302.07121.

[28] Kawar, Bahjat, Elad, Michael, Ermon, Stefano, Song, Jiaming. Denoising Diffusion Restoration Models. 2022. arXiv:2201.11793.

[29] Chung, Hyungjin, Kim, Jeongsol, Mccann, Michael T., Klasky, Marc L., Ye, Jong Chul. Diffusion Posterior Sampling for General Noisy Inverse Problems. 2022. arXiv:2209.14687.

[30] Singhal, Raghav et al.. A General Framework for Inference-time Scaling and Steering of Diffusion Models. 2025. arXiv:2501.06848.

[31] Nichol, Alex, Dhariwal, Prafulla. Improved Denoising Diffusion Probabilistic Models. 2021. arXiv:2102.09672.

[32] Karras, Tero, Aittala, Miika, Aila, Timo, Laine, Samuli. Elucidating the Design Space of Diffusion-Based Generative Models. 2022. arXiv:2206.00364.

[33] Lu, Cheng, Zhou, Yuhao, Bao, Fan, Chen, Jianfei, Li, Chongxuan, Zhu, Jun. DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps. 2022. arXiv:2206.00927.

[34] Rasul, Kashif, Seward, Calvin, Schuster, Ingmar, Vollgraf, Roland. Autoregressive Denoising Diffusion Models for Multivariate Probabilistic Time Series Forecasting. 2021. arXiv:2101.12072.

[35] Tashiro, Yusuke, Song, Jiaming, Song, Yang, Ermon, Stefano. CSDI: Conditional Score-based Diffusion Models for Probabilistic Time Series Imputation. 2021. arXiv:2107.03502.

[36] Kong, Zhifeng, Ping, Wei, Huang, Jiaji, Zhao, Kexin, Catanzaro, Bryan. DiffWave: A Versatile Diffusion Model for Audio Synthesis. 2020. arXiv:2009.09761.

[37] Bai, Shaojie, Kolter, J. Zico, Koltun, Vladlen. An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling. 2018. arXiv:1803.01271.

[38] Cho, Kyunghyun et al.. Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation. 2014. arXiv:1406.1078.

[39] Fawaz, Hassan Ismail et al.. InceptionTime: Finding AlexNet for Time Series Classification. 2019. arXiv:1909.04939.

[40] Dempster, Angus, Petitjean, François, Webb, Geoffrey I.. ROCKET: Exceptionally fast and accurate time series classification using random convolutional kernels. 2019. arXiv:1910.13051.

[41] Dempster, Angus, Schmidt, Daniel F., Webb, Geoffrey I.. MINIROCKET: A Very Fast (Almost) Deterministic Transform for Time Series Classification. 2020. arXiv:2012.08791.

[42] Franceschi, Jean-Yves, Dieuleveut, Aymeric, Jaggi, Martin. Unsupervised Scalable Representation Learning for Multivariate Time Series. 2019. arXiv:1901.10738.

[43] Lopez-Paz, David, Oquab, Maxime. Revisiting Classifier Two-Sample Tests. 2016. arXiv:1610.06545.

[44] Sajjadi, Mehdi S. M., Bachem, Olivier, Lucic, Mario, Bousquet, Olivier, Gelly, Sylvain. Assessing Generative Models via Precision and Recall. 2018. arXiv:1806.00035.

[45] Kynkäänniemi, Tuomas, Karras, Tero, Laine, Samuli, Lehtinen, Jaakko, Aila, Timo. Improved Precision and Recall Metric for Assessing Generative Models. 2019. arXiv:1904.06991.

[46] Esteban, Cristóbal, Hyland, Stephanie L., Rätsch, Gunnar. Real-valued (Medical) Time Series Generation with Recurrent Conditional GANs. 2017. arXiv:1706.02633.

[47] Eclipse SUMO project. FCD Output and Randomness documentation. Accessed 5 November 2026. Floating-car output; random seeds.

[48] C. Guo, G. Pleiss, Y. Sun and K. Q. Weinberger. On Calibration of Modern Neural Networks. *ICML*, PMLR 70, 1321–1330, 2017. Conference proceedings.

[49] S. Kornblith, M. Norouzi, H. Lee and G. Hinton, Similarity of Neural Network Representations Revisited, *Proceedings of ICML*, PMLR 97, 2019, pp. 3519–3529. Proceedings page.

[50] S. Yu, S. Kwak, H. Jang, J. Jeong, J. Huang, J. Shin and S. Xie. Representation Alignment for Generation: Training Diffusion Transformers Is Easier Than You Think. *ICLR*, 2025. arXiv:2410.06940.

[51] Eclipse SUMO project. Vehicle Type Parameter Defaults. Accessed 6 November 2026. Official documentation.

[52] T. Gneiting and A. E. Raftery. Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, 102(477), 359–378, 2007. doi:10.1198/016214506000001437.